reactome

Pathway Analysis Report

This report contains the pathway analysis results for the submitted sample ''. Analysis was performed against Reactome version 95 on 19/12/2025. The web link to these results is:

https://reactome.org/PathwayBrowser/#/ANALYSIS=MjAyNTEyMTkwODUzMzJfNTM0MA%3D%3D

Please keep in mind that analysis results are temporarily stored on our server. The storage period depends on usage of the service but is at least 7 days. As a result, please note that this URL is only valid for a limited time period and it might have expired.

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1. Introduction

Reactome is a curated database of pathways and reactions in human biology. Reactions can be considered as pathway 'steps'. Reactome defines a 'reaction' as any event in biology that changes the state of a biological molecule. Binding, activation, translocation, degradation and classical biochemical events involving a catalyst are all reactions. Information in the database is authored by expert biologists, entered and maintained by Reactome’s team of curators and editorial staff. Reactome content frequently cross-references other resources e.g. NCBI, Ensembl, UniProt, KEGG (Gene and Compound), ChEBI, PubMed and GO. Orthologous reactions inferred from annotation for Homo sapiens are available for 14 non-human species including mouse, rat, chicken, puffer fish, worm, fly and yeast. Pathways are represented by simple diagrams following an SBGN-like format.

Reactome's annotated data describe reactions possible if all annotated proteins and small molecules were present and active simultaneously in a cell. By overlaying an experimental dataset on these annotations, a user can perform a pathway over-representation analysis. By overlaying quantitative expression data or time series, a user can visualize the extent of change in affected pathways and its progression. A binomial test is used to calculate the probability shown for each result, and the p-values are corrected for the multiple testing (Benjamini–Hochberg procedure) that arises from evaluating the submitted list of identifiers against every pathway.

To learn more about our Pathway Analysis, please have a look at our relevant publications:

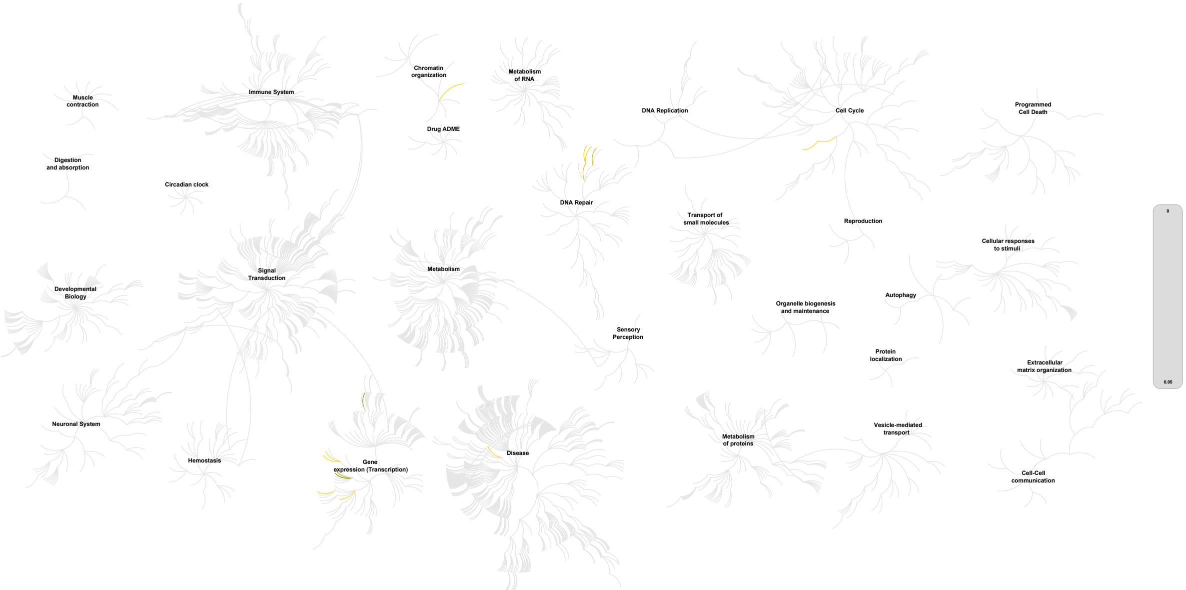
Fabregat A, Sidiropoulos K, Garapati P, Gillespie M, Hausmann K, Haw R, … D’Eustachio P (2016). The reactome pathway knowledgebase. Nucleic Acids Research, 44(D1), D481–D487. https://doi.org/10.1093/nar/gkv1351.

Fabregat A, Sidiropoulos K, Viteri G, Forner O, Marin-Garcia P, Arnau V, … Hermjakob H (2017). Reactome pathway analysis: a high-performance in-memory approach. BMC Bioinformatics, 18.

2. Properties

- This is an **overrepresentation** analysis: A statistical (hypergeometric distribution) test that determines whether certain Reactome pathways are over-represented (enriched) in the submitted data. It answers the question 'Does my list contain more proteins for pathway X than would be expected by chance?' This test produces a probability score, which is corrected for false discovery rate using the Benjamani-Hochberg method.
- 1423 out of 4027 identifiers in the sample were found in Reactome, where 2343 pathways were hit by at least one of them.
- All non-human identifiers have been converted to their human equivalent.
- IntAct interactors were included to increase the analysis background. This greatly increases the size of Reactome pathways, which maximises the chances of matching your submitted identifiers to the expanded pathway, but will include interactors that have not undergone manual curation by Reactome and may include interactors that have no biological significance, or unexplained relevance.
- This report is filtered to show only results for species 'Homo sapiens' and resource 'all resources'.
- The unique ID for this analysis (token) is MjAyNTEyMTkwODUzMzJfNTM0MA%3D%3D. This ID is valid for at least 7 days in Reactome's server. Use it to access Reactome services with your data.

3. Genome-wide overview



This figure shows a genome-wide overview of the results of your pathway analysis. Reactome pathways are arranged in a hierarchy. The center of each of the circular "bursts" is the root of one top-level pathway, for example "DNA Repair". Each step away from the center represents the next level lower in the pathway hierarchy. The color code denotes over-representation of that pathway in your input dataset. Light grey signifies pathways which are not significantly over-represented.

4. Most significant pathways

The following table shows the 25 most relevant pathways sorted by p-value.

Pathway name	Entities				Reactions	
	found	ratio	p-value	FDR*	found	ratio
RNA Polymerase I Promoter Escape	34 / 62	0.002	2.14e-06	0.005	2 / 2	1.26e-04
Regulation of endogenous retroelements by the Human Silencing Hub (HUSH) complex	25 / 48	0.002	2.38e-04	0.173	6 / 8	5.04e-04
HDACs deacetylate histones	33 / 73	0.003	2.60e-04	0.173	5 / 5	3.15e-04
RNA Polymerase I Promoter Opening	22 / 41	0.002	4.14e-04	0.173	1 / 2	1.26e-04
Deposition of new CENPA-containing nucleosomes at the centromere	25 / 53	0.002	4.16e-04	0.173	4 / 4	2.52e-04
Nucleosome assembly	25 / 53	0.002	4.16e-04	0.173	4 / 4	2.52e-04
PRC2 methylates histones and DNA	22 / 46	0.002	7.46e-04	0.265	4 / 4	2.52e-04
Defective pyroptosis	23 / 51	0.002	0.001	0.383	2 / 3	1.89e-04
Recognition and association of DNA glycosylase with site containing an affected purine	20 / 41	0.002	0.002	0.64	4 / 10	6.30e-04
Cleavage of the damaged purine	19 / 44	0.002	0.005	1	2 / 9	5.67e-04
Depurination	20 / 48	0.002	0.011	1	6 / 19	0.001
Cleavage of the damaged pyrimidine	19 / 50	0.002	0.017	1	1 / 20	0.001
Transcriptional regulation by small RNAs	34 / 83	0.003	0.031	1	5 / 5	3.15e-04
Regulation of NPAS4 mRNA translation	6 / 11	4.43e-04	0.036	1	2 / 2	1.26e-04
Small interfering RNA (siRNA) biogenesis	5 / 9	3.62e-04	0.05	1	5 / 5	3.15e-04
Chromatin modifications during the maternal to zygotic transition (MZT)	24 / 72	0.003	0.051	1	11 / 12	7.56e-04
Regulation of endogenous retroelements by Piwi-interacting RNAs (piRNAs)	28 / 88	0.004	0.055	1	2 / 2	1.26e-04
Assembly of the ORC complex at the origin of replication	28 / 77	0.003	0.058	1	10 / 11	6.93e-04
Competing endogenous RNAs (ceRNAs) regulate PTEN translation	8 / 19	7.65e-04	0.061	1	11 / 11	6.93e-04
Post-transcriptional silencing by small RNAs	4 / 7	2.82e-04	0.07	1	3 / 3	1.89e-04
Gap-filling DNA repair synthesis and ligation in TC-NER	20 / 64	0.003	0.076	1	2 / 2	1.26e-04
Gap-filling DNA repair synthesis and ligation in GG-NER	10 / 27	0.001	0.077	1	2 / 2	1.26e-04

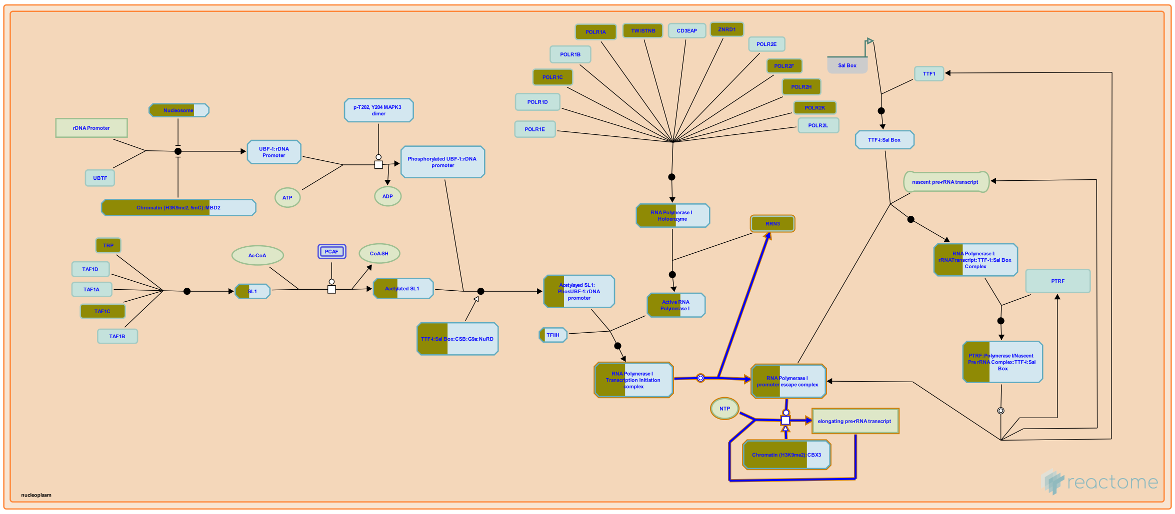
Pathway name	Entities				Reactions	
	found	ratio	p-value	FDR*	found	ratio
DNA methylation	22 / 68	0.003	0.077	1	7 / 7	4.41e-04
p75NTR negatively regulates cell cycle via SC1	5 / 9	3.62e-04	0.137	1	3 / 3	1.89e-04
E2F-enabled inhibition of pre-replication complex formation	4 / 9	3.62e-04	0.137	1	1 / 1	6.30e-05

* False Discovery Rate

5. Pathways details

For every pathway of the most significant pathways, we present its diagram, as well as a short summary, its bibliography and the list of inputs found in it.

1. RNA Polymerase I Promoter Escape (R-HSA-73772)



Cellular compartments: nucleolus.

As the active RNA Polymerase I complex leaves the promoter Rrn3 dissociates from the complex. RNA polymerase I Promoter Clearance is complete and Chain Elongation begins (Milkereit and Tschochner, 1998).
The assembly of the initiation complex on the promoter and the transition from a closed to an open complex is then followed by promoter clearance and transcription elongation by RNA Pol I. Unlike the RNA polymerase II system, RNA polymerase I transcription does not require a form of energy such as ATP for initiation and elongation. Regulatory mechanisms operating at both the level of transcription initiation and elongation probably concurrently to adjust the level of rRNA synthesis to the need of the cell.

References

Milkereit P & Tschochner H (1998). A specialized form of RNA polymerase I, essential for initiation and growth-dependent regulation of rRNA synthesis, is disrupted during transcription. EMBO J., 17, 3692-703.

Peyroche G, Milkereit P, Bischler N, Tschochner H, Schultz P, Sentenac A, ... Riva M (2000). The recruitment of RNA polymerase I on rDNA is mediated by the interaction of the A43 subunit with Rrn3. EMBO J., 19, 5473-82.

Edit history

Date	Action	Author
2003-07-03	Authored	Comai L
2003-07-03	Created	Comai L
2025-11-12	Edited	Gillespie ME

Date	Action	Author
2025-11-15	Modified	Weiser JD

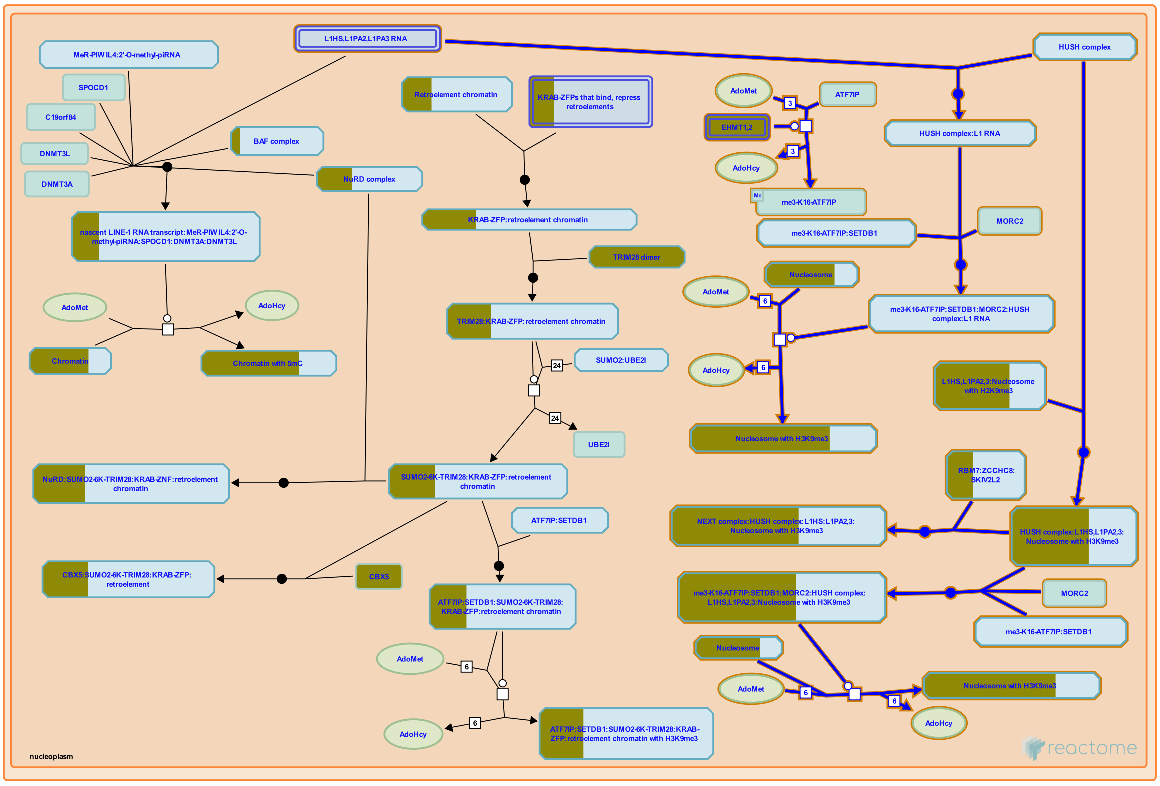
63 submitted entities found in this pathway, mapping to 75 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CBX3	Q13185	GTF2H3	Q13889	GTF2H4	Q92759
H2AC12	Q99878	H2AC14	Q99878	H2AC16	P20671
H2AC18	Q6FI13	H2AC19	Q6FI13	H2AC20	Q16777
H2AC4	P04908, Q99878	H2AC6	Q93077	H2AC7	P20671
H2AC8	P04908	H2BC10	P62807	H2BC11	P06899
H2BC12	O60814, Q93079, Q99877, Q99880	H2BC12L	P57053	H2BC13	Q99880
H2BC14	Q99879	H2BC15	Q93079, Q99877	H2BC17	P23527
H2BC18	Q99877	H2BC21	P06899, Q16778	H2BC3	P06899, P33778
H2BC4	P62807, Q93079	H2BC5	P58876	H2BC6	P62807, Q93079
H2BC7	P62807	H2BC8	P62807, Q93079	H2BC9	P58876, Q93079, Q99877
H3C1	P68431	H3C11	P68431	H3C12	P68431
H3C13	Q71DI3	H3C14	Q71DI3	H3C15	Q71DI3
H3C2	P68431	H3C3	P68431	H3C4	P68431
H3C6	P68431	H3C7	P68431	H3C8	P68431
H4C1	P62805	H4C13	P62805	H4C16	P62805
H4C2	P62805	H4C3	P62805	H4C4	P62805
H4C5	P62805	H4C6	P62805	H4C8	P62805
H4C9	P62805	MT1A	P53803	POLR1C	O15160
POLR1F	Q3B726	POLR1H	Q9P1U0	POLR2F	P61218
POLR2H	P52434	POLR2K	P53803	RPA1	O95602
RRN3	Q9NYV6	TAF1C	Q15572	TBP	P20226

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
POLR1F	Q3B726	Q9NYV6			

2. Regulation of endogenous retroelements by the Human Silencing Hub (HUSH) complex (R-HSA-9843970)



The Human Silencing Hub (HUSH) complex comprises MPHOSPH8 (MPP8), Periphilin (PHPLN1), and TASOR, which appears to act as a scaffold that binds the other subunits (reviewed in Seczynska and Lehner 2023). The HUSH complex preferentially represses transcription of young LINE1 retroelements (Liu et al. 2018, Robbez-Masson et al. 2018) and some HIV integrants (Tchasovnikarova et al. 2015, Zhu et al. 2018, Chougui and Margottin-Goguet 2019).

The HUSH complex creates repressive chromatin in two ways (reviewed in Seczynska and Lehner 2023). Firstly, HUSH can cause spreading of existing heterochromatin by binding existing trimethylated H3K9 and recruiting SETDB1 to trimethylate H3K9 of adjacent nucleosomes (Tchasovnikarova et al. 2015). Secondly, HUSH can initiate heterochromatin by binding nascent transcripts via its PHPLN1 subunit and recruiting SETDB1 to trimethylate lysine-9 of histone H3 (H3K9) at the locus being transcribed (Seczynska et al. 2022).

By an uncharacterized mechanism, the HUSH complex targets long intronless cDNAs, such as those produced by retroelements, as well as unusually long exons of normal cellular genes (Seczynska et al. 2022). Introns somehow protect against silencing by HUSH, though actual splicing is not required (Seczynska et al. 2022). In mice, ZNF638 (NP220, Znf638 gene) recruits the HUSH complex to unintegrated murine leukemia virus (MLV) (Zhu et al. 2018) and in human cells ZNF638 (NP220) and HUSH silence recombinant adeno-associated viruses (Das et al. 2022). The potential recruitment of HUSH by ZNF638 to human retroelements is not yet demonstrated.

References

Seczynska M & Lehner PJ (2023). The sound of silence: mechanisms and implications of HUSH complex function. Trends Genet, 39, 251-267.

Zhu Y, Wang GZ, Cingöz O & Goff SP (2018). NP220 mediates silencing of unintegrated retroviral DNA. Nature, 564, 278-282.

Das A, Vijayan M, Walton EM, Stafford VG, Fiflis DN & Asokan A (2022). Epigenetic Silencing of Recombinant Adeno-associated Virus Genomes by NP220 and the HUSH Complex. J Virol, 96, e0203921.

Tchasovnikarova IA, Timms RT, Matheson NJ, Wals K, Antrobus R, Göttgens B, ... Lehner PJ (2015). GENE SILENCING. Epigenetic silencing by the HUSH complex mediates position-effect variegation in human cells. Science, 348, 1481-1485.

Seczynska M, Bloor S, Cuesta SM & Lehner PJ (2022). Genome surveillance by HUSH-mediated silencing of intronless mobile elements. Nature, 601, 440-445.

Edit history

Date	Action	Author
2023-09-11	Edited	May B
2023-09-11	Authored	May B
2023-09-12	Created	May B
2024-05-14	Reviewed	Fernandes LP, Bourc'his D
2025-11-15	Modified	Weiser JD

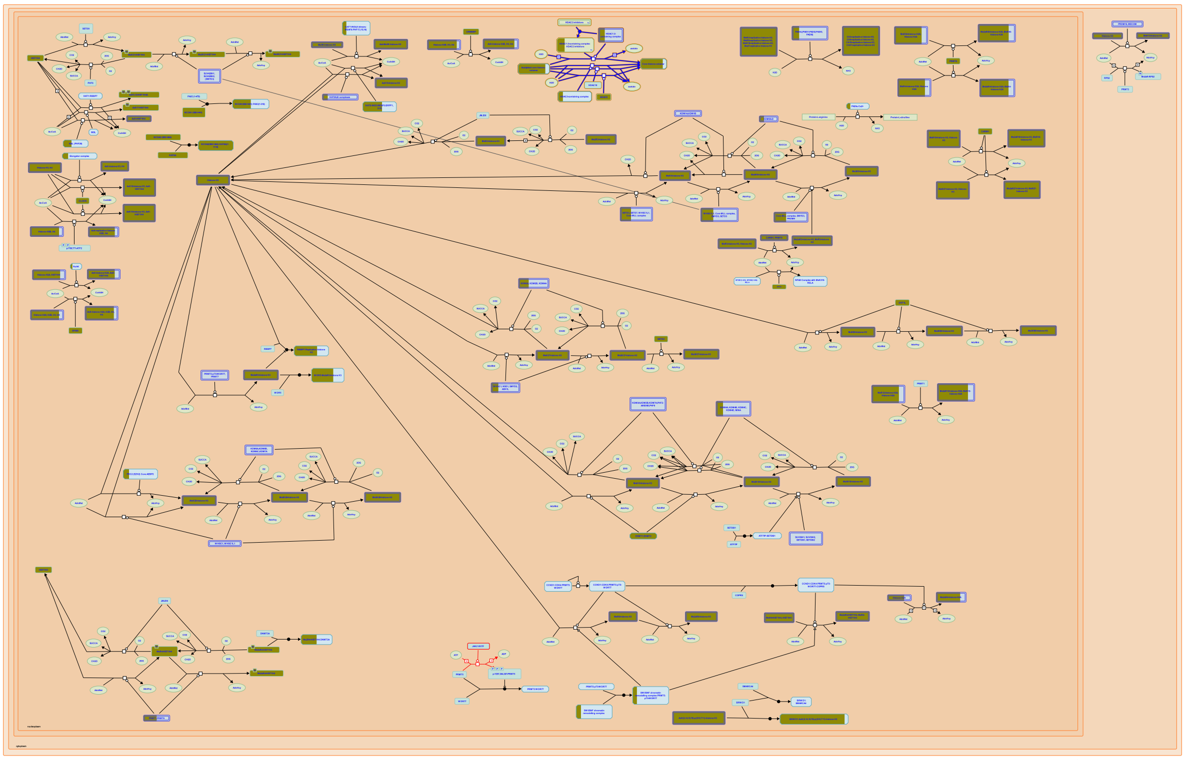
52 submitted entities found in this pathway, mapping to 64 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
EHMT1	Q9H9B1	EHMT2	Q96KQ7	H2AC12	Q99878
H2AC14	Q99878	H2AC16	P20671	H2AC18	Q6FI13
H2AC19	Q6FI13	H2AC20	Q16777	H2AC4	P04908, Q99878
H2AC6	Q93077	H2AC7	P20671	H2AC8	P04908
H2BC10	P62807	H2BC11	P06899	H2BC12	O60814, Q93079, Q99877, Q99880
H2BC12L	P57053	H2BC13	Q99880	H2BC14	Q99879
H2BC15	Q93079, Q99877	H2BC17	P23527	H2BC18	Q99877
H2BC21	P06899, Q16778	H2BC3	P06899, P33778	H2BC4	P62807, Q93079
H2BC5	P58876	H2BC6	P62807, Q93079	H2BC7	P62807
H2BC8	P62807, Q93079	H2BC9	P58876, Q93079, Q99877	H3C1	P68431
H3C11	P68431	H3C12	P68431	H3C13	Q71DI3
H3C14	Q71DI3	H3C15	Q71DI3	H3C2	P68431
H3C3	P68431	H3C4	P68431	H3C6	P68431
H3C7	P68431	H3C8	P68431	H4C1	P62805
H4C13	P62805	H4C16	P62805	H4C2	P62805
H4C3	P62805	H4C4	P62805	H4C5	P62805
H4C6	P62805	H4C8	P62805	H4C9	P62805
RBM7	Q9Y580				

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
VHL	P40337	Q6VMQ6			

3. HDACs deacetylate histones (R-HSA-3214815)



Lysine deacetylases (KDACs), historically referred to as histone deacetylases (HDACs), are divided into the Rpd3/Hda1 metal-dependent 'classical HDAC family' (de Ruijter et al. 2003, Verdin et al. 2003) and the unrelated sirtuins (Milne & Denu 2008). Phylogenetic analysis divides human KDACs into four classes (Gregoretti et al. 2004): Class I includes HDAC1, 2, 3 and 8; Class IIa includes HDAC4, 5, 7 and 9; Class IIb includes HDAC6 and 10; Class III are the sirtuins (SIRT1-7); Class IV has one member, HDAC11 (Gao et al. 2002). Class III enzymes use an NAD+ cofactor to perform deacetylation (Milne & Denu 2008, Yang & Seto 2008), the others classes use a metal-dependent mechanism (Gregoretti et al. 2004) to catalyze the hydrolysis of acetyl-L-lysine side chains in histone and non-histone proteins yielding L-lysine and acetate. X-ray crystal structures are available for four human HDACs; these structures have conserved active site residues, suggesting a common catalytic mechanism (Lombardi et al. 2011). They require a single transition metal ion and are typically studied in vitro as Zn2+-containing enzymes, though in vivo HDAC8 exhibits increased activity when substituted with Fe2+ (Gantt et al. 2006). The structurally-related enzyme acetylpolyamine amidohydrolase (APAH) (Leipe & Landsman 1997) exhibits optimal activity with Mn2+, followed closely by Zn2+ (Sakurada et al. 1996).

HDACs are often part of multi-protein transcriptional complexes that are recruited to gene promoters, regulating transcription without direct DNA binding. With the exception of HDAC8, all class I members can be catalytic subunits of multiprotein complexes (Yang & Seto 2008). HDAC1 and HDAC2 interact to form the catalytic core of several multisubunit complexes including Sin3, nucleosome remodeling deacetylase (NuRD) and corepressor of REST (CoREST) complexes (Grozinger & Schreiber 2002). HDAC3 is part of the silencing mediator of retinoic acid and thyroid hormone receptor (SMRT) complex or the homologous nuclear receptor corepressor (NCoR) (Li et al. 2000, Wen et al. 2000, Zhang et al. 2002, Yoon et al. 2003, Oberoi et al. 2011) which are involved in a wide range of processes including metabolism, inflammation, and circadian rhythms (Mottis et al. 2013).

Class IIa HDACs (HDAC4, -5, -7, and -9) shuttle between the nucleus and cytoplasm (Yang & Seto 2008, Haberland et al. 2009). The nuclear export of class IIa HDACs requires phosphorylation stimulated by calcium or other stimuli. They appear to have been evolutionarily inactivated as enzymes, having acquired a histidine substitution of the tyrosine residue in the active site of the mammalian deacetylase domain (H976 in humans) (Lahm et al. 2007, Schuetz et al. 2008). Instead they function as transcriptional corepressors for the MEF2 family of transcription factors (Yang & Gregoire 2005) .

Histones are the primary substrate for most HDACs except HDAC6 which is predominantly cytoplasmic and acts on alpha-tublin (Hubbert et al. 2002, Zhang et al. 2003, Boyault et al. 2007). HDACs also deacetylate proteins such as p53, E2F1, RelA, YY1, TFIIE, BCL6 and TFIIF (Glozak et al. 2005).

Histone deacetylases are targeted by structurally diverse compounds known as HDAC inhibitors (HDIs) (Marks et al. 2000). These can induce cytodifferentiation, cell cycle arrest and apoptosis of transformed cells (Marks et al. 2000, Bolden et al. 2006). Some HDIs have significant antitumor activity (Marks and Breslow 2007, Ma et al. 2009) and at least two are approved anti-cancer drugs.

The coordinates of post-translational modifications represented and described here follow UniProt standard practice whereby coordinates refer to the translated protein before any further processing. Histone literature typically refers to coordinates of the protein after the initiating methionine has been removed. Therefore the coordinates of post-translated residues in the Reactome database and described here are frequently +1 when compared with the literature.

References

de Ruijter AJ, van Gennip AH, Caron HN, Kemp S & van Kuilenburg AB (2003). Histone deacetylases (HDACs): characterization of the classical HDAC family. Biochem. J., 370, 737-49.

Lombardi PM, Cole KE, Dowling DP & Christianson DW (2011). Structure, mechanism, and inhibition of histone deacetylases and related metalloenzymes. Curr. Opin. Struct. Biol., 21, 735-43.

Edit history

Date	Action	Author
2013-03-12	Edited	Jupe S
2013-03-12	Authored	Jupe S
2013-03-12	Created	Jupe S
2014-05-08	Reviewed	Yang XJ
2025-11-15	Modified	Weiser JD

61 submitted entities found in this pathway, mapping to 77 Reactome entities

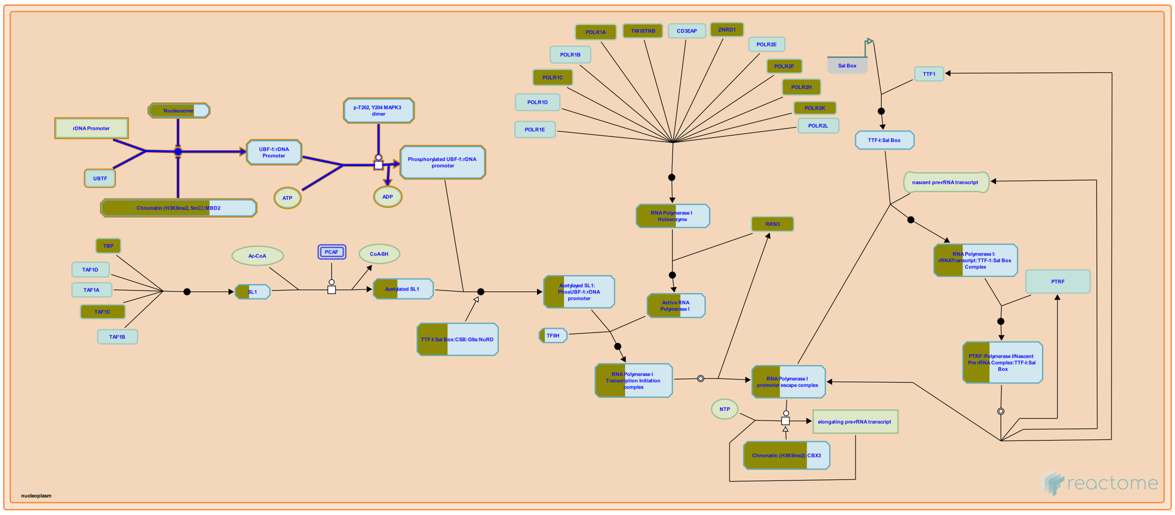
Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
GATAD2B	Q8WXI9	H2AC11	P0C0S8	H2AC12	Q96KK5, Q99878
H2AC13	P0C0S8, Q96KK5	H2AC14	Q99878	H2AC15	P0C0S8
H2AC16	P0C0S8, P20671	H2AC17	P0C0S8	H2AC18	Q6FI13
H2AC19	Q6FI13	H2AC20	Q16777	H2AC21	Q8IUE6
H2AC4	P04908, Q99878	H2AC6	Q93077	H2AC7	P20671
H2AC8	P04908	H2BC10	P62807	H2BC11	P06899
H2BC12	O60814, Q93079, Q99877, Q99880	H2BC13	Q99880	H2BC14	Q99879

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
H2BC15	Q93079, Q99877	H2BC17	P23527	H2BC18	Q5QNW6, Q99877
H2BC21	P06899, Q16778	H2BC3	P06899, P33778	H2BC4	P62807, Q93079
H2BC5	P58876	H2BC6	P62807, Q93079	H2BC7	P62807
H2BC8	P62807, Q93079	H2BC9	P58876, Q93079, Q99877	H3C1	P68431
H3C11	P68431	H3C12	P68431	H3C13	Q71DI3
H3C14	Q71DI3	H3C15	Q71DI3	H3C2	P68431
H3C3	P68431	H3C4	P68431	H3C6	P68431
H3C7	P68431	H3C8	P68431	H4C1	P62805
H4C13	P62805	H4C16	P62805	H4C2	P62805
H4C3	P62805	H4C4	P62805	H4C5	P62805
H4C6	P62805	H4C8	P62805	H4C9	P62805
HDAC2	Q92769	HDAC8	Q9BY41	MTA1	Q13330
MTA2	O94776	PHF21A	Q96BD5	REST	Q13127
TBL1X	O60907				

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
PEX26	Q7Z412	Q969S8			

4. RNA Polymerase I Promoter Opening (R-HSA-73728)



Cellular compartments: nucleolus.

The activity of the upstream binding factor (UBF-1) plays an important role in the regulation of rRNA synthesis. Studies reveal that phosphorylation of UBF-1 is required for its interaction with the RNA polymerase I complex, suggesting that phosphorylation of UBF-1 bound to the rDNA promoter during promoter opening modulates the assembly of the transcription initiation complex.

References

Edit history

Date	Action	Author
2003-07-03	Authored	Comai L
2003-07-03	Created	Comai L
2025-11-12	Edited	Gillespie ME
2025-11-15	Modified	Weiser JD

49 submitted entities found in this pathway, mapping to 61 Reactome entities

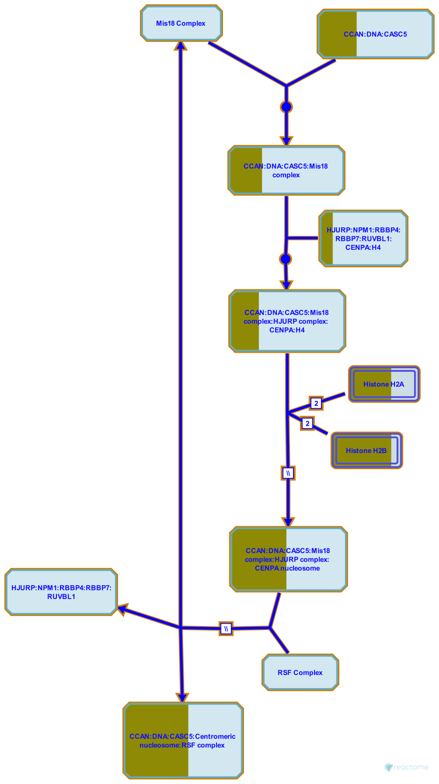
Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
H2AC12	Q99878	H2AC14	Q99878	H2AC16	P20671
H2AC18	Q6FI13	H2AC19	Q6FI13	H2AC20	Q16777
H2AC4	P04908, Q99878	H2AC6	Q93077	H2AC7	P20671
H2AC8	P04908	H2BC10	P62807	H2BC11	P06899
H2BC12	O60814, Q93079, Q99877, Q99880	H2BC12L	P57053	H2BC13	Q99880
H2BC14	Q99879	H2BC15	Q93079, Q99877	H2BC17	P23527
H2BC18	Q99877	H2BC21	P06899, Q16778	H2BC3	P06899, P33778
H2BC4	P62807, Q93079	H2BC5	P58876	H2BC6	P62807, Q93079
H2BC7	P62807	H2BC8	P62807, Q93079	H2BC9	P58876, Q93079, Q99877
H3C1	P68431	H3C11	P68431	H3C12	P68431
H3C13	Q71DI3	H3C14	Q71DI3	H3C15	Q71DI3
H3C2	P68431	H3C3	P68431	H3C4	P68431
H3C6	P68431	H3C7	P68431	H3C8	P68431
H4C1	P62805	H4C13	P62805	H4C16	P62805

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
H4C2	P62805	H4C3	P62805	H4C4	P62805
H4C5	P62805	H4C6	P62805	H4C8	P62805
H4C9	P62805				

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
RUNX2	Q13950	P17480			

5. Deposition of new CENPA-containing nucleosomes at the centromere (R-HSA-606279)



Cellular compartments: nucleoplasm.

Eukaryotic centromeres are marked by a unique form of histone H3, designated CENPA in humans. In human cells newly synthesized CENPA is deposited in nucleosomes at the centromere during late telophase/early G1 phase of the cell cycle. Once deposited, nucleosomes containing CENPA remain stably associated with the centromere and are partitioned equally to daughter centromeres during S phase. A current model proposes that pre-existing CENPA at the centromere drives recruitment of new CENPA, however this has not been proved.

The deposition process requires at least 3 complexes: the Mis18 complex, HJURP complex, and the RSF complex. HJURP binds newly synthesized CENPA-H4 tetramers before deposition and brings them to the centromere for deposition in new CENPA-containing nucleosomes. The exact mechanism of deposition remains unknown.

References

Jansen LE, Black BE, Foltz DR & Cleveland DW (2007). Propagation of centromeric chromatin requires exit from mitosis. J Cell Biol, 176, 795-805.

Foltz DR, Jansen LE, Bailey AO, Yates JR 3rd, Bassett EA, Wood S, ... Cleveland DW (2009). Centromere-specific assembly of CENP-a nucleosomes is mediated by HJURP. Cell, 137, 472-84.

Dunleavy EM, Roche D, Tagami H, Lacoste N, Ray-Gallet D, Nakamura Y, ... Almouzni-Pettinotti G (2009). HJURP is a cell-cycle-dependent maintenance and deposition factor of CENP-A at centromeres. Cell, 137, 485-97.

Shuaib M, Ouararhni K, Dimitrov S & Hamiche A (2010). HJURP binds CENP-A via a highly conserved N-terminal domain and mediates its deposition at centromeres. Proc Natl Acad Sci U S A, 107, 1349-54.

Mellone BG, Zhang W & Karpen GH (2009). Frodos found: Behold the CENP-a "Ring" bearers. Cell, 137, 409-12.

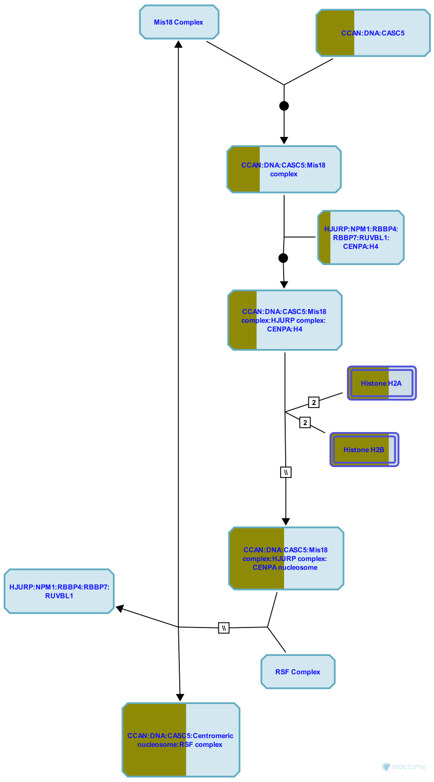
Edit history

Date	Action	Author
2010-04-15	Edited	May B
2010-04-15	Authored	May B
2010-04-17	Created	May B
2010-05-30	Reviewed	Dunleavy EM, Almouzni-Pettinotti G
2010-06-15	Reviewed	Foltz DR
2025-11-15	Modified	Weiser JD

43 submitted entities found in this pathway, mapping to 55 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CENPH	Q9H3R5	CENPK	Q9BS16	CENPP	Q6IPU0
CENPQ	Q7L2Z9	CENPU	Q71F23	H2AC12	Q99878
H2AC14	Q99878	H2AC16	P20671	H2AC18	Q6FI13
H2AC19	Q6FI13	H2AC20	Q16777	H2AC4	P04908, Q99878
H2AC6	Q93077	H2AC7	P20671	H2AC8	P04908
H2BC10	P62807	H2BC11	P06899	H2BC12	O60814, Q93079, Q99877, Q99880
H2BC12L	P57053	H2BC13	Q99880	H2BC14	Q99879
H2BC15	Q93079, Q99877	H2BC17	P23527	H2BC18	Q99877
H2BC21	P06899, Q16778	H2BC3	P06899, P33778	H2BC4	P62807, Q93079
H2BC5	P58876	H2BC6	P62807, Q93079	H2BC7	P62807
H2BC8	P62807, Q93079	H2BC9	P58876, Q93079, Q99877	H4C1	P62805
H4C13	P62805	H4C16	P62805	H4C2	P62805
H4C3	P62805	H4C4	P62805	H4C5	P62805
H4C6	P62805	H4C8	P62805	H4C9	P62805
ITGB3BP	Q13352				

6. Nucleosome assembly (R-HSA-774815)



The formation of centromeric chromatin assembly outside the context of DNA replication involves the assembly of nucleosomes containing the histone H3 variant CenH3 (also called CENP-A).

References

Edit history

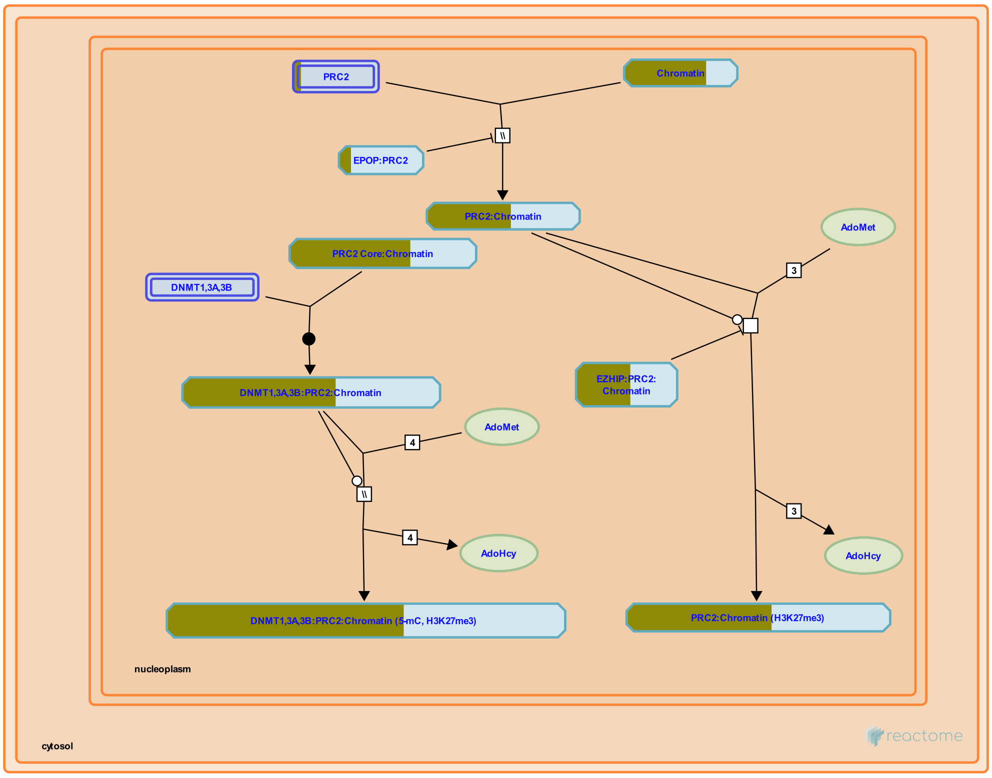
Date	Action	Author
2010-05-21	Authored	Matthews L
2010-05-21	Created	Matthews L
2010-05-24	Edited	Matthews L
2025-11-15	Modified	Weiser JD

43 submitted entities found in this pathway, mapping to 55 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CENPH	Q9H3R5	CENPK	Q9BS16	CENPP	Q6IPU0
CENPQ	Q7L2Z9	CENPU	Q71F23	H2AC12	Q99878
H2AC14	Q99878	H2AC16	P20671	H2AC18	Q6FI13
H2AC19	Q6FI13	H2AC20	Q16777	H2AC4	P04908, Q99878
H2AC6	Q93077	H2AC7	P20671	H2AC8	P04908
H2BC10	P62807	H2BC11	P06899	H2BC12	O60814, Q93079, Q99877, Q99880
H2BC12L	P57053	H2BC13	Q99880	H2BC14	Q99879
H2BC15	Q93079, Q99877	H2BC17	P23527	H2BC18	Q99877
H2BC21	P06899, Q16778	H2BC3	P06899, P33778	H2BC4	P62807, Q93079
H2BC5	P58876	H2BC6	P62807, Q93079	H2BC7	P62807

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
H2BC8	P62807, Q93079	H2BC9	P58876, Q93079, Q99877	H4C1	P62805
H4C13	P62805	H4C16	P62805	H4C2	P62805
H4C3	P62805	H4C4	P62805	H4C5	P62805
H4C6	P62805	H4C8	P62805	H4C9	P62805
ITGB3BP	Q13352				

7. PRC2 methylates histones and DNA (R-HSA-212300)



Cellular compartments: nucleoplasm.

Polycomb group proteins are responsible for the heritable repression of genes during development (Lee et al. 2006, Ku et al. 2008, reviewed in Simon and Kingston 2009, Margueron and Reinberg 2011, Di Croce and Helin 2013). Two major families of Polycomb complexes exist: Polycomb Repressive Complex 1 (PRC1) and Polycomb Repressive Complex 2 (PRC2). PRC1 and PRC2 each appear to comprise sets of distinct complexes that contain common core subunits and distinct accessory subunits (reviewed in Nayak et al. 2011). PRC2, through its component EZH2 or, in some complexes, EZH1 produces the initial molecular mark of repression, the trimethylation of lysine-27 of histone H3 (H3K27me3). How PRC2 is initially recruited to a locus remains unknown, however cytosine-guanine (CpG) motifs and transcripts have been suggested. Different mechanisms may be used at different loci. The trimethylated H3K27 produced by PRC2 is bound by the Polycomb subunit of PRC1. PRC1 ubiquitinates histone H2A and maintains repression.

References

Simon JA & Kingston RE (2009). Mechanisms of polycomb gene silencing: knowns and unknowns. Nat Rev Mol Cell Biol, 10, 697-708.

Margueron R & Reinberg D (2011). The Polycomb complex PRC2 and its mark in life. Nature, 469, 343-9.

Simon JA & Kingston RE (2013). Occupying chromatin: Polycomb mechanisms for getting to genomic targets, stopping transcriptional traffic, and staying put. Mol. Cell, 49, 808-24.

Ku M, Koche RP, Rheinbay E, Mendenhall EM, Endoh M, Mikkelsen TS, ... Bernstein BE (2008). Genomewide analysis of PRC1 and PRC2 occupancy identifies two classes of bivalent domains. PLoS Genet., 4, e1000242.

Di Croce L & Helin K (2013). Transcriptional regulation by Polycomb group proteins. Nat. Struct. Mol. Biol., 20, 1147-55.

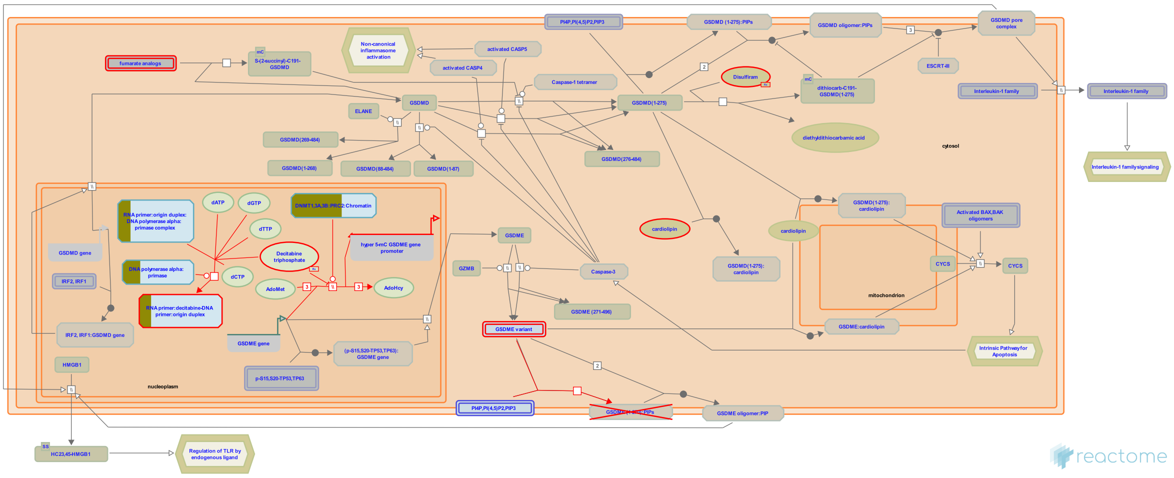
Edit history

Date	Action	Author
2008-02-09	Authored	May B, Gopinathrao G
2008-02-09	Created	May B, Gopinathrao G
2010-04-06	Edited	May B
2014-02-12	Reviewed	Di Croce L
2025-09-09	Revised	Orlic-Milacic M
2025-11-15	Modified	Weiser JD

50 submitted entities found in this pathway, mapping to 62 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
H2AC12	Q99878	H2AC14	Q99878	H2AC16	P20671
H2AC18	Q6FI13	H2AC19	Q6FI13	H2AC20	Q16777
H2AC4	P04908, Q99878	H2AC6	Q93077	H2AC7	P20671
H2AC8	P04908	H2BC10	P62807	H2BC11	P06899
H2BC12	O60814, Q93079, Q99877, Q99880	H2BC12L	P57053	H2BC13	Q99880
H2BC14	Q99879	H2BC15	Q93079, Q99877	H2BC17	P23527
H2BC18	Q99877	H2BC21	P06899, Q16778	H2BC3	P06899, P33778
H2BC4	P62807, Q93079	H2BC5	P58876	H2BC6	P62807, Q93079
H2BC7	P62807	H2BC8	P62807, Q93079	H2BC9	P58876, Q93079, Q99877
H3C1	P68431	H3C11	P68431	H3C12	P68431
H3C13	Q71DI3	H3C14	Q71DI3	H3C15	Q71DI3
H3C2	P68431	H3C3	P68431	H3C4	P68431
H3C6	P68431	H3C7	P68431	H3C8	P68431
H4C1	P62805	H4C13	P62805	H4C16	P62805
H4C2	P62805	H4C3	P62805	H4C4	P62805
H4C5	P62805	H4C6	P62805	H4C8	P62805
H4C9	P62805	SUZ12	Q15022		

8. Defective pyroptosis (R-HSA-9710421)



Diseases: cancer, breast carcinoma, colon adenocarcinoma, gastric adenocarcinoma, head and neck squamous cell carcinoma, lung adenocarcinoma, melanoma.

Pyroptosis is a form of lytic inflammatory programmed cell death that is mediated by the pore-forming gasdermins (GSDMs) (Shi J et al. 2017) to stimulate immune responses through the release of pro-inflammatory interleukin (IL)-1β, IL-18 (mainly in GSDMD-mediated pyroptosis) as well as danger signals such as adenosine triphosphate (ATP) or high mobility group protein B1 (HMGB1) (reviewed in Shi J et al. 2017; Man SM et al. 2017; Tang D et al. 2019; Lieberman J et al. 2019). Pyroptosis protects the host from microbial infection but can also lead to pathological inflammation if overactivated or dysregulated (reviewed in Orning P et al. 2019; Tang L et al. 2020). During infections, the excessive production of cytokines can lead to a cytokine storm, which is associated with acute respiratory distress syndrome (ARDS) and systemic inflammatory response syndrome (SIRS) (reviewed in Tisoncik JR et al. 2012; Karki R et al. 2020; Ragab D et al. 2020). Pyroptosis has a close but complicated relationship to tumorigenesis, affected by tissue type and genetic background. Pyroptosis can trigger potent antitumor immune responses or serve as an effector mechanism in antitumor immunity (Wang Q et al. 2020; Zhou Z et al. 2020; Zhang Z et al. 2020), while in other cases, as a type of proinflammatory death, pyroptosis can contribute to the formation of a microenvironment suitable for tumor cell growth (reviewed in Xia X et al. 2019; Jiang M et al. 2020; Zhang Z et al. 2021).

This Reactome module describes the defective GSDME function caused by cancer-related GSDME mutations (Zhang Z et al. 2020). It also shows epigenetic inactivation of GSDME due to hypermethylation of the GSDME promoter region (Akino K et al. 2007; Kim MS et al. 2008a,b; Croes L et al. 2017, 2018; Ibrahim J et al. 2019). Aberrant promoter methylation is considered to be a hallmark of cancer (Ehrlich M et al. 2002; Dong Y et al. 2014; Lam K et al. 2016; Croes L et al. 2018). Treatment with the DNA methyltransferase inhibitor decitabine (5-aza-2'-deoxycytidine or DAC) may elevate GSDME expression in certain cancer cells (Akino K et al. 2007; Fujikane T et al. 2009; Wang Y et al. 2017).

References

Tang L, Lu C, Zheng G & Burgering BM (2020). Emerging insights on the role of gasdermins in infection and inflammatory diseases. Clin Transl Immunology, 9, e1186.

Zhang Z, Zhang Y, Xia S, Kong Q, Li S, Liu X, ... Lieberman J (2020). Gasdermin E suppresses tumour growth by activating anti-tumour immunity. Nature, 579, 415-420.

Ibrahim J, Op de Beeck K, Fransen E, Croes L, Beyens M, Suls A, ... Van Camp G (2019). Methylation analysis of Gasdermin E shows great promise as a biomarker for colorectal cancer. Cancer Med, 8, 2133-2145.

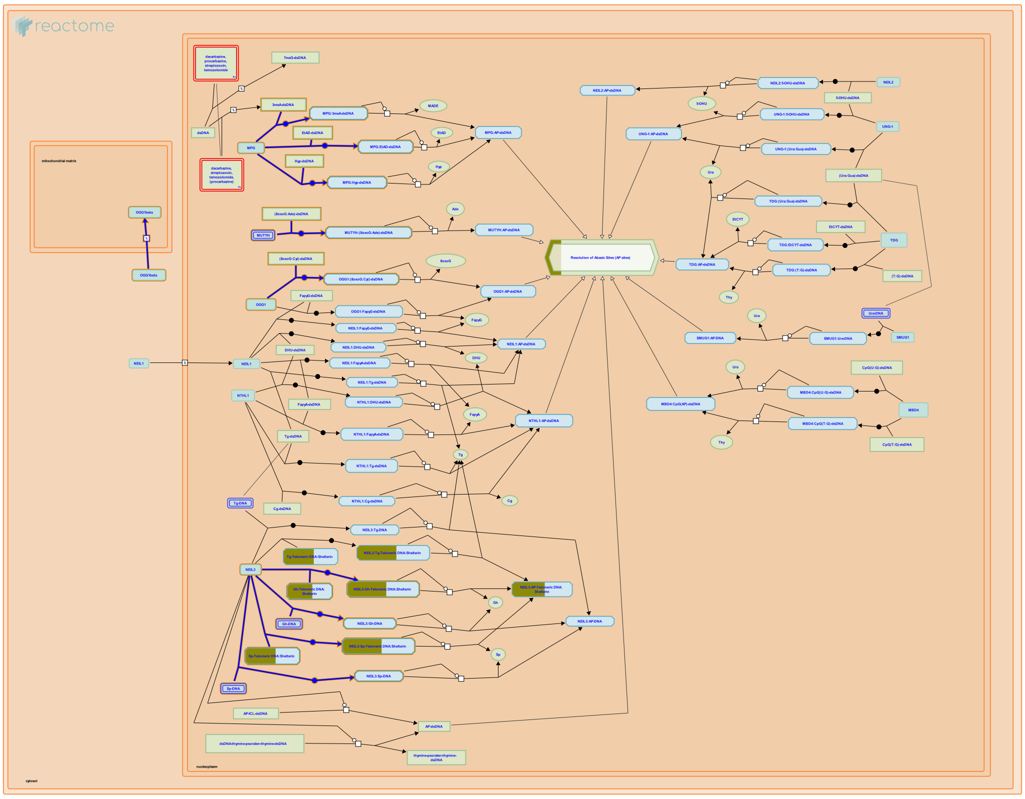
Edit history

Date	Action	Author
2020-11-09	Authored	Shamovsky V
2020-12-30	Created	Shamovsky V
2021-02-17	Edited	Shorser S
2021-02-17	Reviewed	Kanneganti TD, Zhang Z, D'Eustachio P
2021-04-22	Reviewed	Shao F
2023-10-12	Modified	Weiser JD

51 submitted entities found in this pathway, mapping to 63 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
H2AC12	Q99878	H2AC14	Q99878	H2AC16	P20671
H2AC18	Q6FI13	H2AC19	Q6FI13	H2AC20	Q16777
H2AC4	P04908, Q99878	H2AC6	Q93077	H2AC7	P20671
H2AC8	P04908	H2BC10	P62807	H2BC11	P06899
H2BC12	O60814, Q93079, Q99877, Q99880	H2BC12L	P57053	H2BC13	Q99880
H2BC14	Q99879	H2BC15	Q93079, Q99877	H2BC17	P23527
H2BC18	Q99877	H2BC21	P06899, Q16778	H2BC3	P06899, P33778
H2BC4	P62807, Q93079	H2BC5	P58876	H2BC6	P62807, Q93079
H2BC7	P62807	H2BC8	P62807, Q93079	H2BC9	P58876, Q93079, Q99877
H3C1	P68431	H3C11	P68431	H3C12	P68431
H3C13	Q71DI3	H3C14	Q71DI3	H3C15	Q71DI3
H3C2	P68431	H3C3	P68431	H3C4	P68431
H3C6	P68431	H3C7	P68431	H3C8	P68431
H4C1	P62805	H4C13	P62805	H4C16	P62805
H4C2	P62805	H4C3	P62805	H4C4	P62805
H4C5	P62805	H4C6	P62805	H4C8	P62805
H4C9	P62805	SPP1	P49642	SUZ12	Q15022

9. Recognition and association of DNA glycosylase with site containing an affected purine (R-HSA-110330)



Cellular compartments: nucleoplasm.

The recognition and removal of an altered base by a DNA glycosylase is thought to involve the diffusion of the enzyme along the minor grove of the DNA molecule. The enzyme presumably compresses the backbone of the affected DNA strand at the site of damage. This compression is thought to result in an outward rotation of the damaged residue into a "pocket" of the enzyme that recognizes and cleaves the altered base from the backbone (Slupphaug et al. 1996, Parikh et al. 1998).

References

Slupphaug G, Mol CD, Kavli B, Arvai AS, Krokan HE & Tainer JA (1996). A nucleotide-flipping mechanism from the structure of human uracil-DNA glycosylase bound to DNA. Nature, 384, 87-92.

Parikh SS, Mol CD, Slupphaug G, Bharati S, Krokan HE & Tainer JA (1998). Base excision repair initiation revealed by crystal structures and binding kinetics of human uracil-DNA glycosylase with DNA. EMBO J, 17, 5214-26.

Edit history

Date	Action	Author
2004-01-29	Created	Matthews L
2004-02-03	Edited	Matthews L
2004-02-09	Authored	Matthews L
2014-12-04	Revised	Orlic-Milacic M
2014-12-04	Edited	Orlic-Milacic M

Date	Action	Author
2014-12-22	Reviewed	Borowiec JA
2025-11-15	Modified	Weiser JD

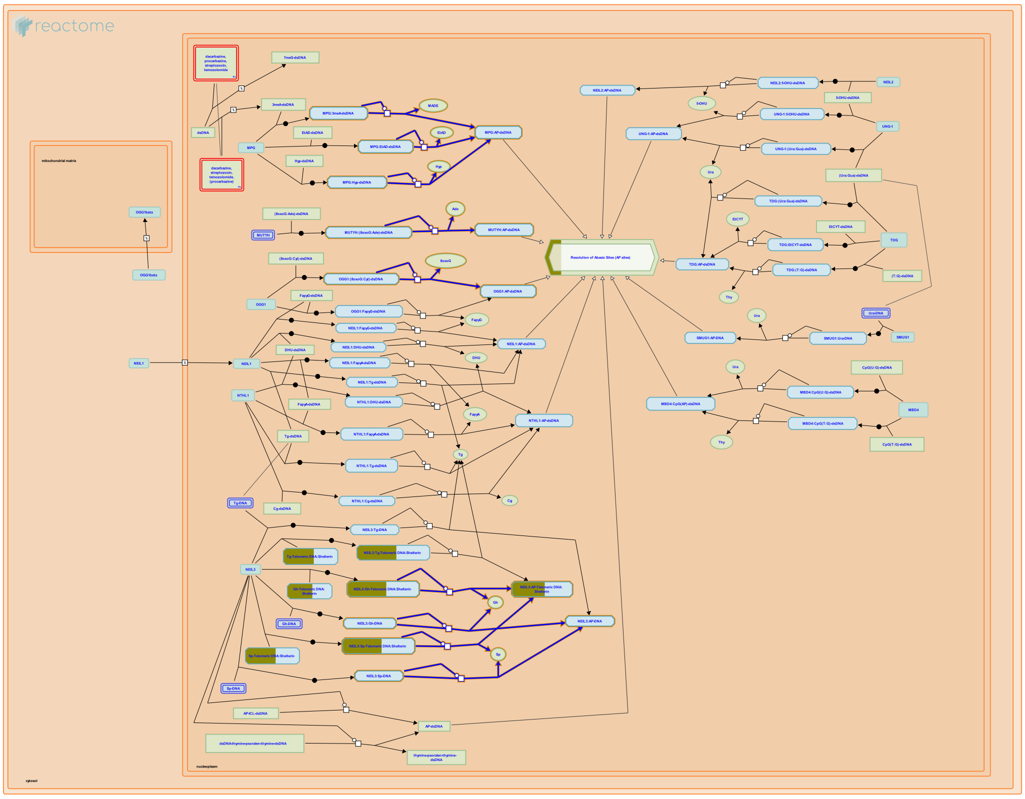
37 submitted entities found in this pathway, mapping to 49 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
H2AC12	Q99878	H2AC14	Q99878	H2AC16	P20671
H2AC18	Q6FI13	H2AC19	Q6FI13	H2AC20	Q16777
H2AC4	P04908, Q99878	H2AC6	Q93077	H2AC7	P20671
H2AC8	P04908	H2BC10	P62807	H2BC11	P06899
H2BC12	O60814, Q93079, Q99877, Q99880	H2BC12L	P57053	H2BC13	Q99880
H2BC14	Q99879	H2BC15	Q93079, Q99877	H2BC17	P23527
H2BC18	Q99877	H2BC21	P06899, Q16778	H2BC3	P06899, P33778
H2BC4	P62807, Q93079	H2BC5	P58876	H2BC6	P62807, Q93079
H2BC7	P62807	H2BC8	P62807, Q93079	H2BC9	P58876, Q93079, Q99877
H4C1	P62805	H4C13	P62805	H4C16	P62805
H4C2	P62805	H4C3	P62805	H4C4	P62805
H4C5	P62805	H4C6	P62805	H4C8	P62805
H4C9	P62805				

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
PTGES3	Q15185	Q8TAT5			

10. Cleavage of the damaged purine (R-HSA-110331)



Cellular compartments: nucleoplasm.

Damaged purines are cleaved from the sugar-phosphate backbone by purine-specific glycosylases (Saparbaev and Laval 1994, Lindahl and Wood 1999).

References

Saparbaev M & Laval J (1994). Excision of hypoxanthine from DNA containing dIMP residues by the Escherichia coli, yeast, rat, and human alkylpurine DNA glycosylases. Proc. Natl. Acad. Sci. U.S.A., 91, 5873-7.

Lindahl T & Wood RD (1999). Quality control by DNA repair. Science, 286, 1897-905.

Edit history

Date	Action	Author
2004-01-29	Created	Matthews L
2004-02-03	Edited	Matthews L
2004-02-03	Authored	Matthews L
2014-12-04	Revised	Orlic-Milacic M
2014-12-04	Edited	Orlic-Milacic M
2014-12-22	Reviewed	Borowiec JA
2025-11-15	Modified	Weiser JD

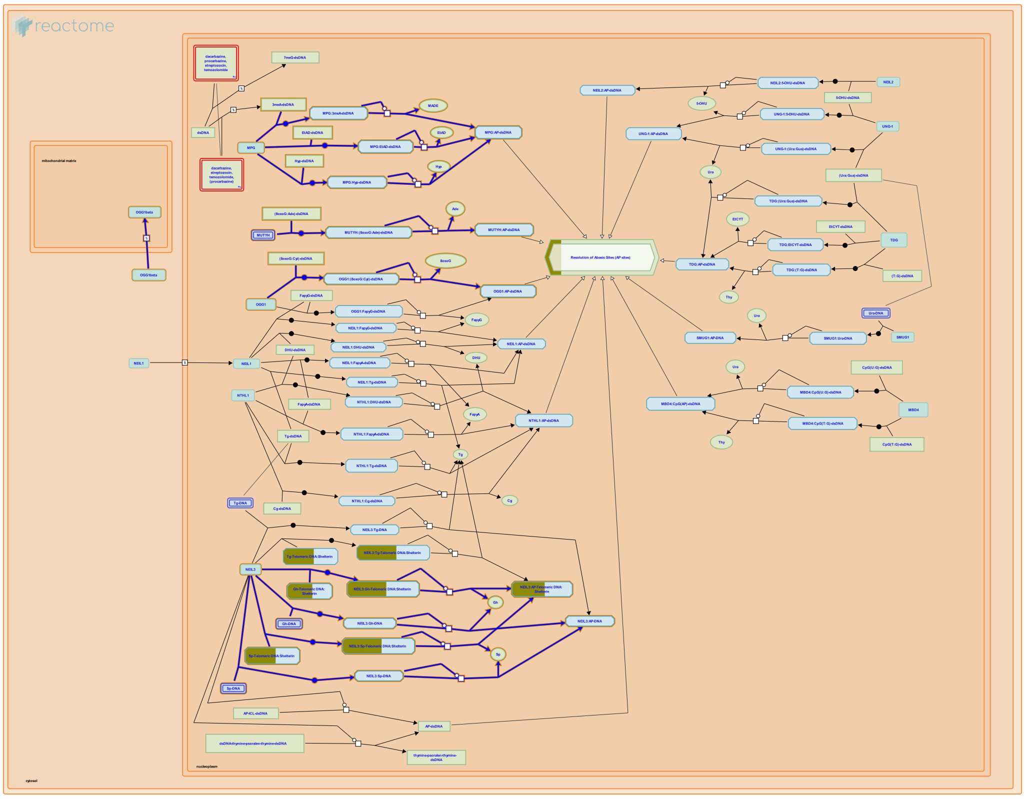
37 submitted entities found in this pathway, mapping to 49 Reactome entities

Input	UniProt Id
H2AC12	Q99878
H2AC18	Q6FI13
H2AC4	P04908, Q99878
H2AC8	P04908
H2BC12	O60814, Q93079, Q99877, Q99880
H2BC14	Q99879
H2BC18	Q99877
H2BC4	P62807, Q93079
H2BC7	P62807
H4C1	P62805
H4C2	P62805
H4C5	P62805
H4C9	P62805

Input	UniProt Id
H2AC14	Q99878
H2AC19	Q6FI13
H2AC6	Q93077
H2BC10	P62807
H2BC12L	P57053
H2BC15	Q93079, Q99877
H2BC21	P06899, Q16778
H2BC5	P58876
H2BC8	P62807, Q93079
H4C13	P62805
H4C3	P62805
H4C6	P62805

Input	UniProt Id
H2AC16	P20671
H2AC20	Q16777
H2AC7	P20671
H2BC11	P06899
H2BC13	Q99880
H2BC17	P23527
H2BC3	P06899, P33778
H2BC6	P62807, Q93079
H2BC9	P58876, Q93079, Q99877
H4C16	P62805
H4C4	P62805
H4C8	P62805

11. Depurination (R-HSA-73927)



Cellular compartments: nucleoplasm.

Depurination of a damaged nucleotide is mediated by a purine-specific DNA glycosylase. The glycosylase cleaves the N-C1' glycosidic bond between the damaged DNA base and the deoxyribose sugar, generating a free base and an abasic i.e. apurinic/apyrimidinic (AP) site (Slupphaug et al. 1996, Parikh et al. 1998).

References

Slupphaug G, Mol CD, Kavli B, Arvai AS, Krokan HE & Tainer JA (1996). A nucleotide-flipping mechanism from the structure of human uracil-DNA glycosylase bound to DNA. Nature, 384, 87-92.

Parikh SS, Mol CD, Slupphaug G, Bharati S, Krokan HE & Tainer JA (1998). Base excision repair initiation revealed by crystal structures and binding kinetics of human uracil-DNA glycosylase with DNA. EMBO J, 17, 5214-26.

Edit history

Date	Action	Author
2004-02-03	Edited	Matthews L
2004-02-03	Created	Matthews L
2004-02-09	Authored	Matthews L
2014-12-04	Revised	Orlic-Milacic M
2014-12-04	Edited	Orlic-Milacic M
2014-12-22	Reviewed	Borowiec JA
2025-11-15	Modified	Weiser JD

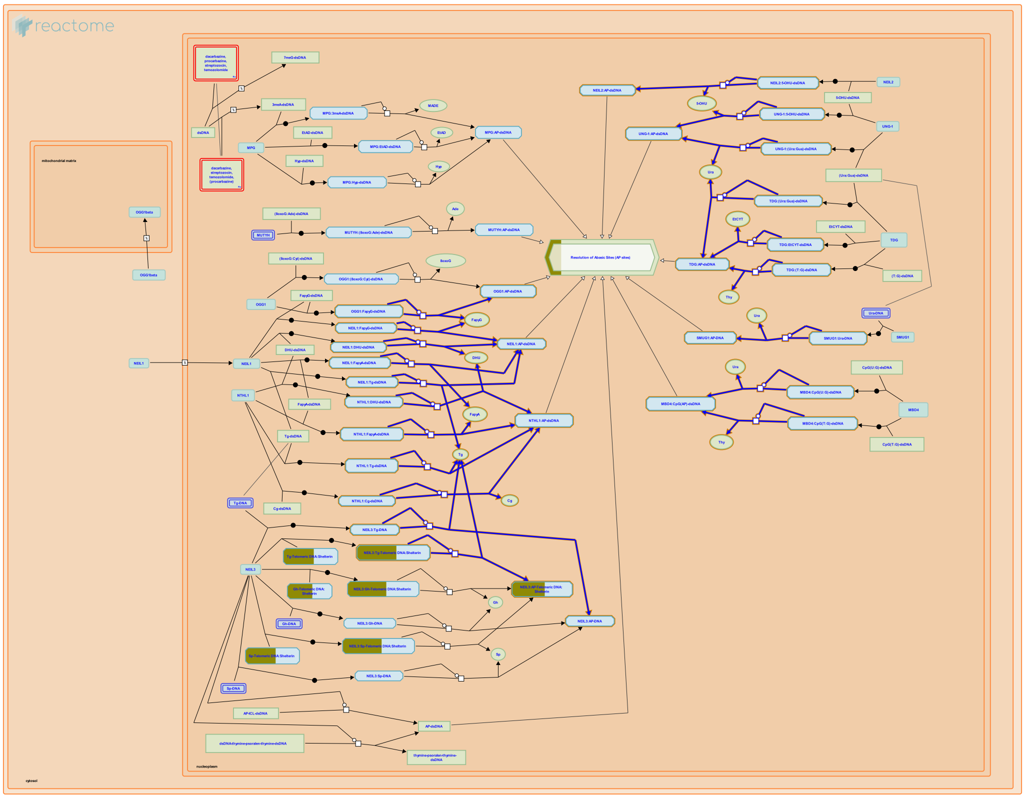
37 submitted entities found in this pathway, mapping to 49 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
H2AC12	Q99878	H2AC14	Q99878	H2AC16	P20671
H2AC18	Q6FI13	H2AC19	Q6FI13	H2AC20	Q16777
H2AC4	P04908, Q99878	H2AC6	Q93077	H2AC7	P20671
H2AC8	P04908	H2BC10	P62807	H2BC11	P06899
H2BC12	O60814, Q93079, Q99877, Q99880	H2BC12L	P57053	H2BC13	Q99880
H2BC14	Q99879	H2BC15	Q93079, Q99877	H2BC17	P23527
H2BC18	Q99877	H2BC21	P06899, Q16778	H2BC3	P06899, P33778
H2BC4	P62807, Q93079	H2BC5	P58876	H2BC6	P62807, Q93079
H2BC7	P62807	H2BC8	P62807, Q93079	H2BC9	P58876, Q93079, Q99877
H4C1	P62805	H4C13	P62805	H4C16	P62805
H4C2	P62805	H4C3	P62805	H4C4	P62805
H4C5	P62805	H4C6	P62805	H4C8	P62805
H4C9	P62805				

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
PTGES3	Q15185	Q8TAT5			

12. Cleavage of the damaged pyrimidine (R-HSA-110329)



Cellular compartments: nucleoplasm.

Damaged pyrimidines are cleaved by pyrimide-specific glycosylases (Lindahl and Wood 1999).

References

Lindahl T & Wood RD (1999). Quality control by DNA repair. Science, 286, 1897-905.

Edit history

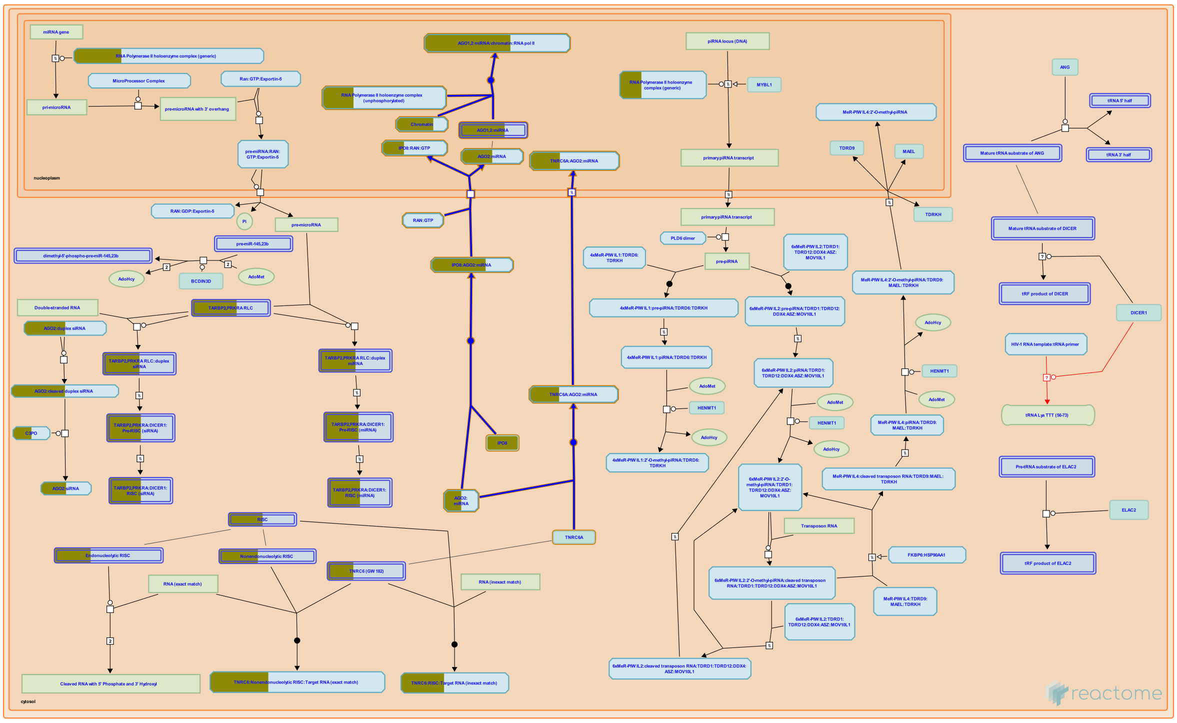
Date	Action	Author
2004-01-29	Created	Matthews L
2004-02-03	Edited	Matthews L
2004-02-03	Authored	Matthews L
2014-12-04	Revised	Orlic-Milacic M
2014-12-04	Edited	Orlic-Milacic M
2014-12-22	Reviewed	Borowiec JA
2025-11-15	Modified	Weiser JD

37 submitted entities found in this pathway, mapping to 49 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
H2AC12	Q99878	H2AC14	Q99878	H2AC16	P20671
H2AC18	Q6FI13	H2AC19	Q6FI13	H2AC20	Q16777
H2AC4	P04908, Q99878	H2AC6	Q93077	H2AC7	P20671
H2AC8	P04908	H2BC10	P62807	H2BC11	P06899

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
H2BC12	O60814, Q93079, Q99877, Q99880	H2BC12L	P57053	H2BC13	Q99880
H2BC14	Q99879	H2BC15	Q93079, Q99877	H2BC17	P23527
H2BC18	Q99877	H2BC21	P06899, Q16778	H2BC3	P06899, P33778
H2BC4	P62807, Q93079	H2BC5	P58876	H2BC6	P62807, Q93079
H2BC7	P62807	H2BC8	P62807, Q93079	H2BC9	P58876, Q93079, Q99877
H4C1	P62805	H4C13	P62805	H4C16	P62805
H4C2	P62805	H4C3	P62805	H4C4	P62805
H4C5	P62805	H4C6	P62805	H4C8	P62805
H4C9	P62805				

13. Transcriptional regulation by small RNAs (R-HSA-5578749)



Recent evidence indicates that small RNAs participate in transcriptional regulation in addition to post-transcriptional silencing. Components of the RNAi machinery (ARGONAUTE1 (AGO1, EIF2C1), AGO2 (EIF2C2), AGO3 (EIF2C3), AGO4 (EIF2C4), TNRC6A, and DICER) are observed associated with microRNAs (miRNAs) in both the cytosol and the nucleus (Robb et al. 2005, Weinmann et al. 2009, Doyle et al. 2013, Nishi et al. 2013, Gagnon et al. 2014). The AGO:miRNA complexes are imported into the nucleus by IMPORTIN-8 (IPO8, IMP8, RANBP8) and also by an unknown importin while associated with the nuclear shuttling protein TNRC6A (reviewed in Schraivogel and Meister 2014).

Within the nucleus, AGO2, TNRC6A, and DICER may associate in a complex (Gagnon et al. 2014). Nuclear AGO1 and AGO2 in complexes with small RNAs are observed to activate transcription (RNA activation, RNAa) or repress transcription (Transcriptional Gene Silencing, TGS) of genes that contain sequences matching the small RNAs (reviewed in Malecova and Morris 2010, Huang and Li 2012, Gagnon and Corey 2012, Huang and Li 2014, Salmanidis et al. 2014, Stroynowska-Czerwinska et al. 2014). TGS is associated with methylation of cytosine in DNA and methylation of histone H3 at lysine-9 and lysine-27 (Castanotto et al. 2005, Suzuki et al. 2005, Kim et al. 2006, Weinberg et al. 2006, Kim et al. 2008, reviewed in Malecova and Morris 2010, Li et al. 2014); RNAa is associated with methylation of histone H3 at lysine-4 (Huang et al. 2012, reviewed in Li et al. 2014). Small RNAs in the nucleus have also been shown to play roles in alternative splicing (Liu et al., 2012, Ameyar-Zazoua et al., 2012) and DNA damage repair (Wei et al., 2012; Francia et al., 2012). Nevertheless, elucidation of the detailed mechanisms of small RNA action requires further research.

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Salmanidis M, Pillman K, Goodall G & Bracken C (2014). Direct transcriptional regulation by nuclear microRNAs. Int. J. Biochem. Cell Biol..

Edit history

Date	Action	Author
2014-05-30	Edited	May B
2014-05-30	Authored	May B
2014-05-31	Created	May B
2014-08-25	Reviewed	Li LC
2014-09-06	Reviewed	Stroynowska-Czerwinska A
2024-05-23	Reviewed	May B
2025-11-15	Modified	Weiser JD

57 submitted entities found in this pathway, mapping to 70 Reactome entities

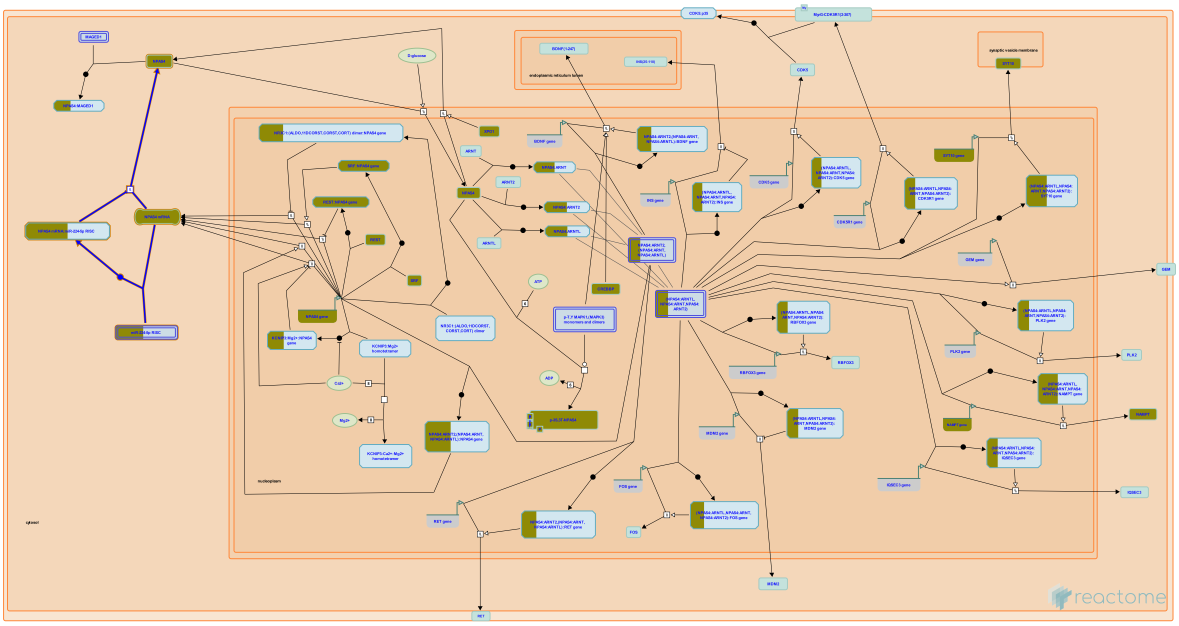
Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
AGO1	Q9UKV8, Q9UL18	AGO2	Q9UKV8	AGO3	Q9UL18
H2AC12	Q99878	H2AC14	Q99878	H2AC16	P20671
H2AC18	Q6FI13	H2AC19	Q6FI13	H2AC20	Q16777
H2AC4	P04908, Q99878	H2AC6	Q93077	H2AC7	P20671
H2AC8	P04908	H2BC10	P62807	H2BC11	P06899
H2BC12	O60814, Q93079, Q99877, Q99880	H2BC12L	P57053	H2BC13	Q99880
H2BC14	Q99879	H2BC15	Q93079, Q99877	H2BC17	P23527
H2BC18	Q99877	H2BC21	P06899, Q16778	H2BC3	P06899, P33778
H2BC4	P62807, Q93079	H2BC5	P58876	H2BC6	P62807, Q93079
H2BC7	P62807	H2BC8	P62807, Q93079	H2BC9	P58876, Q93079, Q99877
H3C1	P68431	H3C11	P68431	H3C12	P68431
H3C13	Q71DI3	H3C14	Q71DI3	H3C15	Q71DI3
H3C2	P68431	H3C3	P68431	H3C4	P68431
H3C6	P68431	H3C7	P68431	H3C8	P68431
H4C1	P62805	H4C13	P62805	H4C16	P62805
H4C2	P62805	H4C3	P62805	H4C4	P62805
H4C5	P62805	H4C6	P62805	H4C8	P62805
H4C9	P62805	IPO8	O15397	MT1A	P53803
POLR2F	P61218	POLR2H	P52434	POLR2K	P53803

Interactors found in this pathway (9)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
AGO1	Q9UL18	O15397, Q8NDV7	AGO2	Q9UKV8	O15397, Q8NDV7
AGO3	Q9H9G7	O15397, Q8NDV7	SMAD3	P84022	O15397

Input	UniProt Id	Interacts with
TRIM28	Q13263	O15397
ZNF460	Q14592	O15397
ZNF74	Q16587	O15397
ZNF264	O43296	O15397
ZNF7	P17097	O15397

14. Regulation of NPAS4 mRNA translation (R-HSA-9768778)



The 3'UTR of NPAS4 mRNA is highly conserved between mouse, rat and human (>95% identity across ~700 bp of the 3'UTR sequence). Of the three putative microRNAs, miR-224, miR-203 and miR-132/212, that were predicted to target the 3'UTR of NPAS4 mRNA by multiple computational algorithms, miR-224 and miR-203 were able to significantly downregulate expression of the NPAS4 3’UTR reporter construct (Bersten et al. 2014). Downregulation of Npas4 expression by miR-224 was also shown in mouse (Choy et al. 2017). Npas4 was also reported as a target gene of miR-1 (Forget et al. 2021), miR-142 (Ji et al. 2019) and miR-744 (Choy et al. 2017). Only microRNAs that were reported to target Npas4 mRNA by at least two studies are shown in the diagram. The circulatory RNA circ_0003420 was also reported to target NPAS4 mRNA and contribute to its degradation (Xiong et al. 2021).

References

Bersten DC, Wright JA, McCarthy PJ & Whitelaw ML (2014). Regulation of the neuronal transcription factor NPAS4 by REST and microRNAs. Biochim Biophys Acta, 1839, 13-24.

Forget B, Garcia EM, Godino A, Rodriguez LD, Kappes V, Poirier P, ... Caboche J (2021). Cell-type- and region-specific modulation of cocaine seeking by micro-RNA-1 in striatal projection neurons. Mol Psychiatry.

Ji LL, Ye Y, Nie PY, Peng JB, Fu CH, Wang ZY & Tong L (2019). Dysregulation of miR-142 results in anxiety-like behaviors following single prolonged stress. Behav Brain Res, 365, 157-163.

Choy FC, Klarić TS, Koblar SA & Lewis MD (2017). miR-744 and miR-224 Downregulate Npas4 and Affect Lineage Differentiation Potential and Neurite Development During Neural Differentiation of Mouse Embryonic Stem Cells. Mol Neurobiol, 54, 3528-3541.

Xiong H, Wang H & Yu Q (2021). Circular RNA circ_0003420 mediates inflammation in sepsis-induced liver damage by downregulating neuronal PAS domain protein 4. Immunopharmacol Immunotoxicol, 43, 271-282.

Edit history

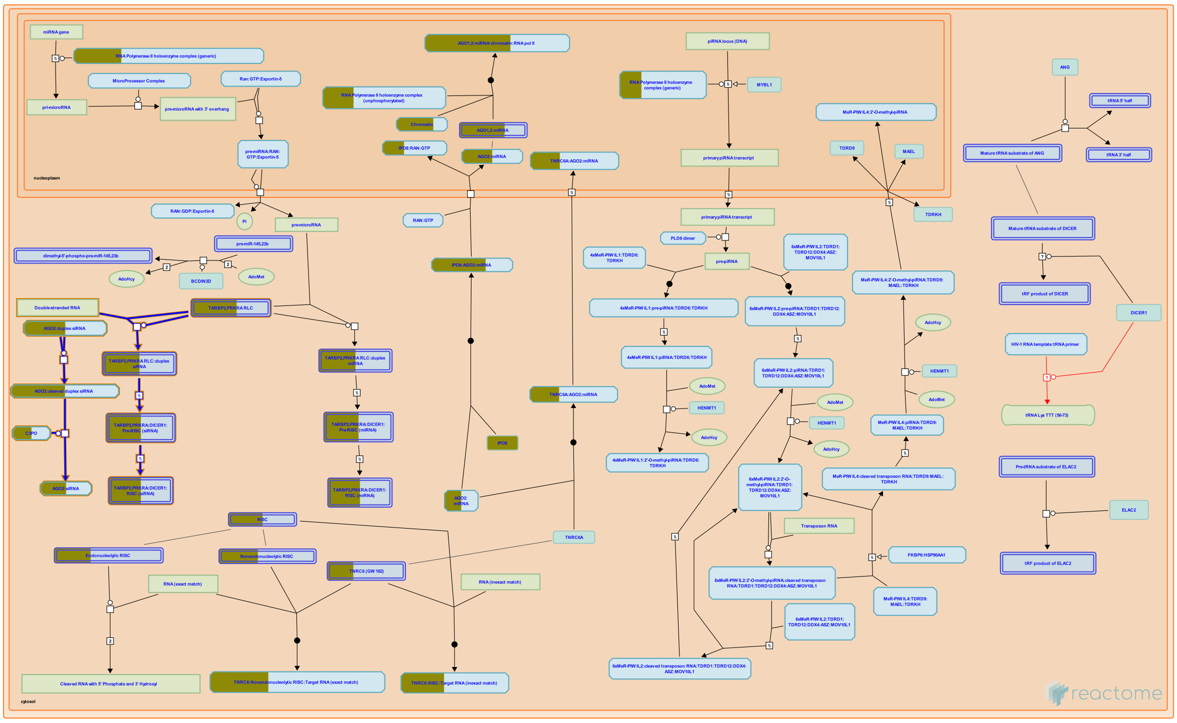
Date	Action	Author
2022-03-14	Created	Orlic-Milacic M
2022-04-08	Authored	Orlic-Milacic M
2022-07-29	Reviewed	Lin Y
2022-08-02	Edited	Orlic-Milacic M
2023-03-08	Modified	Matthews L

5 submitted entities found in this pathway, mapping to 10 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
AGO1	Q9H9G7, Q9UKV8, Q9UL18	AGO2	Q9H9G7, Q9UKV8	AGO3	Q9H9G7, Q9UL18
NPAS4	Q8IUM7	TNRC6B	Q9UPQ9		

Input	Ensembl Id
NPAS4	ENST00000311034

15. Small interfering RNA (siRNA) biogenesis (R-HSA-426486)



Cellular compartments: cytosol.

Small interfering RNAs (siRNAs) are 21-25 nucleotide single-stranded RNAs produced by cleavage of longer double-stranded RNAs by the enzyme DICER1 within the RISC loading complex containing DICER1, an Argonaute protein, and either TARBP2 or PRKRA (PACT). Typically the long double-stranded substrates originate from viruses or repetitive elements in the genome and the two strands of the substrate are exactly complementary.

After cleavage by DICER1 the 21-25 nucleotide double-stranded product is loaded into an Argonaute protein (humans contain 4 Argonautes) and rendered single-stranded by a mechanism that is not well characterized.

siRNA-loaded AGO2 is predominantly located at the cytosolic face of the rough endoplasmic reticulum and has also been observed in the nucleus.

References

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Tomari Y & Zamore PD (2005). Perspective: machines for RNAi. Genes Dev, 19, 517-29.

Kim VN, Han J & Siomi MC (2009). Biogenesis of small RNAs in animals. Nat Rev Mol Cell Biol, 10, 126-39.

Stalder L, Heusermann W, Sokol L, Trojer D, Wirz J, Hean J, ... Meisner-Kober NC (2013). The rough endoplasmic reticulum is a central nucleation site of siRNA-mediated RNA silencing. EMBO J., 32, 1115-27.

Gagnon KT, Li L, Chu Y, Janowski BA & Corey DR (2014). RNAi factors are present and active in human cell nuclei. Cell Rep, 6, 211-21.

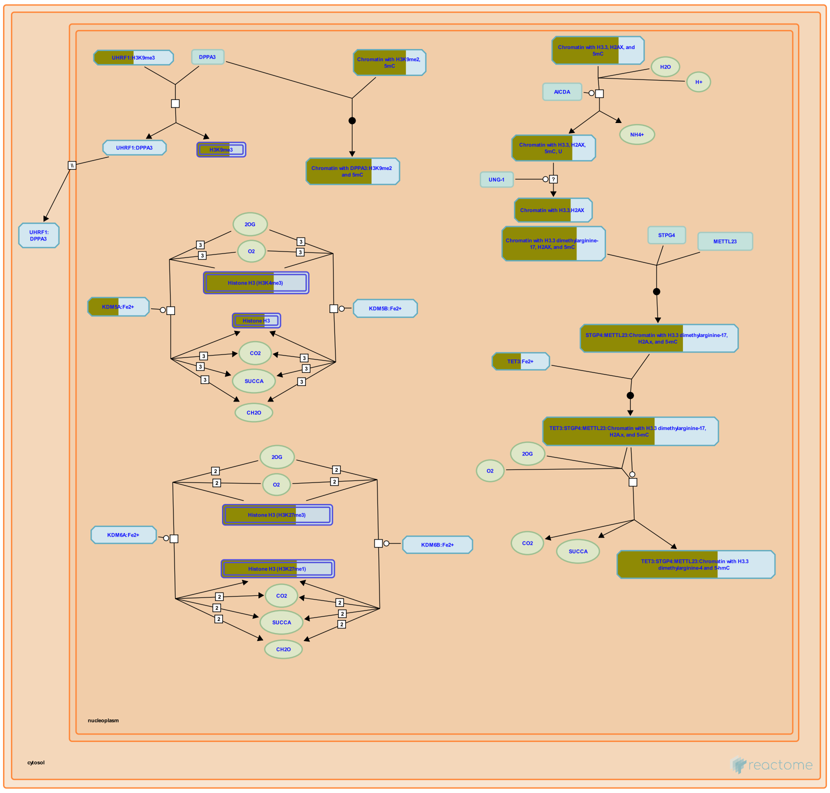
Edit history

Date	Action	Author
2009-06-10	Edited	May B
2009-06-10	Authored	May B
2009-06-17	Created	May B
2012-02-11	Reviewed	Tomari Y
2025-11-15	Modified	Weiser JD

5 submitted entities found in this pathway, mapping to 9 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
AGO1	Q9H9G7, Q9UKV8, Q9UL18	AGO2	Q9H9G7, Q9UKV8	AGO3	Q9H9G7, Q9UL18
PRKRA	O75569	TSNAX	Q99598		

16. Chromatin modifications during the maternal to zygotic transition (MZT) (R-HSA-9821002)



Chromatin in the zygotic pronuclei transitions to a more open and accessible conformation by DNA demethylation and changes to histone modifications. As development proceeds through the cleavage stages to the blastocyst, chromatin continues to become more accessible until DNA methylation and a more restrictive chromatin conformation are re-established after implantation of the embryo in the uterus.

In the oocyte, H3K9me2 produced by EHMT2 (G9a, KMT1C) and H3K9me3 produced by SETDB1 (KMT1E) are transmitted to the female pronucleus of the zygote and protect maternal DNA from active demethylation (inferred from mouse zygotes in Zeng et al. 2019, reviewed in de Macedo et al. 2021). DPPA3 binds H3K9me2, preventing the 5-methylcytosine oxidase TET3 from being recruited to chromatin (inferred from mouse homologs in Nakamura et al. 2007, Wossidlo et al. 2011, Nakamura et al. 2012). DPPA3 also displaces UHRF1 from chromatin, preventing the maintenance DNA methylase DNMT1 from being recruited to chromatin and thus allowing passive DNA demethylation to occur in the female genome (inferred from mouse homologs in Funaki et al. 2014, Li et al. 2018, Du et al. 2019, Mulholland et al. 2020).

In the male pronucleus of the zygote, AICDA (AID) deaminates cytosine residues and long patch repair replaces the mismatches and adjacent 5-methylcytidine residues with cytidine (Santos et al. 2013, Franchini et al. 2014). After this initial demethylation, TET3 is recruited to chromatin by METTL23 and STGP4 (GSE) (inferred from mouse homologs in Hatanaka et al. 2017) where it oxidizes remaining 5-methylcytidine to 5-hydroxymethylcytidine, which is removed by base excision repair and replaced with cytidine (inferred from mouse homologs in Gu et al. 2011, Iqbal et al. 2011, Wossidlo et al. 2011, Santos et al. 2013, Amouroux et al. 2016, Hatanaka et al. 2017).

The repressive mark H3K27me3 decreases in 2-cell embryos near developmentally related genes (Xia et al. 2019). The H3K27me3 demethylases KDM6B (inferred from bovine embryos in Chung et al. 2017, Canovas et al. 2012) and KDM6A (inferred from mouse embryos in Bai et al. 2019) appear to play a role in the decrease of H3K27me3, as downregulation of them impairs H3K27me3 loss, zygotic genome activation, and embryonic development. Embryonic development also requires H3K36me3, a permissive mark located in transcribed gene bodies that is produced in the oocyte by SETD2 (inferred from mouse embryos in Xu et al. 2019).

In mouse oocytes, H3K4me3 occurs in unusually broad regions that span genes Dahl et al. 2016, Zhang et al. 2016). These broad regions persist in the zygote and into the 2-cell stage. In the late 2-cell stage the more usual patterns of H3K4me3 are established as sharp peaks of H3K4me3 near the transcription start sites and stop sites of genes. The histone methyltransferase KMT2B is at least partly responsible for establishing the broad regions of H3K4me3 in the oocyte and the histone demethylases KDM5B and KDM5A remove the broad H3K4me3 in the late 2-cell stage embryo (inferred from mouse homologs in Dahl et al. 2016, reviewed in Eckerseley-Maslin et al. 2018).

In human oocytes and zygotes, however, broad regions of H3K4me3 are not observed across genes but are located across distal, CpG-rich domains which have partial DNA methylation (Xia et al. 2019). At the 8-cell stage, expression of KDM5B increases and the H3K4me3 at the distal domains is lost as zygotic genome activation occurs, suggesting a role for KDM5B in loss of H3K4me3 (Xia et al. 2019).

References

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Zeng TB, Han L, Pierce N, Pfeifer GP & Szabó PE (2019). EHMT2 and SETDB1 protect the maternal pronucleus from 5mC oxidation. Proc Natl Acad Sci U S A, 116, 10834-10841.

Xu Q, Xiang Y, Wang Q, Wang L, Brind'Amour J, Bogutz AB, ... Xie W (2019). SETD2 regulates the maternal epigenome, genomic imprinting and embryonic development. Nat Genet, 51, 844-856.

Eckersley-Maslin MA, Alda-Catalinas C & Reik W (2018). Dynamics of the epigenetic landscape during the maternal-to-zygotic transition. Nat Rev Mol Cell Biol, 19, 436-450.

Edit history

Date	Action	Author
2022-11-27	Edited	May B
2022-11-27	Authored	May B
2022-11-27	Created	May B
2023-08-08	Reviewed	Xie W
2023-08-17	Modified	Matthews L

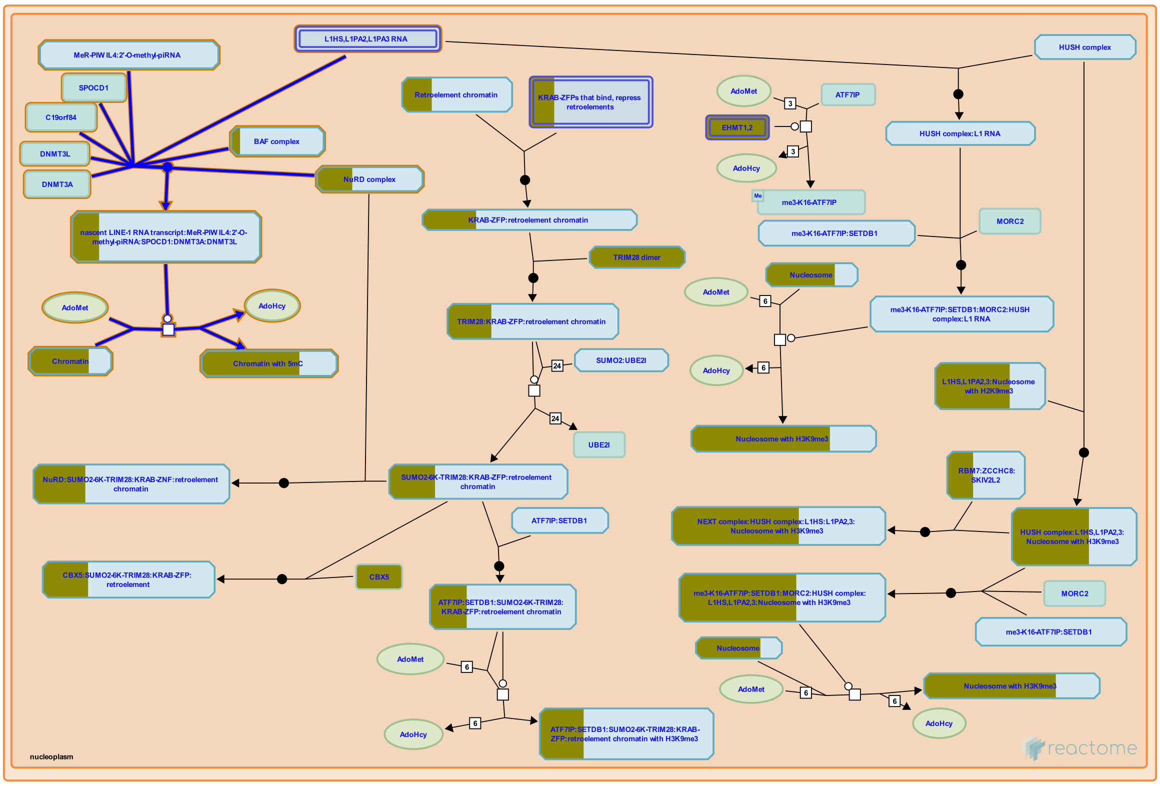
52 submitted entities found in this pathway, mapping to 64 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
H2AC12	Q99878	H2AC14	Q99878	H2AC16	P20671
H2AC18	Q6FI13	H2AC19	Q6FI13	H2AC20	Q16777
H2AC4	P04908, Q99878	H2AC6	Q93077	H2AC7	P20671
H2AC8	P04908	H2BC10	P62807	H2BC11	P06899
H2BC12	O60814, Q93079, Q99877, Q99880	H2BC12L	P57053	H2BC13	Q99880
H2BC14	Q99879	H2BC15	Q93079, Q99877	H2BC17	P23527
H2BC18	Q99877	H2BC21	P06899, Q16778	H2BC3	P06899, P33778
H2BC4	P62807, Q93079	H2BC5	P58876	H2BC6	P62807, Q93079
H2BC7	P62807	H2BC8	P62807, Q93079	H2BC9	P58876, Q93079, Q99877
H3C1	P68431	H3C11	P68431	H3C12	P68431
H3C13	Q71DI3	H3C14	Q71DI3	H3C15	Q71DI3
H3C2	P68431	H3C3	P68431	H3C4	P68431
H3C6	P68431	H3C7	P68431	H3C8	P68431
H4C1	P62805	H4C13	P62805	H4C16	P62805
H4C2	P62805	H4C3	P62805	H4C4	P62805
H4C5	P62805	H4C6	P62805	H4C8	P62805
H4C9	P62805	KDM5A	P29375	RBP2	P29375
TET3	O43151				

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
DYNLL1	P63167	Q6W0C5			

17. Regulation of endogenous retroelements by Piwi-interacting RNAs (piRNAs) (R-HSA-9845323)



PIWI-interacting RNAs (piRNAs) are short RNAs of 24-31 nucleotides that are produced by cleavage of longer RNAs and amplification by a "ping-pong" mechanism involving rounds of strand hybridization and cleavage (reviewed in Sun et al. 2022). The piRNAs are loaded onto PIWI proteins (PIWIL1, PIWIL2, PIWIL4) that are then guided by base-pairing of the piRNAs to nascent and mature transcripts, where the PIWI:piRNA complexes initiate transcriptional and post-transcriptional silencing, respectively (reviewed in Czech et al. 2018, Onishi et al. 2021, Wang et al. 2023).

In mice, sources of piRNAs include transposon RNAs, long non-coding RNAs, exon transcripts, and RNAs from unannotated regions of the genome (Aravin et al. 2007, 2008). Two populations of piRNAs are observed during mouse development: pre-pachytene piRNAs and pachytene piRNAs (Aravin et al. 2008, Gan et al. 2011). Pre-pachytene piRNAs are present prenatally in prospermatogonial cells and postnatally in spermatogonial cells. Pachytene piRNAs are present in more mature postnatal spermatocytes and spermatids. The piRNAs derived from retroelements comprise about half (Aravin et al. 2008) or less (Gan et al. 2011) of the total pre-pachytene piRNAs and the portion falls sharply from pre-pachytene to pachytene (Gan et al. 2011).

In mice, PIWIL2 (MILI, Piwil2 gene) is first detected in primordial germ cells that have reached the genital ridge and expression persists through meiosis in adults. PIWIL4 (MIWI2, PIWIl4 gene) is present in mouse germ cells during de novo DNA methylation shortly before and after birth. PIWIL1 (MIWI, PIWIL1 gene) is present during later stages of meiosis after birth (Aravin et al. 2008). PIWIL4 and PIWIL2 participate in re-methylation of the genome in mouse germ cells and consequent repression of transposable elements (Carmell et al. 2007, Aravin et al. 2008, Kuramochi-Miyagawa et al. 2008, Zoch et al. 2020). In mice, the piRNAs bound by PIWIL4 bind nascent transcripts of transposable elements and connects to the de novo DNA methylation machinery via SPOCD1 and C19ORF84 to direct DNA methylation to the transposable elements (Zoch et al. 2020, 2024). Mutations in SPOCD1 are associated with infertility in men (Zoch et al. 2024)

References

Gan H, Lin X, Zhang Z, Zhang W, Liao S, Wang L & Han C (2011). piRNA profiling during specific stages of mouse spermatogenesis. RNA, 17, 1191-203.

Aravin AA, Sachidanandam R, Bourc'his D, Schaefer C, Pezic D, Toth KF, ... Hannon GJ (2008). A piRNA pathway primed by individual transposons is linked to de novo DNA methylation in mice . Mol. Cell, 31, 785-99.

Carmell MA, Girard A, van de Kant HJ, Bourc'his D, Bestor TH, de Rooij DG & Hannon GJ (2007). MIWI2 is essential for spermatogenesis and repression of transposons in the mouse male germline. Dev. Cell, 12, 503-14.

Kuramochi-Miyagawa S, Watanabe T, Gotoh K, Totoki Y, Toyoda A, Ikawa M, ... Nakano T (2008). DNA methylation of retrotransposon genes is regulated by Piwi family members MILI and MIWI2 in murine fetal testes. Genes Dev., 22, 908-17.

Zoch A, Auchynnikava T, Berrens RV, Kabayama Y, Schöpp T, Heep M, ... O'Carroll D (2020). SPOCD1 is an essential executor of piRNA-directed de novo DNA methylation. Nature, 584, 635-639.

Edit history

Date	Action	Author
2023-09-24	Edited	May B
2023-09-24	Authored	May B
2023-10-02	Created	May B
2024-05-14	Modified	May B
2024-05-14	Reviewed	Fernandes LP, Bourc'his D

55 submitted entities found in this pathway, mapping to 67 Reactome entities

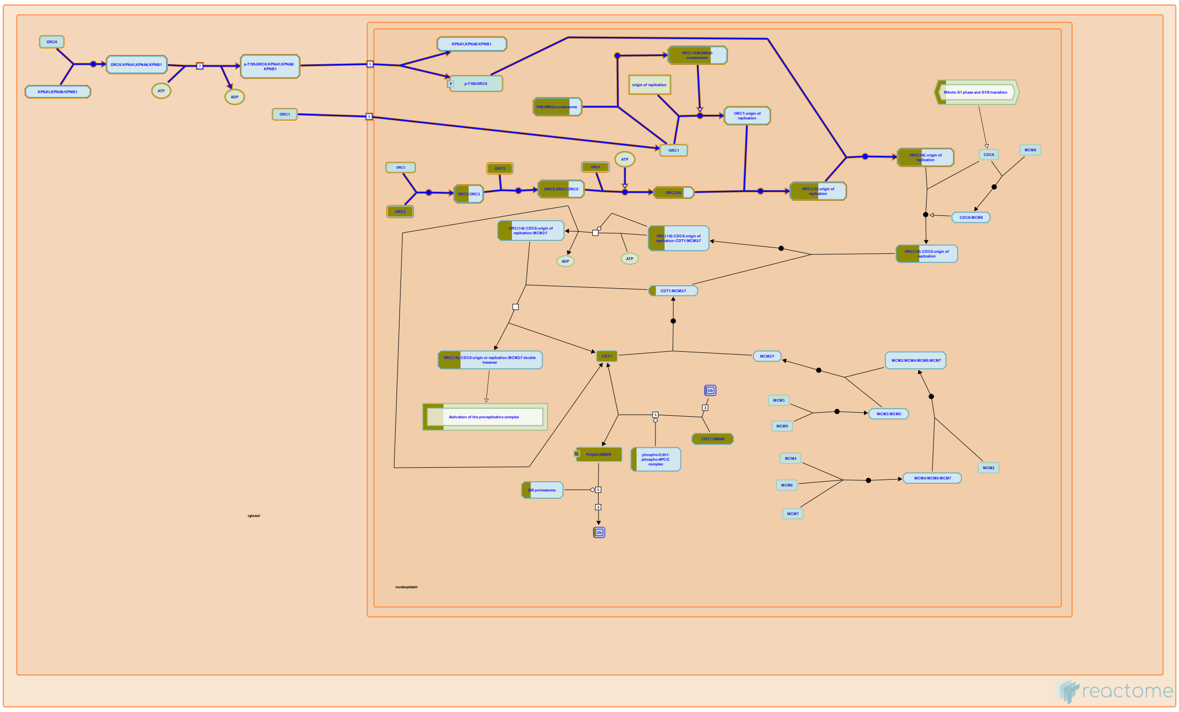
Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
BCL7A	Q4VC05	GATAD2B	Q8WXI9	H2AC12	Q99878
H2AC14	Q99878	H2AC16	P20671	H2AC18	Q6FI13
H2AC19	Q6FI13	H2AC20	Q16777	H2AC4	P04908, Q99878
H2AC6	Q93077	H2AC7	P20671	H2AC8	P04908
H2BC10	P62807	H2BC11	P06899	H2BC12	O60814, Q93079, Q99877, Q99880
H2BC12L	P57053	H2BC13	Q99880	H2BC14	Q99879
H2BC15	Q93079, Q99877	H2BC17	P23527	H2BC18	Q99877

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
H2BC21	P06899, Q16778	H2BC3	P06899, P33778	H2BC4	P62807, Q93079
H2BC5	P58876	H2BC6	P62807, Q93079	H2BC7	P62807
H2BC8	P62807, Q93079	H2BC9	P58876, Q93079, Q99877	H3C1	P68431
H3C11	P68431	H3C12	P68431	H3C13	Q71DI3
H3C14	Q71DI3	H3C15	Q71DI3	H3C2	P68431
H3C3	P68431	H3C4	P68431	H3C6	P68431
H3C7	P68431	H3C8	P68431	H4C1	P62805
H4C13	P62805	H4C16	P62805	H4C2	P62805
H4C3	P62805	H4C4	P62805	H4C5	P62805
H4C6	P62805	H4C8	P62805	H4C9	P62805
HDAC2	Q92769	MTA1	Q13330	MTA2	O94776
SMARCD2	Q92925				

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
RSL24D1	Q9UHA3	Q9UJW3			

18. Assembly of the ORC complex at the origin of replication (R-HSA-68616)



Cellular compartments: nucleoplasm.

Human ORC1 can associate with DNA origin of replication sites independently of other origin of replication complex (ORC) subunits (Hoshina et al. 2013; Eladl et al. 2021). ORC1 localizes to condensed chromosomes during early mitosis (M phase) and serves as a nucleating center for the assembly of the ORC and, subsequently, the pre-replication complex. ORC1 remains associated with late replication origins throughout late G1. Upon S phase entry, ORC1 undergoes ubiquitin-mediated degradation, leading to dissociation of the ORC from chromatin (Kara et al. 2015).

Most human replication origins contain guanine (G)-rich sequences which may form G-quadruplex (G4) structures (Besnard et al. 2012) and these G4 structures may mediate the recognition of replication origins by ORC1 (Hoshina et al. 2013; Eladl et al. 2021). Besides binding to nucleosome-free replication origin DNA, ORC1 interacts with neighboring nucleosomes (Hizume et al. 2013), in particular with nucleosomes containing histone H4 dimethylated at lysine 21 (H4K20me2 mark), which is enriched at replication origins. Binding of ORC1 to H4K20me2 facilitates ORC1 binding to replication origins and ORC chromatin loading (Kuo et al. 2012, Zhang et al. 2015).

ORC1 binding sites are universally associated with transcription start sites (TSSs) of coding and non-coding RNAs. Replication origins associated with moderate to high transcription level TSSs (belonging to coding RNAs) fire in early S phase, while those associated with low transcription level TSSs (belonging to non-coding RNAs) fire throughout the S phase (Dellino et al. 2013).

ORC2 forms a heterodimer with ORC3, which is a prerequisite for the association of ORC5 and, subsequently, ORC4 (Ranjan and Gossen 2006; Siddiqui and Stillman 2007). ORC1 binds to the ORC(2-5) complex in the nucleus to form a stable ORC(1-5) complex (Radichev et al. 2006; Ghosh et al. 2011). ORC1 is necessary for the association of the ORC(2-5) complex to chromatin (Radichev et al. 2006). The ORC(2-5) complex exhibits a tightly autoinhibited conformation, with the winged-helix domain (WHD) of ORC2 completely blocking the central DNA-binding channel. Binding of ORC1 remodels the WHD of ORC2, moving it away from the central channel and partially relieving the autoinhibition (Cheng et al. 2020, Jaremko et al. 2020). ORC6 associates with the ORC(1-5) complex to form the ORC(1-6) complex (Ghosh et al. 2011). The association of ORC6 with the ORC(1-5) complex is weak and it frequently does not co-immunoprecipitate with the other ORC(1-5) subunits. ORC4 is the only ORC(1-5) subunit that was shown to directly bind to ORC6 (Radichev et al. 2006). Some ORC6 mutations reported in Meier-Gorlin syndrome were shown to interfere with ORC6 incorporation into the ORC (Balasov et al. 2015).

References

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Eladl A, Yamaoki Y, Hoshina S, Horinouchi H, Kondo K, Waga S, ... Katahira M (2021). Investigation of the Interaction of Human Origin Recognition Complex Subunit 1 with G-Quadruplex DNAs of Human <i>c-myc</i> Promoter and Telomere Regions. Int J Mol Sci, 22.

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Besnard E, Babled A, Lapasset L, Milhavet O, Parrinello H, Dantec C, ... Lemaitre JM (2012). Unraveling cell type-specific and reprogrammable human replication origin signatures associated with G-quadruplex consensus motifs. Nat Struct Mol Biol, 19, 837-44.

Hizume K, Yagura M & Araki H (2013). Concerted interaction between origin recognition complex (ORC), nucleosomes and replication origin DNA ensures stable ORC-origin binding. Genes Cells, 18, 764-79.

Edit history

Date	Action	Author
2003-06-05	Authored	Davey MJ, O'Donnell M
2003-06-05	Created	Davey MJ, O'Donnell M
2021-07-30	Revised	Kusic-Tisma J
2021-07-30	Authored	Kusic-Tisma J
2021-08-16	Edited	Orlic-Milacic M
2025-11-15	Modified	Weiser JD

52 submitted entities found in this pathway, mapping to 64 Reactome entities

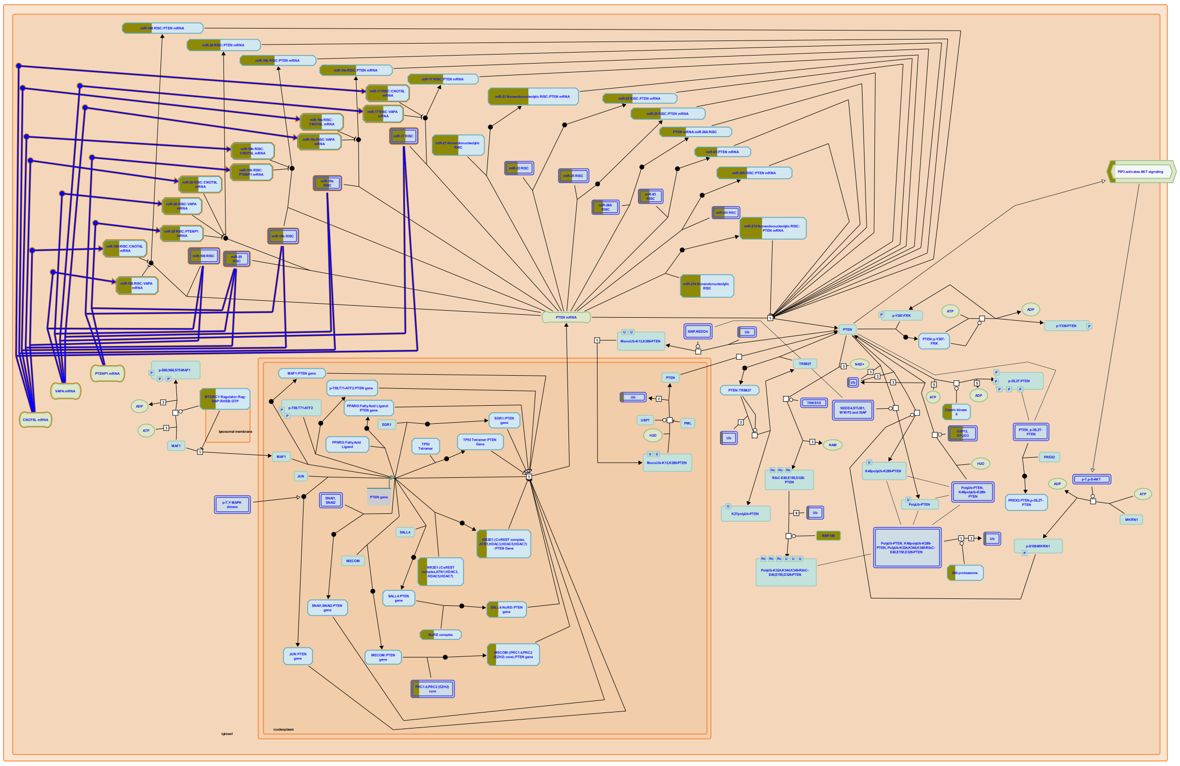
Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
H2AC12	Q99878	H2AC14	Q99878	H2AC16	P20671
H2AC18	Q6FI13	H2AC19	Q6FI13	H2AC20	Q16777

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
H2AC4	P04908, Q99878	H2AC6	Q93077	H2AC7	P20671
H2AC8	P04908	H2BC10	P62807	H2BC11	P06899
H2BC12	O60814, Q93079, Q99877, Q99880	H2BC12L	P57053	H2BC13	Q99880
H2BC14	Q99879	H2BC15	Q93079, Q99877	H2BC17	P23527
H2BC18	Q99877	H2BC21	P06899, Q16778	H2BC3	P06899, P33778
H2BC4	P62807, Q93079	H2BC5	P58876	H2BC6	P62807, Q93079
H2BC7	P62807	H2BC8	P62807, Q93079	H2BC9	P58876, Q93079, Q99877
H3C1	P68431	H3C11	P68431	H3C12	P68431
H3C13	Q71DI3	H3C14	Q71DI3	H3C15	Q71DI3
H3C2	P68431	H3C3	P68431	H3C4	P68431
H3C6	P68431	H3C7	P68431	H3C8	P68431
H4C1	P62805	H4C13	P62805	H4C16	P62805
H4C2	P62805	H4C3	P62805	H4C4	P62805
H4C5	P62805	H4C6	P62805	H4C8	P62805
H4C9	P62805	ORC2	Q13416	ORC4	O43929
ORC5	O43913				

Interactors found in this pathway (7)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
APLNR	P35414	O43913	LHX4	Q969G2	Q9Y5N6
MPP2	Q08050	Q13416	NPTN	Q9Y639	O43929
ORC2	Q13416	O43929, Q9UBD5, Q13415, Q9Y5N6, O43913	ORC4	O43929	Q13416, Q9UBD5, Q13415, Q9Y5N6, O43913
ORC5	O43913	O43929, Q13416, Q9UBD5, Q13415			

19. Competing endogenous RNAs (ceRNAs) regulate PTEN translation (R-HSA-8948700)



Coding and non-coding RNAs can prevent microRNAs from binding to PTEN mRNA. These RNAs are termed competing endogenous RNAs or ceRNAs. Transcripts of the pseudogene PTENP1 and mRNAs transcribed from SERINC1, VAPA and CNOT6L genes exhibit this activity (Poliseno et al. 2010, Tay et al. 2011, Tay et al. 2014). SERINC1 mRNA will be annotated in this context when additional experimental details become available.

References

Poliseno L, Salmena L, Zhang J, Carver B, Haveman WJ & Pandolfi PP (2010). A coding-independent function of gene and pseudogene mRNAs regulates tumour biology. Nature, 465, 1033-8.

Tay Y, Kats L, Salmena L, Weiss D, Tan SM, Ala U, ... Pandolfi PP (2011). Coding-independent regulation of the tumor suppressor PTEN by competing endogenous mRNAs. Cell, 147, 344-57.

Tay Y, Rinn J & Pandolfi PP (2014). The multilayered complexity of ceRNA crosstalk and competition. Nature, 505, 344-52.

Edit history

Date	Action	Author
2016-08-11	Authored	Salmena L, Carracedo A
2016-11-03	Authored	Orlic-Milacic M
2016-11-16	Created	Orlic-Milacic M
2017-05-09	Edited	Orlic-Milacic M
2023-03-08	Modified	Matthews L

8 submitted entities found in this pathway, mapping to 12 Reactome entities

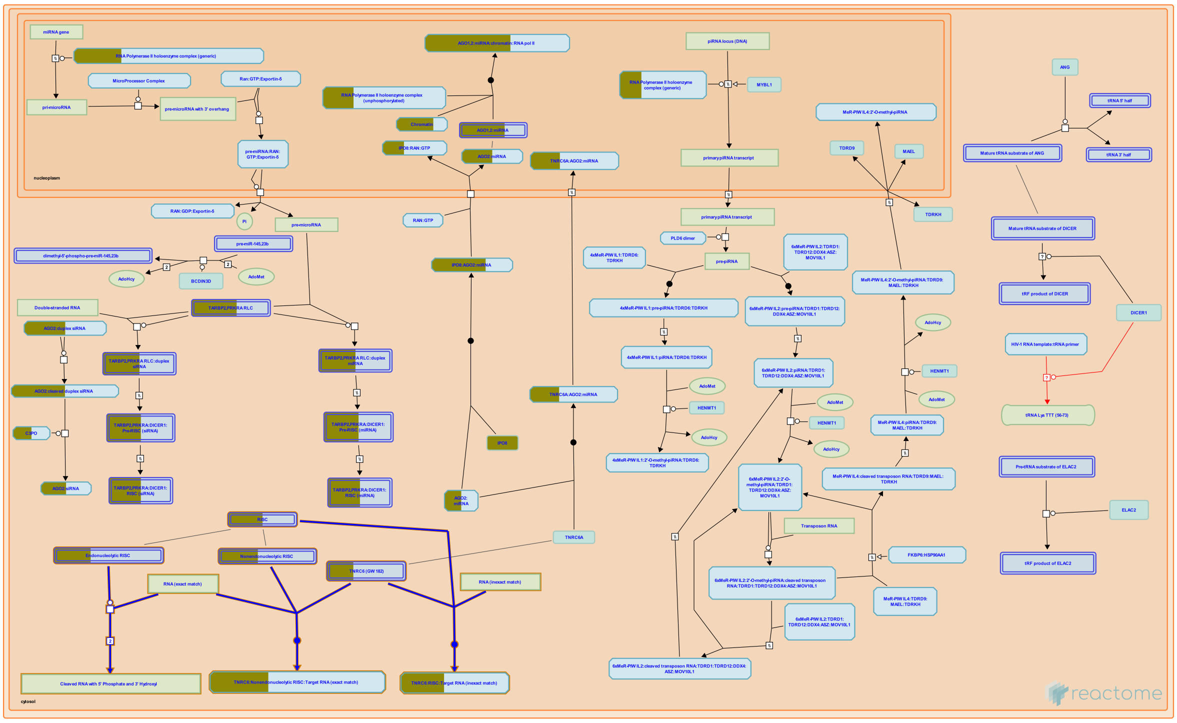
Input	UniProt Id
AGO1	Q9H9G7, Q9UKV8, Q9UL18
AGO3	Q9H9G7, Q9UL18

Input	UniProt Id
AGO2	Q9H9G7, Q9UKV8
TNRC6B	Q9UPQ9

Input	miRBase Id
MIR17	MI0000071
MIR19B1	MI0000074

Input	miRBase Id
MIR19A	MI0000073
MIR20A	MI0000076

20. Post-transcriptional silencing by small RNAs (R-HSA-426496)



Cellular compartments: cytosol.

Small RNAs act with components of the RNA-induced silencing complex (RISC) to post-transcriptionally repress expression of mRNAs (reviewed in Nowotny and Yang 2009, Chua et al. 2009). Two mechanisms exist: 1) cleavage of target RNAs by complexes containing Argonaute2 (AGO2, EIF2C2) and a guide RNA that exactly matches the target mRNA and 2) inhibition of translation of target RNAs by complexes containing AGO2 and an inexactly matching guide RNA or by complexes containing a nonendonucleolytic Argonaute (AGO1 (EIF2C1), AGO3 (EIF2C3), or AGO4 (EIF2C4)) and a guide RNA of exact or inexact match. Small interfering RNAs (siRNAs) and microRNAs (miRNAs) can serve as guide RNAs in both types of mechanism.

RNAi also appears to direct chromatin modifications that cause transcriptional gene silencing (reviewed in Verdel et al. 2009).

References

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Maroney PA, Yu Y, Fisher J & Nilsen TW (2006). Evidence that microRNAs are associated with translating messenger RNAs in human cells. Nat Struct Mol Biol, 13, 1102-7.

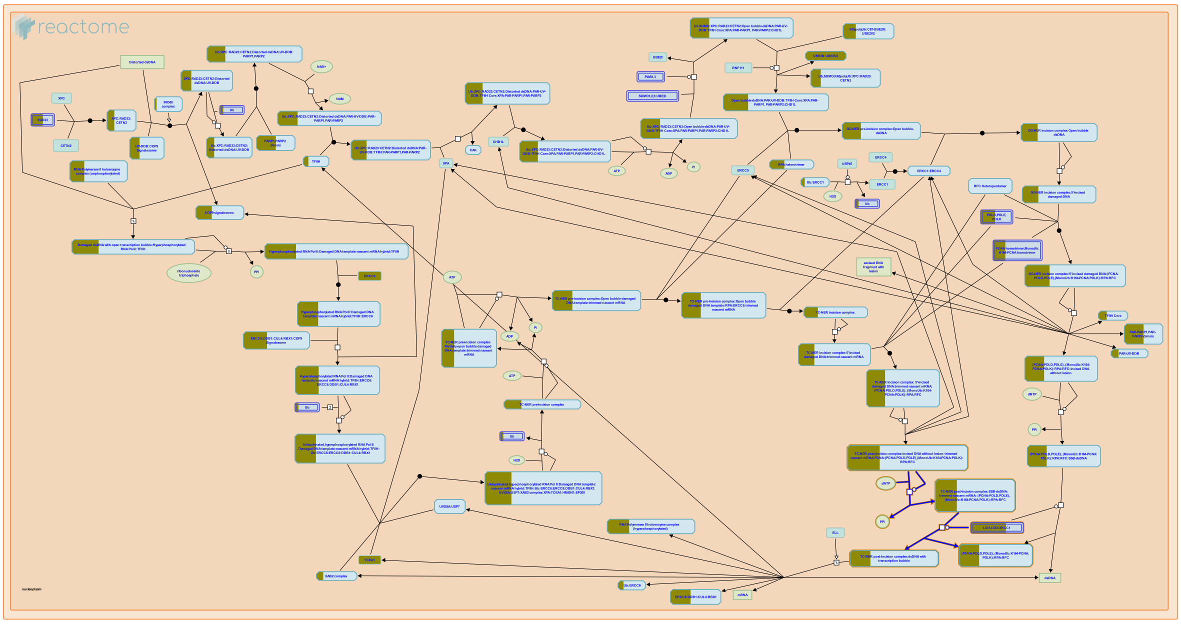
Edit history

Date	Action	Author
2009-06-10	Edited	May B
2009-06-10	Authored	May B
2009-06-17	Created	May B
2012-02-11	Reviewed	Tomari Y
2025-11-15	Modified	Weiser JD

4 submitted entities found in this pathway, mapping to 8 Reactome entities

Input	UniProt Id	Input	UniProt Id
AGO1	Q9H9G7, Q9UKV8, Q9UL18	AGO2	Q9H9G7, Q9UKV8
AGO3	Q9H9G7, Q9UL18	TNRC6B	Q9UPQ9

21. Gap-filling DNA repair synthesis and ligation in TC-NER (R-HSA-6782210)



Cellular compartments: nucleoplasm.

In transcription-coupled nucleotide excision repair (TC-NER), similar to global genome nucleotide excision repair (GG-NER), DNA polymerases delta or epsilon, or the Y family DNA polymerase kappa, fill in the single stranded gap that remains after dual incision. DNA ligases LIG1 or LIG3, subsequently seal the single stranded nick by ligating the 3' end of the newly synthesized patch with the 5' end of incised DNA (Staresincic et al. 2009, Ogi et al. 2010, Arakawa et al. 2012, Paul-Konietzko et al. 2015).

References

Staresincic L, Fagbemi AF, Enzlin JH, Gourdin AM, Wijgers N, Dunand-Sauthier I, ... Schärer OD (2009). Coordination of dual incision and repair synthesis in human nucleotide excision repair. EMBO J., 28, 1111-20.

Ogi T, Limsirichaikul S, Overmeer RM, Volker M, Takenaka K, Cloney R, ... Lehmann AR (2010). Three DNA polymerases, recruited by different mechanisms, carry out NER repair synthesis in human cells. Mol. Cell, 37, 714-27.

Paul-Konietzko K, Thomale J, Arakawa H & Iliakis G (2015). DNA Ligases I and III Support Nucleotide Excision Repair in DT40 Cells with Similar Efficiency. Photochem Photobiol, 91, 1173-80.

Arakawa H, Bednar T, Wang M, Paul K, Mladenov E, Bencsik-Theilen AA & Iliakis G (2012). Functional redundancy between DNA ligases I and III in DNA replication in vertebrate cells. Nucleic Acids Res, 40, 2599-610.

Edit history

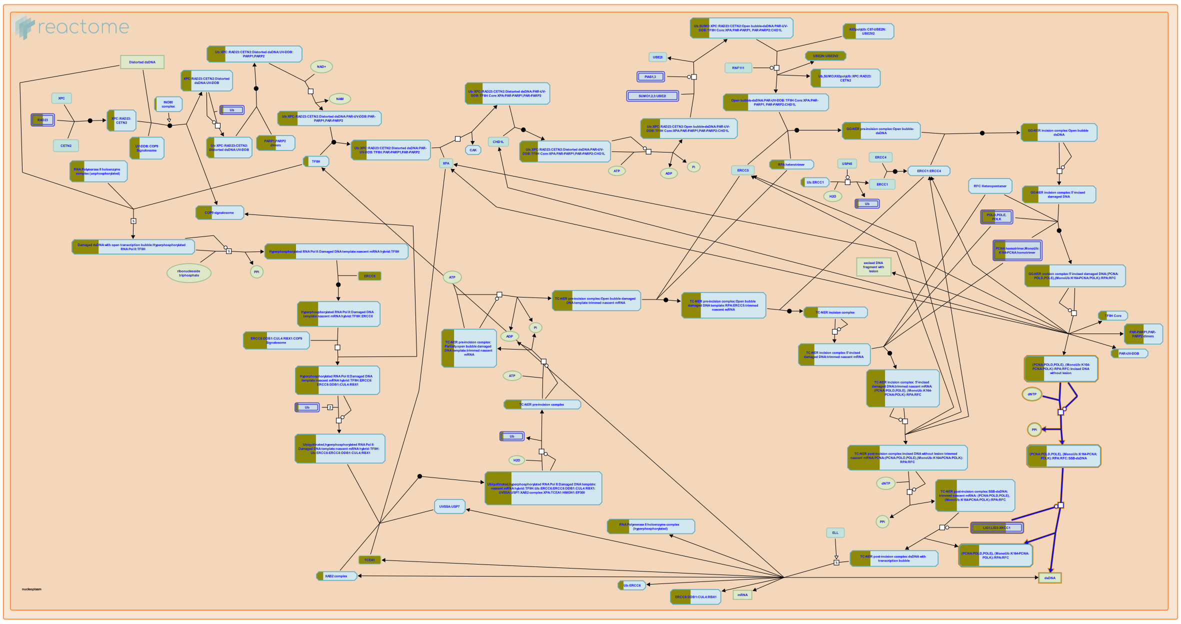
Date	Action	Author
2004-01-29	Authored	Hoeijmakers JH
2015-06-05	Created	Orlic-Milacic M
2015-06-16	Revised	Orlic-Milacic M

Date	Action	Author
2015-06-16	Edited	Orlic-Milacic M
2015-06-16	Authored	Orlic-Milacic M
2015-08-03	Reviewed	Fousteri M
2024-02-22	Edited	Orlic-Milacic M

20 submitted entities found in this pathway, mapping to 21 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ERCC6	Q03468	ERCC8	Q13216	GTF2H3	Q13889
GTF2H4	Q92759	LIG1	P18858	MT1A	P53803
PCNA	P12004	POLD1	P28340	POLD3	Q15054
POLE2	P56282	POLE4	Q9NR33	POLR2F	P61218
POLR2H	P52434	POLR2K	P53803	PRPF19	Q9UMS4
RBX1	P62877	RPA1	P27694	RPS27A	P62979, P62987
TCEA1	P23193	XRCC1	P18887		

22. Gap-filling DNA repair synthesis and ligation in GG-NER (R-HSA-5696397)



Cellular compartments: nucleoplasm.

Global genome nucleotide excision repair (GG-NER) is completed by DNA repair synthesis that fills the single stranded gap created after dual incision of the damaged DNA strand and excision of the ~27-30 bases long oligonucleotide that contains the lesion. DNA synthesis is performed by DNA polymerases epsilon or delta, or the Y family DNA polymerase kappa (POLK), which are loaded to the repair site after 5' incision (Staresincic et al. 2009, Ogi et al. 2010). DNA ligases LIG1 or LIG3 ligate the newly synthesized stretch of oligonucleotides to the incised DNA strand (Arakawa et al. 2012, Paul-Konietzko et al. 2015).

References

Staresincic L, Fagbemi AF, Enzlin JH, Gourdin AM, Wijgers N, Dunand-Sauthier I, ... Schärer OD (2009). Coordination of dual incision and repair synthesis in human nucleotide excision repair. EMBO J., 28, 1111-20.

Ogi T, Limsirichaikul S, Overmeer RM, Volker M, Takenaka K, Cloney R, ... Lehmann AR (2010). Three DNA polymerases, recruited by different mechanisms, carry out NER repair synthesis in human cells. Mol. Cell, 37, 714-27.

Arakawa H, Bednar T, Wang M, Paul K, Mladenov E, Bencsik-Theilen AA & Iliakis G (2012). Functional redundancy between DNA ligases I and III in DNA replication in vertebrate cells. Nucleic Acids Res, 40, 2599-610.

Paul-Konietzko K, Thomale J, Arakawa H & Iliakis G (2015). DNA Ligases I and III Support Nucleotide Excision Repair in DT40 Cells with Similar Efficiency. Photochem Photobiol, 91, 1173-80.

Edit history

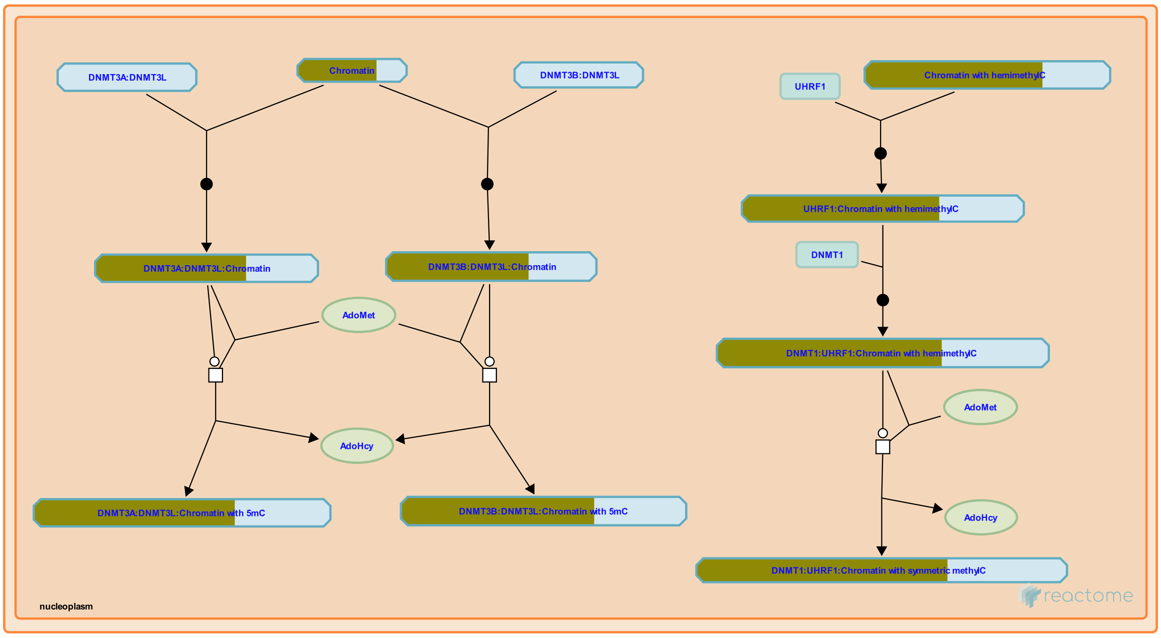
Date	Action	Author
2004-01-29	Authored	Hoeijmakers JH
2004-02-02	Authored	Gopinathrao G
2015-05-28	Created	Orlic-Milacic M

Date	Action	Author
2015-06-16	Revised	Orlic-Milacic M
2015-06-16	Edited	Orlic-Milacic M
2015-06-16	Authored	Orlic-Milacic M
2015-08-03	Reviewed	Fousteri M
2024-02-22	Edited	Orlic-Milacic M

9 submitted entities found in this pathway, mapping to 10 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
LIG1	P18858	PCNA	P12004	POLD1	P28340
POLD3	Q15054	POLE2	P56282	POLE4	Q9NR33
RPA1	P27694	RPS27A	P62979, P62987	XRCC1	P18887

23. DNA methylation (R-HSA-5334118)



Cellular compartments: nucleoplasm.

Methylation of cytosine is catalyzed by a family of DNA methyltransferases (DNMTs): DNMT1, DNMT3A, and DNMT3B transfer methyl groups from S-adenosylmethionine to cytosine, producing 5-methylcytosine and homocysteine (reviewed in Klose and Bird 2006, Ooi et al. 2009, Jurkowska et al. 2011, Moore et al. 2013). (DNMT2 appears to methylate RNA rather than DNA.) DNMT1, the first enzyme discovered, preferentially methylates hemimethylated CG motifs that are produced by replication (template strand methylated, synthesized strand unmethylated). Thus it maintains existing methylation through cell division. DNMT3A and DNMT3B catalyze de novo methylation at unmethylated sites that include both CG dinucleotides and non-CG motifs.

DNA from adult humans contains about 0.76 to 1.00 mole percent 5-methylcytosine (Ehrlich et al. 1982, reviewed in Klose and Bird 2006, Ooi et al. 2009, Moore et al. 2013). Methylation of DNA occurs at cytosines that are mainly located in CG dinucleotides. CG dinucleotides are unevenly distributed in the genome. Promoter regions tend to have a high CG-content, forming so-called CG-islands (CGIs), while the CG-content in the remaining part of the genome is much lower. CGIs tend to be unmethylated, while the majority of CGs outside CGIs are methylated. Methylation in promoters and first exons tends to repress transcription while methylation in gene bodies (regions of genes downstream of the promoter and first exon) correlates with transcription (reviewed in Ehrlich and Lacey 2013, Kulis et al. 2013). Proteins such as MeCP2 and MBDs specifically bind 5-methylcytosine and may recruit other factors.

Mammalian development has two major episodes of genome-wide demethylation and remethylation (reviewed in Zhou 2012, Guibert and Weber 2013, Hackett and Surani 2013, Dean 2014). In mice about 1 day after fertilization the paternal genome is actively demethylated by TET proteins together with thymine DNA glycosylase and the maternal genome is demethylated by passive dilution during replication, however methylation at imprinted sites is maintained. The genome has its lowest methylation level about 3.5 days post-fertilization. Remethylation occurs by 6.5 days post-fertilization. The second demethylation-remethylation event occurs in primordial germ cells of the developing embryo about 12.5 days post-fertilization. DNMT3A and DNMT3B, together with the non-catalytic DNMT3L, play major roles in the remethylation events (reviewed in Chen and Chan 2014). How the methyltransferases are directed to particular regions of the genome remains an area of active research. The mechanisms at each locus may differ in detail but a connection between histone modifications and DNA methylation has been observed (reviewed in Rose and Klose 2014).

References

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Hackett JA & Surani MA (2013). DNA methylation dynamics during the mammalian life cycle. Philos. Trans. R. Soc. Lond., B, Biol. Sci., 368, 20110328.

Moore LD, Le T & Fan G (2013). DNA methylation and its basic function. Neuropsychopharmacology, 38, 23-38.

Chen BF & Chan WY (2014). The de novo DNA methyltransferase DNMT3A in development and cancer. Epigenetics, 9.

Rose NR & Klose RJ (2014). Understanding the relationship between DNA and histone lysine methylation. Biochim. Biophys. Acta.

Edit history

Date	Action	Author
2014-02-21	Edited	May B
2014-02-21	Authored	May B
2014-02-22	Created	May B
2014-07-24	Reviewed	Martín-Subero JI, Beekman R
2024-05-16	Modified	May B

49 submitted entities found in this pathway, mapping to 61 Reactome entities

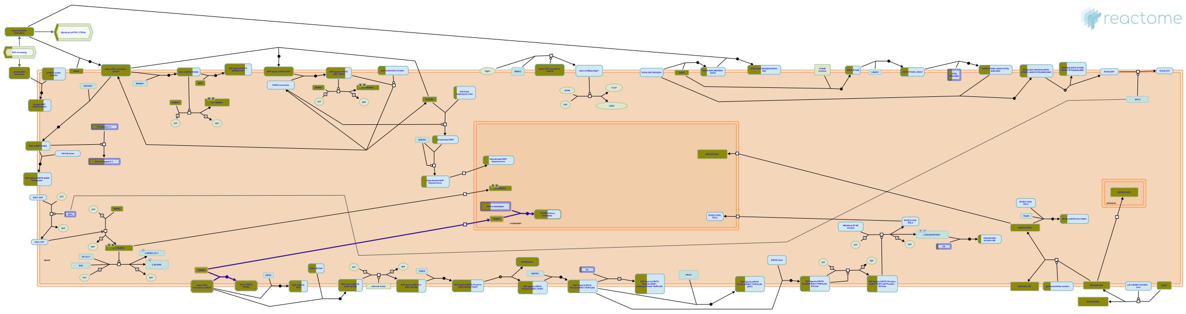
Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
H2AC12	Q99878	H2AC14	Q99878	H2AC16	P20671
H2AC18	Q6FI13	H2AC19	Q6FI13	H2AC20	Q16777
H2AC4	P04908, Q99878	H2AC6	Q93077	H2AC7	P20671
H2AC8	P04908	H2BC10	P62807	H2BC11	P06899
H2BC12	O60814, Q93079, Q99877, Q99880	H2BC12L	P57053	H2BC13	Q99880
H2BC14	Q99879	H2BC15	Q93079, Q99877	H2BC17	P23527
H2BC18	Q99877	H2BC21	P06899, Q16778	H2BC3	P06899, P33778
H2BC4	P62807, Q93079	H2BC5	P58876	H2BC6	P62807, Q93079

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
H2BC7	P62807	H2BC8	P62807, Q93079	H2BC9	P58876, Q93079, Q99877
H3C1	P68431	H3C11	P68431	H3C12	P68431
H3C13	Q71DI3	H3C14	Q71DI3	H3C15	Q71DI3
H3C2	P68431	H3C3	P68431	H3C4	P68431
H3C6	P68431	H3C7	P68431	H3C8	P68431
H4C1	P62805	H4C13	P62805	H4C16	P62805
H4C2	P62805	H4C3	P62805	H4C4	P62805
H4C5	P62805	H4C6	P62805	H4C8	P62805
H4C9	P62805				

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
DCAF5	Q96JK2	P26358			

24. p75NTR negatively regulates cell cycle via SC1 (R-HSA-193670)



SC1 (Schwann Cell factor 1; also called PR domain zinc finger protein 4, PRDM4) interacts with an NGF:p75NTR complex and signals cell cycle arrest by regulating the levels of cyclin E.

References

Lopez-Sanchez N & Frade JM (2002). Control of the cell cycle by neurotrophins: lessons from the p75 neurotrophin receptor. Histol Histopathol, 17, 1227-37.

Edit history

Date	Action	Author
2006-10-10	Authored	Annibali D, Nasi S
2007-02-23	Created	Jassal B
2008-05-20	Edited	Jassal B
2008-05-20	Reviewed	Friedman WJ
2008-05-28	Reviewed	Chao MV
2023-03-08	Modified	Matthews L

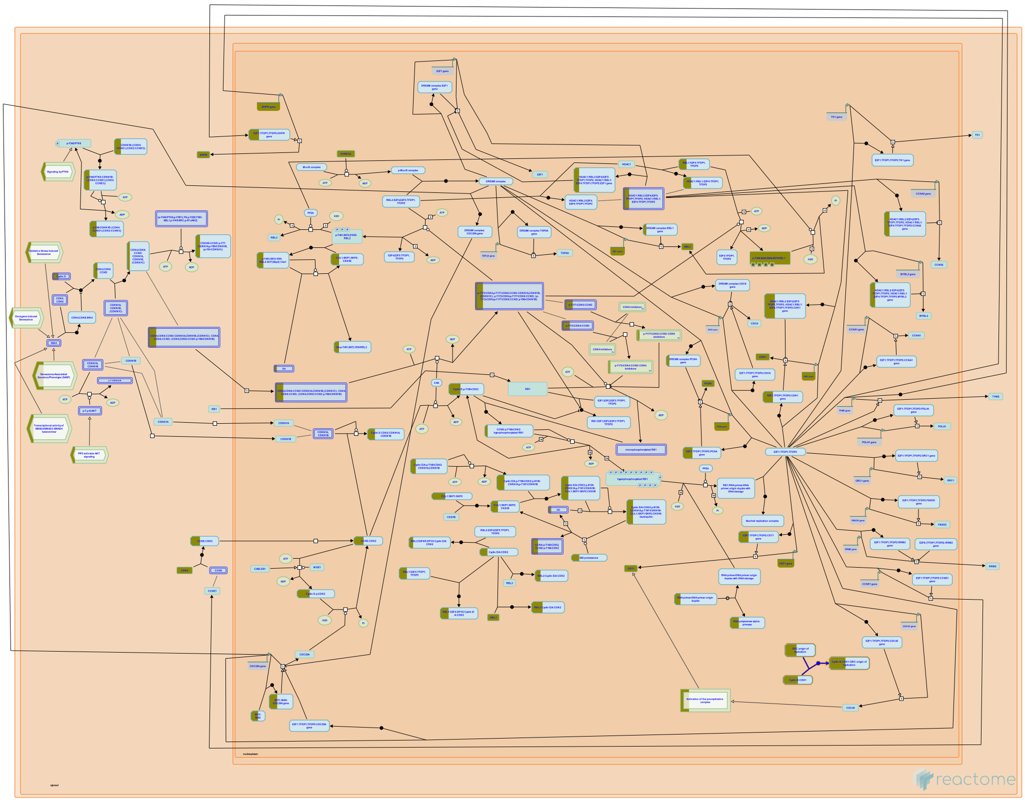
4 submitted entities found in this pathway, mapping to 4 Reactome entities

Input	UniProt Id	Input	UniProt Id
HDAC2	Q92769	NGF	P01138
NGFR	P08138	PRDM4	Q9UKN5

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
CTNNBL1	Q8WYA6	Q9UKN5			

25. E2F-enabled inhibition of pre-replication complex formation (R-HSA-113507)



Cellular compartments: nucleoplasm.

Under specific conditions, Cyclin B, a mitotic cyclin, can inhibit the functions of pre-replicative complex. E2F1 activates Cdc25A protein which regulates Cyclin B in a positive manner. Cyclin B/Cdk1 function is restored which leads to the disruption of pre-replicative complex. This phenomenon has been demonstrated by Bosco et al (2001) in Drosophila.

References

Kennedy BK, Barbie DA, Classon M, Harlow E & Dyson NJ (2000). Nuclear organization of DNA replication in primary mammalian cells. Genes Dev, 14, 2855-68.

Edit history

Date	Action	Author
2004-05-26	Authored	Gopinathrao G
2004-06-16	Created	Bosco G

4 submitted entities found in this pathway, mapping to 4 Reactome entities

Input	UniProt Id	Input	UniProt Id
CDK1	P06493	ORC2	Q13416
ORC4	O43929	ORC5	O43913

6. Identifiers found

Below is a list of the input identifiers that have been found or mapped to an equivalent element in Reactome, classified by resource.

1423 of the submitted entities were found, mapping to 1665 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
A1CF	Q9NQ94	A2M	P01023	AAAS	Q9NRG9
AACS	Q86V21	AADAT	Q8N5Z0	AAK1	Q2M2I8
AAMP	Q13685	AASDHPPT	Q9NRN7	AATF	Q9NY61
ABCA7	Q8IZY2	ABCB5	Q75027	ABCC9	O60706
ABCD3	O14678	ABL2	P42684	ACAA2	P42765
ACAD10	Q6JQN1	ACBD7	Q8N6N7	ACP6	Q9NPH0
ACSL3	O95573	ACSL5	Q9ULC5	ACSM3	Q68CK6
ACSM6	Q6P461	ACTA2	P62736	ACTC1	P68032
ACTG2	P63267	ADA2	Q9NZK5	ADAM12	O43184
ADAM20	O43506	ADAMTS7	Q9UKP4	ADAT2	Q7Z6V5
ADCY10	Q96PN6	ADCY5	O95622	ADH4	P08319
ADO	Q96SZ5	ADORA2A	P29274	ADSS2	P30520
AFG3L2	Q9Y4W6	AGL	P35573	AGO1	Q9UKV8, Q9UL18
AGO2	Q9UKV8	AGO3	Q9UL18	AGPAT5	Q9NUQ2
AGPS	O00116	AGRN	O00468	AIFM2	Q9BRQ8
AK6	Q9Y3D8	ALDH18A1	P54886-1, P54886-2	ALG6	Q9Y672
ALMS1	Q8TCU4	ALYREF	Q86V81	AMD1	Q01432
ANAPC4	Q9UJX5	ANGPT2	O15123	ANKRD9	Q96BM1
ANKZF1	Q9H8Y5	ANO10	Q9NW15	AP1S1	P56377, P61966
AP1S2	P56377, Q96PC3	AP1S3	Q96PC3	AP2A1	O95782
APEX1	P27695	APH1A	Q96BI3	APH1B	Q8WW43
APIP	Q96GX9	APLNR	P35414	APOBEC4	Q8WW27
APOF	Q13790	AQP10	Q96PS8	ARAP1	Q96P48
ARF5	P84077	ARHGAP11B	Q3KRB8, Q6P4F7	ARHGAP21	Q5T5U3
ARHGAP30	Q7Z6I6	ARHGAP35	Q9NRY4	ARHGAP4	P98171
ARHGAP45	Q92619	ARHGEF11	O15085	ARHGEF9	O43307
ARID2	Q68CP9	ARID3A	Q99856	ARL3	P36405
ARPC2	O15144	ARSB	P15848	ARSG	Q96EG1
ART4	Q93070	ASAH1	Q13510	ASAP1	Q9ULH1
ASB7	Q8WXK1, Q9H672	ASH1L	Q9NR48	ASMT	P46597
ASNS	P08243	ASPN	Q9BXN1	ATAT1	Q5SQI0
ATF5	Q9Y2D1	ATG12	O94817	ATG4C	Q96DT6
ATOX1	O00244	ATP10A	O60312	ATP10B	O94823
ATP2A3	Q93084	ATP2B1	P20020	ATP5ME	P56385
ATP5MF	P56134	ATP5MJ	P56378	ATP5PF	P18859
ATP6AP2	O75787	ATP6V0A2	Q9Y487	ATP6V1G1	O75348, Q96LB4
ATP8B1	O43520	ATP9B	O43861	ATXN7	O15265
AUTS2	Q8WXX7	AVEN	Q9NQS1	AWAT1	Q58HT5
AZIN1	O14977	B3GALT2	O43825	B3GALT5	Q9Y2C3
B3GNT2	Q9NY97	BAG4	O95429	BAX	Q07812
BBOX1	O75936	BBS10	Q8TAM1	BCL7A	Q4VC05

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
BCR	P11274	BET1	O15155	BET1L	Q9NYM9
BGN	P21810	BICD1	Q96G01	BICD2	Q8TD16
BICRAL	Q6AI39	BID	P55957	BIRC2	Q13489, Q13490
BMP2K	Q2M2I8	BMPR1B	O00238	BNIP2	Q12982
BPGM	P07738	BPNT1	P49441	BRAF	P15056
BRAP	Q7Z569	BRD4	O60885	BRPF3	Q9ULD4
BST2	Q10589	BTBD1	Q9H0C5	BTD	P43251
BTN2A1	Q7KYR7	C1D	Q13901	C1GALT1C1	Q96EU7
C6orf120	Q7Z4R8	CA4	P22748	CA5B	P35218, Q9Y2D0
CAMLG	P49069	CAP2	P40123	CAPNS1	P04632, Q96L46
CAPZA2	P47755	CARNMT1	Q8N4J0	CASP2	P42575
CASP8	Q14790	CASQ2	O14958	CASTOR2	A6NHX0
CATSPER2	Q96P56	CATSPERG	Q6ZRH7	CBL	P22681
CBLB	Q13191	CBX3	Q13185	CBX5	P45973
CC2D2A	Q9P2K1	CCAR2	Q8N163	CCDC12	Q8WUD4
CCDC59	Q9P031	CCL27	Q9Y4X3	CCND2	P30279
CCNK	O75909	CCR9	P51686	CCZ1	P86791
CD200R1L	Q8TD46	CD44	P16070	CD55	P08174
CD9	P21926	CDC37L1	Q7L3B6	CDC40	O60508
CDC42BPA	Q5VT25	CDK1	P06493	CDK11A	Q9UQ88
CDK12	Q14004, Q9NYV4	CDT1	Q9H211	CENPH	Q9H3R5
CENPK	Q9BS16	CENPP	Q6IPU0	CENPQ	Q7L2Z9
CENPU	Q71F23	CEP70	Q8NHQ1	CEP89	Q96ST8
CEP97	Q8IW35	CEPT1	Q9Y6K0	CER1	O95813
CERS5	Q8N5B7	CERT1	Q9Y5P4-2	CFB	P00751
CHGA	P10645	CHMP2B	Q9UQN3	CHRDL1	Q9BU40
CHRNG	P07510	CHST11	Q9NPF2	CHTF8	P0CG13
CHTOP	Q9Y3Y2	CIAO3	Q9H6Q4	CIITA	P33076
CKB	P12277	CLASP1	Q7Z460	CLDN18	P56856
CLDN20	P56880	CLDN22	Q8N7P3	CLEC4A	Q9UMR7
CLOCK	O15516	CMC2	Q9NRP2	CMKLR1	Q99788
CMPK1	P30085	CMTM6	Q9NX76	CNIH1	O95406
CNOT10	Q9H9A5	CNOT4	O95628	CNOT7	Q9UIV1
CNTF	P26441	CNTN3	Q9P232	COG7	P83436
COL10A1	Q03692	COL12A1	Q99715	COL14A1	Q05707
COL18A1	P53420	COL3A1	P02461	COL4A3	Q01955
COL8A1	P27658	COMMD10	Q9Y6G5	COMMD2	Q86X83
COMMD8	Q9NX08	COPS6	Q7L5N1	COPS7B	Q9H9Q2
COPS8	Q99627	COPZ2	Q9P299	COQ10A	Q96MF6
CORO1A	P31146	COX14	Q96I36	COX16	P57103
COX6B1	P14854	COX6C	P09669	COX7A1	P24310
COX7C	P15954	CPE	P16870	CPOX	P36551
CPT2	P23786	CRCP	O75575	CREB3L2	Q70SY1
CREBBP	Q92793	CRIPTO3	P51864	CRK	P46108
CRMP1	Q14194	CRTC2	Q53ET0	CSF1R	P07333
CSNK1G2	P78368	CSNK2A2	P19784	CSTB	P04080
CSTF1	Q05048	CSTF2T	Q9H0L4	CSTF3	Q12996
CTBP1	Q13363	CTDP1	Q9Y5B0	CTNNBL1	Q8WYA6
CUL5	Q93034	CUTC	Q9NTM9	CWF19L1	O15111

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CXCR6	O00574	CYB5R2	Q6BCY4	CYBA	P13498
CYP19A1	P11511	CYP3A5	P20815	CYP4A22	P13584, Q5TCH4
CYRIB	Q9NUQ9	DARS2	Q6PI48	DBF4	Q9UBU7
DBI	P07108	DBN1	Q16643	DCAF11	Q8TEB1
DCAF5	Q96JK2	DCAKD	Q8WVC6	DCK	P27707
DCP1A	Q9NPI6	DCUN1D2	Q6PH85	DCUN1D5	Q9BTE7
DDIT3	P35638	DDR1	Q08345	DDR2	Q16832
DDX21	Q9NR30	DEK	P51580	DENND2D	Q9H6A0
DERPC	O00219	DGKH	Q86XP1	DGUOK	Q16854
DHCR24	Q15392	DHDDS	Q86SQ9	DHFR	P00374
DHX35	Q9H5Z1	DHX37	Q8IY37	DKK1	O94907
DLAT	P10515	DMAC1	Q96GE9	DMAC2L	Q99766
DMP1	Q13316	DNAJB6	O75190	DNAJB9	Q9UBS3
DNAJC19	Q96DA6	DNAJC24	Q6P3W2	DNAJC7	Q99615
DNMBP	Q6XZF7	DOT1L	Q8TEK3	DPAGT1	Q9H3H5
DPM1	O60762	DPY30	Q9C005	DPYSL2	Q16555
DR1	Q01658	DSCC1	Q9BVC3	DSPP	Q9NZW4
DST	Q03001	DTNBP1	Q96EV8	DUOX1	Q9NRD8, Q9NRD9
DUSP5	Q16690	DUT	P33316-2	DYNC2LI1	Q8TCX1
DYNLL1	P63167	DYNLT1	P63172	DYRK1A	Q13627
E2F8	A0AVK6	ECM1	Q16610	EDAR	Q9UNE0
EEF1A2	Q05639	EEF1B2	P24534	EFNA5	P52803
EHMT1	Q9H9B1	EHMT2	Q96KQ7	EIF1AX	P47813
EIF2B2	P49770	EIF3B	P55884	EIF3J	O75822
EIF4E	P06730	ELOVL5	Q9NYP7	ELP5	Q8TE02
EMC4	Q5J8M3	ENGASE	Q8NFI3	ENPP4	Q9Y6X5
EP300	Q09472	EPB41L1	Q9H4G0	EPC1	Q9H2F5
EPN1	Q9Y6I3	EPN2	O95208	EPPIN-WFDC6	O95925
ERC1	Q96FL8	ERCC6	Q03468	ERCC8	Q13216
ESR2	Q92731	ETNK1	Q9HBU6	ETV6	P41212
EXOC4	Q96A65	EXOC5	O00471	EXOSC7	Q15024
EXT1	Q16394	EXTL2	Q9UBQ6	EYA3	Q99504
F2	P00734	F2RL1	P25116	F2RL2	O00254
FAM114A2	Q9NRY5	FAM3C	Q92520	FAM98B	Q52LJ0
FAN1	Q9Y2M0	FANCE	Q9HB96	FANCF	Q9NPI8
FAR1	Q8WVX9	FASN	P49327	FBF1	Q8TES7
FBXL19	Q6PCT2	FBXL20	Q96IG2	FBXL5	Q9UKA1
FBXL7	Q9UJT9	FBXO11	Q86XK2	FBXO30	Q8TB52
FBXO32	Q969P5	FBXW11	Q9UKB1	FCF1	Q9Y324
FGF1	P05230	FGF14	Q92915	FGF5	P12034-1
FGFR3	P22607	FIP1L1	Q6UN15	FKBP8	Q14318
FMNL3	Q8IVF7	FMO3	P31513	FNTA	P49354
FNTB	P49356	FOSL1	P15407	FOXO3B	O43524
FPGT	O14772	FPR3	P25089	FRS2	Q8WU20
FZD4	Q9ULV1	FZD7	O75084	GAB1	Q13480
GAB2	Q9UQC2	GABPA	Q06546	GABRR2	P28476
GAK	O14976	GALM	Q96C23	GALNT2	Q10471
GAN	Q9H2C0	GAP43	P17677	GATAD2B	Q8WXI9
GBP3	Q9H0R5	GCNT7	Q6ZNI0	GCSH	P23434

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
GEMIN7	Q9H840	GET1	O00258	GGCT	O75223
GGH	Q92820	GGT7	Q9UJ14	GHDC	Q8N2G8
GHRL	Q9UBU3-1, Q9UBU3-2	GID4	Q8IVV7	GJA9	Q9UKL4
GK3	Q14409	GLA	P06280	GLI2	P10070
GLI3	P10071	GLMN	Q92990	GLO1	Q04760
GLRA1	P23415	GLRA2	P23416	GLRX	P35754
GLRX3	O76003	GLS	O94925	GMNN	O75496
GNA12	Q03113	GNAS	P63092, Q5JWF2	GNAT2	P19087
GNMT	Q14749	GNPDA1	P46926, Q8TDQ7	GOPC	Q9HD26
GORASP2	Q9H8Y8	GP6	Q9HCN6	GPC6	Q9Y625
GPER1	Q99527	GPR18	Q14330	GPR183	P32249
GPRIN1	Q7Z2K8	GPX2	P18283	GREM1	O60565
GREM2	Q9H772	GRIN2B	Q13224	GRIN3A	Q8TCU5
GRK1	Q15835	GRK7	Q8WTQ7	GRM7	Q14831
GSK3B	P49841	GSPT1	P15170, Q8IYD1	GSPT2	Q8IYD1
GSTM1	P09488	GSTM4	Q03013	GTF2B	Q00403
GTF2E1	P29083	GTF2H3	Q13889	GTF2H4	Q92759
H1-1	Q02539	H1-2	P16403	H1-3	P16402
H1-4	P10412	H1-5	P16401	H2AC11	P0C0S8
H2AC12	Q99878	H2AC13	P0C0S8, Q96KK5	H2AC14	Q99878
H2AC15	P0C0S8	H2AC16	P20671	H2AC17	P0C0S8
H2AC18	Q6FI13	H2AC19	Q6FI13	H2AC20	Q16777
H2AC21	Q8IUE6	H2AC4	P04908, Q99878	H2AC6	Q93077
H2AC7	P20671	H2AC8	P04908	H2AZ1	P0C0S5
H2BC10	P62807	H2BC11	P06899	H2BC12	O60814, Q93079, Q99877, Q99880
H2BC12L	P57053	H2BC13	Q99880	H2BC14	Q99879
H2BC15	Q93079, Q99877	H2BC17	P23527	H2BC18	Q99877
H2BC21	P06899, Q16778	H2BC3	P06899, P33778	H2BC4	P62807, Q93079
H2BC5	P58876	H2BC6	P62807, Q93079	H2BC7	P62807
H2BC8	P62807, Q93079	H2BC9	P58876, Q93079, Q99877	H3C1	P68431
H3C11	P68431	H3C12	P68431	H3C13	Q71DI3
H3C14	Q71DI3	H3C15	Q71DI3	H3C2	P68431
H3C3	P68431	H3C4	P68431	H3C6	P68431
H3C7	P68431	H3C8	P68431	H4C1	P62805
H4C13	P62805	H4C16	P62805	H4C2	P62805
H4C3	P62805	H4C4	P62805	H4C5	P62805
H4C6	P62805	H4C8	P62805	H4C9	P62805
HAS2	Q92819	HAUS2	Q9NVX0	HAUS3	Q68CZ6
HBE1	P02100	HCN4	Q9Y3Q4	HDAC2	Q92769
HDAC4	P56524	HDAC8	Q9BY41	HELZ2	Q9BYK8
HEXB	P07686	HGSNAT	Q68CP4	HHIP	Q96QV1
HIF3A	Q9Y2N7	HIGD2A	Q9BW72	HINT1	P49773
HIP1R	O75146	HIPK2	Q9H2X6	HIRA	P54198
HLA-A	P04439	HLA-DPB1	P04440	HLA-DRA	P01903
HMGB1	P09429	HMGCS1	P54868	HMGCS2	P54868
HNRNPA0	Q13151	HNRNPL	P14866	HOXC8	P31273
HP	P00738	HPGDS	O60760	HPR	P77335

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
HPRT1	P00492	HS6ST1	O60243	HSPA1L	P34931
HTD2	P86397	HTR2B	P41595	HTR3A	P46098
HUS1	O60921	IAPP	P10997	IFITM2	P13164
IFITM3	Q01628	IFNGR1	P15260	IFT140	Q96RY7
IFT20	Q8IY31	IFT43	Q96FT9	IFT80	Q9P2H3
IFT81	Q8WYA0	IGF1	P05019	IGFBP3	P17936
IGFBP7	Q16270	IL1B	P01583	ILK	Q13418
IMPA1	P29218	IMPACT	Q9P2X3	IMPDH2	P12268
INHBA	P08476	INO80D	Q53TQ3	INSIG1	O15503
INTS13	Q9NVM9	INTS8	Q75QN2	INTU	Q9ULD6
IPMK	Q8NFU5	IPO8	O15397	IRF2	P14316
ISCA1	Q9BUE6	ISCA2	Q86U28	ISG20	Q96AZ6
ITGA11	Q9UKX5	ITGAE	P38570	ITGB3BP	Q13352
ITGB8	P26012	ITGBL1	O95965	ITM2B	Q9Y287
ITPK1	Q13572	IVD	P26440	KANSL1	Q7Z3B3
KANSL2	Q9H9L4	KANSL3	Q9P2N6	KAT6A	Q92794
KAT6B	Q8WYB5	KBTBD6	Q86V97	KBTBD7	Q8WVZ9
KBTBD8	Q8NFY9	KCNAB2	Q13303	KCND3	Q9UK17
KCNIP4	Q6PIL6	KCNJ9	Q92806	KCNMB1	Q16558
KCNS2	Q9ULS6	KCNV2	Q8TDN2	KCTD13	Q8WZ19
KCTD7	Q96MP8	KDM4A	O75164	KDM5A	P29375
KHDRBS1	Q07666	KHSRP	Q92945	KIAA1549	Q9HCM3
KIF1C	O43896	KIF24	Q5T7B8	KIF26A	Q9ULI4
KIF2C	Q99661	KIF4B	O95239, Q2VIQ3	KIFC1	Q9BW19
KIN	O60870	KLB	Q86Z14	KLC3	Q6P597
KLHDC3	Q9BQ90	KLHL11	Q9NVR0	KLHL12	Q53G59
KLK1	P20151	KLRB1	Q12918	KMT2A	Q03164
KMT2C	Q8NEZ4	KMT2D	O14686	KRT34	P02533
KRT8	P13647	KRTAP1-5	Q07627, Q9BYS1	KRTAP2-3	P0C7H8
KSR2	Q6VAB6, Q8WXI2	KYAT3	Q6YP21	L2HGDH	Q9H9P8
LAMTOR3	Q9UHA4	LAMTOR5	O43504	LARP1	Q6PKG0
LATS1	O95835	LBR	Q14739	LCLAT1	Q6UWP7
LEAP2	Q969E1	LEO1	Q8WVC0	LFNG	Q8NES3
LHX4	Q969G2	LIG1	P18858	LILRA6	Q6PI73
LIMK1	P53667	LIPA	P18054	LMBRD1	Q9NUN5
LMNB1	P20700	LMO2	P25791	LOX	P28300
LOXL3	P58215	LOXL4	Q96JB6	LPAR1	Q92633
LPCAT1	Q8NF37	LRAT	O95237	LRRK1	Q96J92
LRRTM2	O43300	LSM5	Q9Y4Y9	LSR	Q86X29
LSS	P48449	LTBP2	Q14767	LYPLA1	O75608
LYRM7	Q5U5X0	MAD1L1	Q9Y6D9	MAD2L1	Q13257
MAGOH	P61326	MAGT1	Q9H0U3	MAMLD1	Q13495
MAN2B1	O00754	MAP1B	P46821	MAP3K7	O43318
MAPK8	P45983	MARCKS	P29966	MASP2	O00187-1
MAX	P61244	MBD5	Q9P267	MBOAT1	Q6ZNC8
MCL1	Q07820	MDM4	O15151	MEAF6	Q9HAF1
MED13	Q9UHV7	MED18	Q9BUE0	MEF2A	Q02078
MEPE	Q9NQ76	METAP2	P50579	METTL14	Q9HCE5
MFAP4	P55083	MFAP5	Q13361	MFN1	Q8IWA4

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
MGST3	O14880	MICOS10	Q5TGZ0	MICU2	Q8IYU8
MINPP1	Q9UNW1	MIP	P26623	MMP12	P39900
MMP17	Q9ULZ9	MMP2	P08253	MMP3	P08254
MMS19	Q96T76	MPIG6B	O95866	MPL	P40238
MPP2	Q08050	MPST	P25325	MPV17	P39210
MR1	Q8N490-2	MRPL1	Q9BYD6	MRPL18	Q9H0U6
MRPL20	Q9BYC9	MRPL46	Q9H2W6	MRPL48	Q96GC5
MRPL50	Q8N5N7	MRPL51	Q4U2R6	MRPL58	Q14197
MRPL9	Q9BYD2	MRPS14	O60783	MRPS15	P82914
MRPS18C	Q9Y3D5	MRPS21	P82921	MRPS23	Q9Y3D9
MRPS24	Q96EL2	MRPS33	Q9Y291	MRPS35	P82673, Q9Y2Q9
MRPS9	P82933	MT-ATP8	P03928	MT-ND3	P03897
MT1A	P53803	MT2A	P02795	MTA1	Q13330
MTA2	O94776	MTERF3	Q96E29	MTF1	Q9H5Q4
MTHFS	P49914	MTMR3	Q96QG7	MTPAP	Q9NVV4
MUC19	Q7Z5P9	MUC6	Q6W4X9	MVB12B	Q9H7P6
MX2	P20591, P20592	MYD88	Q99836	MYH11	P35749
MYL11	Q96A32	MYL12A	P19105	MYL2	P10916
MYL4	P12829	MYL6	P60660	MYL7	Q01449
MYO5A	Q9Y4I1	MZT1	Q08AG7	NADK	O95544
NAE1	Q13564	NAMPT	P43490	NBN	O60934
NCAN	O14594	NCAPD2	Q15021	NCAPG2	Q86XI2
NCF2	P19878	NCK1	O43639, P16333	NCOA1	Q15788
NCOA2	Q15596	NCOA3	Q9Y6Q9	NCOA6	Q14686
NDE1	Q9NXR1	NDST2	P52849	NDUFA1	O15239
NDUFA12	Q9UI09	NDUFA2	O43678	NDUFA5	Q16718
NDUFA8	P51970	NDUFAF3	Q9BU61	NDUFB2	O95178
NDUFB9	Q9Y6M9	NDUFS3	O75489	NDUFV3	P56181
NEDD1	Q8NHV4	NEDD8	Q15843	NEK1	Q96PY6
NEK3	Q9UHY1	NEK9	Q8TD19	NEO1	O43861
NES	Q6P1A2	NF1	P21359	NFATC3	Q12968
NFIB	O00712	NFIC	P08651	NFIX	Q14938
NFU1	Q9UMS0	NFYC	Q13952	NGF	P01138
NGFR	P08138	NIPAL4	Q0D2K0	NKIRAS1	Q9NYS0
NKIRAS2	Q9NYR9	NOC2L	Q9Y3T9	NOL4L	Q86VX2
NOL9	Q5SY16	NOP14	P78316	NOSTRIN	Q8IVI9
NOTCH2NLB	P0DPK3	NPAS4	Q8IUM7	NPFF	O15130
NPIPB3	Q92617	NPTN	Q9Y639	NR1D1	P20393
NR2C2	P49116	NRP1	O14786	NSUN4	Q96CB9
NT5C3A	Q9H0P0-4	NT5C3B	Q969T7	NTM	Q9P121
NUAK1	O60285	NUBP1	P53384	NUDT11	Q96G61
NUDT12	Q9BQG2	NUDT15	Q9NV35	NUDT21	O43809
NUP133	Q8WUM0	NUP214	P35658, Q96HA1	NUP35	Q8NFH5
NUP54	Q7Z3B4	NUS1	Q96E22	OAT	P04181
OBI1	Q5W0B1	ODC1	P11926	OGN	P20774
OMD	Q99983	OMG	P23515	OPHN1	O60890
OR10A2	Q9H208	OR10A4	Q9H209	OR10A5	Q9H207
OR10AD1	Q8NGE0	OR13G1	Q8NGZ3	OR13H1	Q8NG92
OR14A2	Q96R54	OR1C1	Q15619	OR2A1	Q8NGT9

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
OR2A42	Q8NGT9	OR2D2	Q9H210	OR2D3	Q8NGH3
OR6A2	O95222	OR6X1	Q8NH79	ORC2	Q13416
ORC4	O43929	ORC5	O43913	ORMDL1	Q9P0S3
ORMDL2	Q53FV1	OSR1	Q8TAX0	OST4	P0C6T2
OSTC	Q9NRP0	OSTM1	Q86WC4	OVCH1	Q7RTY9
P2RY12	Q9H244	P2RY13	Q9BPV8	P2RY14	Q15391
PAAF1	Q9BRP4	PADI3	Q9ULW8	PAIP1	Q9H074
PALS1	Q8N3R9	PAM16	Q9Y3D7	PAPOLB	P51003
PAPPA	Q13219	PARP14	Q460N5	PARP2	Q9UGN5
PARP6	Q2NL67	PATL1	Q86TB9	PCBD1	P61457
PCDH7	O60245	PCLAF	Q15004	PCNA	P12004
PCOLCE2	Q9UKZ9	PDCD1LG2	Q9BQ51	PDCL	Q13371
PDE6D	O43924	PDHB	P11177	PDZD11	Q5EBL8
PELI2	Q9HAT8	PELP1	Q8IZL8	PER2	O15055
PERP	Q96FX8	PES1	O00541	PET100	P0DJ07
PET117	Q6UWS5	PEX13	Q92968	PEX26	Q7Z412
PEX3	P56589	PFAS	O15067	PFDN4	Q9NQP4
PFDN5	Q99471	PFDN6	O15212	PFKFB3	Q16875
PFKFB4	Q16877	PFN2	P07737, P35080	PGA4	P0DJD7
PHB1	P35232	PHC3	Q8NDX5	PHF21A	Q96BD5
PHF5A	Q7RTV0	PIGK	Q92643	PIGN	O95427
PIGU	Q9H490	PIGY	Q3MUY2	PIK3R2	O00459
PILRA	Q9UKJ1	PIP5K1A	Q99755	PITX2	Q99697
PKN1	Q16512	PKP3	Q9Y446	PKP4	Q99569
PLA2G4A	P47712	PLCG1	P19174	PLD2	O14939
PLEKHA3	Q9HB20	PLEKHA4	Q9H4M7	PLEKHG2	Q9H7P9
PLET1	Q6UQ28	PLN	P26678	PLOD2	O00469
PLPP4	Q5VZY2	PLPP5	Q8NEB5	PLPP6	Q8IY26
PLPPR3	Q6T4P5	PMCH	P20382	PMS1	P54278
PMS2	P54278	PNPLA6	Q8IY17	PNPT1	Q8TCS8
PNRC2	Q9NPJ4	PODXL2	Q9NZ53	POLD1	P28340
POLD3	Q15054	POLE2	P56282	POLE4	Q9NR33
POLG	P54098	POLR1C	O15160	POLR1F	Q3B726
POLR1H	Q9P1U0	POLR2F	P61218	POLR2H	P52434
POLR2K	P53803	POLR3K	Q9Y2Y1	POM121	Q96HA1
POM121C	A8CG34, Q96HA1	POMK	Q9H5K3	POP5	Q969H6
POR	P16435	POTEE	Q6S8J3	PPA2	Q15181, Q9H2U2
PPIA	P62937	PPM1F	P49593	PPME1	Q9Y570
PPP1CB	P62140	PPP1R10	Q96QC0	PPP1R12B	O60237
PPP2R2D	Q66LE6	PPP2R5E	Q16537	PPP3CC	P16298
PPP3R1	P63098	PPP4R2	Q9NY27	PPRC1	Q5VV67
PRCC	Q92733	PRDM1	O75626	PRDM4	Q9UKN5
PRG4	Q96GM1	PRKAR2A	P13861	PRKRA	O75569
PRLR	P16471	PRMT6	Q96LA8	PRPF19	Q9UMS4
PRPF3	O43395	PRR4	Q96NY8	PRX	Q9BXM0, Q9BXM0-1
PSG1	P11464	PSG5	Q15238	PSMA7	O14818
PSMB4	P28070	PSMB5	P28074	PSMB7	Q99436
PSMC1	P62191	PSMC6	P62333	PSMG4	Q5JS54
PTGES3	Q15185	PTK7	P42679	PTPN1	P18031

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
PTPN14	Q15678	PTPRA	P18433	PXYLP1	Q8TE99
PYGM	P11217	PYGO1	Q9Y3Y4	PYURF	Q96I23
QPCT	Q16769	RAB18	Q9NP72	RAB22A	Q9UL26
RAB3D	O95716	RAB41	Q5JT25, Q86YS6	RAB4A	P20338
RABGAP1	Q9Y3P9	RABGGTB	P53611	RAC3	P60763
RAD23B	P54727	RAD52	P43351	RAD9B	Q6WBX8
RALA	P11233	RANBP9	Q96S59	RAP1B	P61224
RAPGEF6	Q9Y4G8	RARS2	Q5T160	RASAL2	Q9UJF2
RBFOX2	O43251	RBL1	P28749	RBM17	Q96I25
RBM28	Q9NW13	RBM4B	P52272	RBM7	Q9Y580
RBP2	P29375	RBX1	P62877	RCC1	P18754
RCL1	Q9Y2P8	RCN1	Q15293	REST	Q13127
RFFL	O75771	RGS20	O76081	RHOQ	P17081
RIOK1	Q9BRS2	RMI1	Q9H9A7	RNF138	Q8WVD3
RNF146	Q9NTX7	RNF214	Q86Y13	RNF8	O76064
RNLS	Q5VYX0	RNPC3	Q96LT9	RPA1	O95602
RPA4	Q13156	RPE	Q2QD12, Q96AT9	RPIA	P49247
RPL22	P35268	RPL22L1	Q6P5R6	RPL26L1	Q9UNX3
RPL27	P61353	RPL30	P62888	RPL31	P62899
RPL36AL	Q969Q0	RPL37	P61927	RPL38	P63173
RPL41	P62945	RPL5	P46777	RPL9	P32969
RPLP2	P05387	RPP25	Q9BUL9	RPP38	P78345
RPRD2	Q5VT52	RPS10	P46783	RPS11	P62280
RPS12	P25398	RPS13	P62277	RPS15A	P62244
RPS23	P62266	RPS25	P62851	RPS27	P42677, Q71UM5
RPS27A	P62979, P62987	RPS4X	P62701, Q8TD47	RRAGA	Q5VZM2, Q7L523
RRAGB	Q5VZM2	RRBP1	Q9P2E9	RRN3	Q9NYV6
RRP8	O43159	RUNX1	Q01196	RUNX2	Q13950-1, Q13950-2
RYBP	Q8N488	RYR2	Q92736	RYR3	Q15413
S100A11	P31949	S1PR2	O95136	SAP130	Q9H0E3
SAT2	Q96QD8	SBF1	O95248	SC5D	O75845
SCAMP1	O15126	SCAP	Q12770	SCCPDH	Q8NBX0
SCFD1	Q8WVM8	SCLT1	Q96NL6	SCLY	Q96I15
SCMH1	Q96GD3-2	SCN8A	Q9UQD0	SCO1	O75880
SCOC	Q9UIL1	SCRIB	Q14160	SCUBE3	Q8IX30
SDHAF4	Q5VUM1	SEC11C	Q9BY50	SEC24B	O95487
SEC61A2	Q9H9S3	SEL1L	Q9UBV2	SEM1	P60896
SEMG1	P04279	SENP3	Q9H4L4	SENP5	Q96HI0
SEPTIN7	Q16181	SERP1	Q9Y6X1	SERPINB2	P05120
SERPINB8	P50452	SERPIND1	P05546	SETD2	Q9BYW2
SETD6	Q8TBK2	SF3A2	Q15428	SGCD	Q92629
SGPL1	O95470	SHPRH	Q149N8	SHQ1	Q6PI26
SIGMAR1	Q99720	SIN3A	Q96ST3	SIPA1L1	O43166
SKP1	P63208	SLC10A1	Q14973	SLC12A6	Q9UHW9
SLC16A1	P53985	SLC1A1	P43005	SLC1A4	P43007
SLC22A1	O15245	SLC22A3	O75751	SLC25A14	O95258
SLC25A15	Q9Y619	SLC25A2	Q9BXI2	SLC25A29	Q8N8R3
SLC25A44	Q96H78	SLC26A2	P50443	SLC27A5	Q9Y2P5
SLC29A3	Q9BZD2	SLC2A2	P11168	SLC31A1	O15431

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
SLC33A1	O00400	SLC35A3	Q9Y2D2	SLC35D1	Q9NTN3
SLC38A9	Q8NBW4	SLC39A1	Q9NP94, Q9NY26	SLC39A5	Q6ZMH5
SLC39A7	Q92504	SLC41A1	Q8IVJ1	SLC45A3	Q96JT2
SLC4A5	Q9BY07	SLC4A7	Q9Y6M7	SLCO2A1	Q92959
SMAD2	Q15796	SMAD3	P84022	SMARCD2	Q92925
SMDT1	Q9H4I9	SMG7	Q92540	SNAPC3	Q92966
SNRNP25	Q9BV90	SNRNP27	Q8WVK2	SNRNP70	P08621
SNTG2	Q9NY99	SNX12	Q8WY07	SNX3	O60493
SOCS4	Q8WXH5	SOD2	P04179	SOX4	Q06945
SPCS3	P61009	SPINK6	Q6UWN8	SPN	P16150
SPON1	Q9HCB6	SPP1	P49642	SPPL2A	Q8TCT8
SPRED3	Q2MJR0	SPTLC1	O15269	SPTSSA	Q969W0
SREBF1	P36956	SREBF2	Q12772	SRF	P11831
SRGAP1	Q7Z6B7	SRI	P30626	SRP14	P37108
SRPRB	Q9Y5M8	SRRM2	Q9UQ35	SST	P61278
SSTR1	P30872	SSU72	Q9NP77	ST6GALNAC2	Q9UJ37
ST7	Q9Y561	STEEP1	Q9H5V9	STK24	Q9Y6E0
STMN1	P16949	STMN2	Q93045	STRC	Q7RTU9
STT3B	Q8TCJ2	STX18	Q9P2W9	STXBP2	Q15833
SUN1	O94901	SUPT7L	O94864	SUZ12	Q15022
SV2A	Q7L0J3	SYN3	O14994	SYT10	Q6XYQ8
TACO1	Q9BSH4	TAF13	Q15543	TAF1C	Q15572
TARS2	Q9BW92	TAS2R19	P59542	TAS2R20	P59543
TAS2R3	Q9NYW6	TAS2R30	P59541	TAS2R31	P59538
TAS2R4	Q9NYW5	TAS2R40	P59535, P59541	TAS2R43	P59537
TAS2R46	P59540	TAS2R5	Q9NYW4	TAS2R50	P59544
TAS2R60	P59551	TBC1D16	Q8TBP0	TBCA	O75347
TBCD	Q9BTW9	TBL1X	O60907	TBP	P20226
TCEA1	P23193	TCF7L2	Q9NQB0	TDP2	O95551
TEAD4	Q15561, Q15562	TENT4A	Q5XG87	TET2	Q6N021
TET3	O43151	TEX2	Q8IWB9	TFAM	P09429
TFAP2E	Q6VUC0	TFB1M	Q8WVM0	TFG	Q92734
TGIF1	Q15583	THBS1	P07996	THBS3	P49746
THSD1	Q9NS62	THUMPD1	Q9NXG2	TICRR	Q7Z2Z1
TIMM23	O14925	TIMM23B	O14925	TIMM44	O43615
TIMM8B	Q9Y5J9	TIMP3	P35625	TIRAP	P58753
TJP1	Q07157	TLE4	Q04727	TMC6	Q7Z403
TMED7	Q9Y3B3	TMEM30A	Q9NV96	TMEM67	Q5HYA8
TMPO	P42167	TMX3	Q96JJ7	TNFAIP8	O95379
TNFAIP8L1	Q8WVP5	TNFAIP8L3	Q5GJ75	TNFSF4	P23510
TNKS1BP1	Q9C0C2	TNRC6B	Q9UPQ9	TOMM22	Q9NS69
TOMM40	O96008	TOMM7	Q9P0U1	TOP1	P11387
TOR1A	O14656	TPH2	Q8IWU9	TPM1	P09493
TPRKB	Q9Y3C4	TRAF3IP1	Q8TDR0	TRAF7	Q6Q0C0
TRAM1	Q15629	TRAPPC6B	O75865, Q86SZ2	TRIM28	Q13263
TRIM32	Q13049	TRIM39	Q9HCM9	TRIM45	Q9H8W5
TRIM6	Q9C030	TRMT10C	Q7L0Y3	TRMT13	Q9NUP7
TRNT1	Q96Q11	TSEN2	Q8NCE0	TSG101	Q99816
TSNAX	Q99598	TTLL5	Q6EMB2	TTN	Q8WZ42

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
TUBA1B	P68363	TUBA4B	Q9H853	TUBB1	Q9H4B7
TULP4	Q9NRJ4	TUSC3	Q13454	TWF1	Q12792
TWIST1	Q15672	U2AF2	P26368	UBAC1	Q9BSL1
UBE2D1	P51668	UBE2D4	Q9Y2X8	UBE2E3	Q969T4
UBE2H	P62256	UBE2N	P61088	UBE2R2	Q712K3
UBE2V2	Q15819	UBE2W	Q96B02	UBL5	Q9BZL1
UCHL3	P15374	UCK2	Q9BZX2	UCP2	P55851, P55916
UGDH	O60701	UGGT2	Q9NYU1	ULBP1	Q9BZM5
ULBP2	Q9BZM5	UNC79	Q9P2D8	UPF1	Q92900
UQCC1	Q9NVA1	UQCC5	Q8WVI0	UQCC6	Q69YU5
UQCR10	Q9UDW1	UQCRHL	A0A096LP55, P07919	UQCRQ	O14949
USE1	Q9NZ43	USP1	O94782	USP13	Q92995
USP19	O94966	USP24	Q9UPU5	USP31	Q86UV5
UTP11	Q9Y3A2	UTP18	Q9Y5J1	UTP3	Q9NQZ2
UTRN	P46939	UVRAG	Q9P2Y5	VARS1	P26640
VARS2	P26640	VASP	Q2MJR0	VAV2	P52735
VBP1	P61758	VDAC2	P45880	VEGFC	P49767
VHL	P40337	VIT	O75030	VPS26A	O75436
VPS39	Q96JC1	VPS52	Q8N1B4	VPS53	Q5VIR6
WARS2	Q9UGM6	WBP4	O75554	WDR45B	Q5MNZ6
WDR70	Q9NW82	WNK1	Q9H4A3	WNT2B	Q93097
WNT3	P56703	WNT4	O14905, P56705	WTAP	Q15007
XBP1	P17861-2	XDH	Q06278	XPO1	O14980
XRCC1	P18887	XRCC3	O43542	XYLB	O75191
YAP1	P46937	YEATS4	O95619	ZCCHC7	Q8N3Z6
ZCRB1	Q8TBF4	ZDHHC20	Q5W0Z9	ZDHHC9	Q9Y397
ZEB1	P37275	ZFAND6	Q6FIF0	ZKSCAN3	Q9BRR0
ZNF133	P52736	ZNF135	P52742, Q6ZN57	ZNF20	P17024
ZNF226	Q9NZL3	ZNF264	O43296	ZNF268	Q96BR6
ZNF302	Q9NR11	ZNF320	A2RRD8	ZNF337	Q9Y3M9
ZNF350	Q9GZX5	ZNF397	Q96MU6	ZNF460	Q14592
ZNF490	Q9ULM2	ZNF496	Q96IT1	ZNF521	Q96K83
ZNF543	Q08ER8	ZNF557	Q8N988, Q96NG5	ZNF559	Q9BR84
ZNF585A	Q6P3V2	ZNF585B	Q52M93	ZNF586	Q9NXT0
ZNF597	Q96LX8	ZNF625	Q96I27	ZNF7	Q96MU6
ZNF70	Q9UC06	ZNF701	Q9NV72	ZNF702P	Q9H963
ZNF724	A8MTY0	ZNF74	Q16587	ZNF740	Q8NDX6
ZNF750	Q32MQ0	ZNF770	Q6IQ21	ZNF786	Q8N393
ZNF8	Q86UQ0	ZNF805	O43296	ZNF816	Q0VGE8
ZNF875	P10072	ZSCAN22	Q68DY9	ZW10	O43264

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
AAMP	ENSG00000127837	ACTA2	ENST00000224784	AIFM2	ENSG00000042286
ARID3A	ENSG00000116017	ASAH1	ENSG00000104763	ASNS	ENSG00000070669
ATF5	ENSG00000169136, ENST00000423777	ATP6AP2	ENSG00000182220	ATP6V1G1	ENSG00000136888
BAX	ENSG00000087088	BID	ENSG00000015475	BST2	ENSG00000130303
CBX5	ENSG00000094916	CCNK	ENSG00000090061	CD44	ENST00000428726
CDK1	ENSG00000170312	CDT1	ENSG00000167513	CER1	ENST00000380911

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
CIITA	ENSG00000179583	CLOCK	ENSG00000134852	COX7A1	ENSG00000161281
CPT2	ENSG00000157184	CSF1R	ENSG00000182578	DDIT3	ENSG00000175197, ENST00000346473
DHFR	ENSG00000228716	DKK1	ENSG00000107984	DNAJB6	ENSG00000105993
DNAJB9	ENSG00000128590	ELOVL5	ENSG00000012660	EXTL2	ENSG00000162694
FAN1	ENSG00000075618	FASN	ENSG00000169710	FBXO32	ENSG00000156804
GABPA	ENSG00000154727	GBP3	ENSG00000117226	GHRL	ENSG00000157017
GLI2	ENSG00000074047	GPRIN1	ENSG00000169258	GREM1	ENSG00000166923
GRIN2B	ENSG00000273079	HBE1	ENSG00000213931	HHIP	ENSG00000164161
HLA-A	ENSG00000206503	HLA-DPB1	ENSG00000223865	HLA-DRA	ENSG00000204287
HMGCS1	ENSG00000112972	HMGCS2	ENSG00000134240	HSPA1L	ENSG00000206383, ENSG00000226704, ENSG00000234258, ENSG00000236251
IAPP	ENSG00000121351	IFITM2	ENSG00000185201	IFITM3	ENSG00000142089
IGFBP3	ENSG00000146674	IGFBP7	ENSG00000163453	IL1B	ENSG00000125538, ENST00000263341
IRF2	ENSG00000168310	ISG20	ENSG00000172183	ITGBL1	ENSG00000198542
KDM5A	ENST00000399788	KLHDC3	ENSG00000124702	KLK1	ENST00000301420
KRT8	ENST00000293308	LFNG	ENSG00000106003	LIG1	ENSG00000105486
LMNB1	ENSG00000113368	LSS	ENSG00000160285	MCL1	ENSG00000143384
MED13	ENST00000397786	MIR21	ENSG00000284190	MMP2	ENSG00000087245
MMP3	ENSG00000149968	MRPL18	ENSG00000112110	MT-ATP8	ENST00000361851
MT-ND3	ENST00000361227	MT-TS1	ENST00000387416	MT2A	ENSG00000125148
MX2	ENSG00000183486	MYO5A	ENSG00000197535	NAMPT	ENSG00000105835
NOTCH2NLB	ENSG00000286019, ENST00000593495	NOTCH2NLR	ENSG00000286106, ENST00000624419	NPAS4	ENST00000311034
NR1D1	ENSG00000126368	OR10A2	ENSG00000170790	OR10A4	ENSG00000170782
OR10A5	ENSG00000166363	OR10AD1	ENSG00000172640	OR13G1	ENSG00000197437
OR13H1	ENSG00000171054	OR14A2	ENSG00000241128	OR1C1	ENSG00000221888
OR2A1	ENSG00000212807, ENSG00000221970	OR2A42	ENSG00000212807, ENSG00000221970	OR2D2	ENSG00000166368
OR2D3	ENSG00000178358	OR6A2	ENSG00000184933	OR6X1	ENSG00000221931
OSR1	ENSG00000143867	PCNA	ENSG00000132646	PER2	ENSG00000132326
PERP	ENSG00000112378	PITX2	ENSG00000164093	PMS2	ENSG00000122512
PPIA	ENSG00000196262	PRDM1	ENSG00000057657	PRX	ENSG00000105227
PTPN1	ENSG00000196396	PTPN14	ENSG00000152104	RALA	ENSG00000006451
RAP1B	ENSG00000127314	RBL1	ENSG00000080839	RNU11	ENSG00000274978, ENST00000387069
RUNX1	ENSG00000159216, ENST00000344691	RUNX2	ENSG00000124813	SC5D	ENSG00000109929
SERP1	ENSG00000120742	SERPINB2	ENSG00000197632	SLC2A2	ENSG00000163581
SOCS4	ENSG00000180008	SOD2	ENSG00000291237	SPP1	ENSG00000118785
SREBF1	ENSG00000072310	SRF	ENSG00000112658	SRPRB	ENSG00000144867
SST	ENSG00000157005	STMN1	ENSG00000117632	STT3B	ENSG00000163527
SUZ12	ENSG00000178691	SYT10	ENSG00000110975	TET2	ENST00000265149
TFAM	ENSG00000108064	TFB1M	ENSG00000029639	THBS1	ENSG00000137801
TJP1	ENSG00000104067	TRIM45	ENSG00000134253	TRIM6	ENSG00000121236
TWIST1	ENSG00000122691	WNT4	ENSG00000162552	XBP1	ENSG00000100219
YAP1	ENST00000282441	ZEB1	ENSG00000148516	ZNF521	ENSG00000198795

Input	miRBase Id	Input	miRBase Id	Input	miRBase Id
MIR17	MI0000071	MIR18A	MI0000072	MIR19A	MI0000073
MIR19B1	MI0000074	MIR20A	MI0000076	MIR21	MI0000077
MIR32	MI0000090				

Interactors (1628)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
A1CF	Q9NQ94	Q13077	A2M	P01023	Q8NBL1
AAGAB	Q6PD74	O00186, Q12846	AAK1	Q2M2I8	Q9NQ11
AAMDC	Q9H7C9	P54252	AAMP	Q13685	Q9BYF1
AASDHPPT	Q9NRN7	Q9UQK1	AATF	Q9NY61	O96017
ABCD3	P28288	P33897	ABL2	P42684	P19174
ACAA2	P42765	Q9BQP7	ACAD10	Q6JQN1	O95861
ACBD7	Q8N6N7	Q9NWQ8	ACRV1	P26436	P50897
ACSL3	O95573	P05919	ACSL5	Q9ULC5	Q14973
ACTA2	P62736	Q15392	ACTC1	P68032	P23528
ACTR6	Q9GZN1	Q15274	ADAM12	O43184-2	O95633
ADAT2	Q7Z6V5	Q96GM5	ADORA2A	P29274	P14416
ADSS2	P30520	Q8NB12	AFF1	P51825	P04608
AFG2A	Q8NB90	O14980	AFG3L2	Q9Y4W6	P42858
AGO1	Q9UL18	O15397, Q8NDV7	AGO2	Q9UKV8	O15397, Q8NDV7
AGO3	Q9H9G7	O15397, Q8NDV7	AGPAT5	Q9NUQ2	Q96BA8
AGR2	O95994	Q9UBH0	AGRN	O00468	O15530
AIPL1	Q9NZN9	O00267	AK6	Q9UIJ7	P02649
AKAP17A	Q02040	O75152	AKAP7	Q9P0M2	P17612
ALDH16A1	Q8IZ83	Q9Y315	ALG6	Q9Y672	Q9BVK2
ALMS1	Q8TCU4	O75923	ALYREF	Q86V81	Q92900
AMD1	P17707	P18509	ANAPC4	Q9UJX5	P51955
ANGPT2	O15123, O15123-1	Q02763	ANGPTL7	O43827	P17987
ANKRD13C	Q8N6S4	P56539	ANKRD13C-DT	Q9XRX5, Q9XRX5-2	Q13287
ANKRD17	O75179	P55075	ANKRD36	A1A5B0	Q08379
ANKRD36B	Q8N2N9-4	Q99757	ANKRD50	Q9ULJ7	Q9UJV9
AP1S1	P61966-2	P37173	AP1S2	P56377	P56539
AP2A1	O95782	P29353	AP3D1	O14617	Q96EV8
APEX1	P27695	Q09472	APH1A	Q96BI3	O43561-2
APH1B	Q8WW43	P05067	APIP	Q96GX9	Q96GX9
APLNR	P35414	O43913	APOBEC4	Q8WW27	P55212
APTX	Q7Z2E3	P04637	AQP10	Q96PS8	P15151
ARAP1	Q96P48	P25090	ARF5	P84085	P46059
ARFIP1	P53367	O60906	ARGLU1	Q9NWB6	P52298
ARHGAP21	Q5T5U3	P29692	ARHGAP35	Q9NRY4	P05120
ARHGAP45	Q92619	P63000	ARHGEF11	O15085	P61586
ARHGEF9	O43307	Q9NW38	ARID2	Q68CP9	P52294
ARID3A	Q99856	P23025	ARL15	Q9NXU5	O43924
ARL3	P36405	P36404	ARMC1	Q9NVT9	Q16143
ARPC2	O15144	Q16236	ARR3	P36575	P32121, P49407
ARSB	P15848	P01185	ARSG	Q96EG1	Q8NBK3
ART4	Q9ULX3	P51530	ASAH1	Q13510, Q13510-2	Q13285
ASAP1	Q9ULH1	P19174	ASAP3	Q8TDY4	Q86UT5
ASB7	Q9H672	Q93034, Q9UBF6	ASH1L	Q9NR48	P05067

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
ASMT	P46597	P46597	ASNSD1	Q9NWL6	Q14160
ATAT1	Q5SQI0	P61981	ATF5	Q9Y2D1	P49715
ATF7	P17544	P18847	ATG12	O94817	Q9H1Y0, Q9H0Y0, Q9NT62, Q676U5
ATG4C	Q96DT6	O00401	ATOX1	O00244	P35670
ATP2A3	Q93084	P0DTC5	ATP2B1	P20020	Q14160
ATP5ME	P56385	Q9H1C4	ATP5MF	P56134	P01106
ATP5PF	P18859	Q9H2H9	ATP6AP2	O75787	Q15904
ATP6V0A2	Q9Y487	Q15904	ATP6V1G1	O75348	P0C6X7
ATPAF1	Q5TC12	P06576	ATXN2	Q99700, Q99700-5	P11940
ATXN2L	Q8WWM7	P11940	ATXN7	O15265	Q8N2S1
AUTS2	Q8WXX7	Q09472	B3GALT2	O43825	O94822
B3GALT5	Q9Y2C3	P81534	BACH2	Q9BYV9	O60675
BAG4	O95429	O75496	BAX	Q07812	Q9ULZ3
BAZ2B	Q9UIF8	Q96GX9	BBOX1	O75936	Q9UHD2
BBS10	Q8TAM1	Q8IWZ6	BCAS3	Q9H6U6	P08670
BCL2L12	Q9HB09	Q07812	BCL7A	Q4VC05	Q9H8M2
BCR	P11274	Q96CW1	BET1	O15155	P29274
BET1L	Q9NYM9	Q9P2W9	BEX2	Q9BXY8	P19883
BICD1	Q96G01	P49841	BICD2	Q8TD16	P53350
BICRAL	Q6AI39	Q96GM5	BID	P70444, P55957, P70444-PRO_0000223236	Q07812
BIRC2	Q13490	Q13546, Q9Y572	BLCAP	P62952	O95237
BLOC1S2	Q6QNY1	A1L4H1	BLTP3A	Q6BDS2	Q9Y3M2
BMP2K	Q9NSY1	O95630, Q96FJ0	BNIP2	Q12982	Q9NRZ7
BNIP3	Q12983	O60313	BOLA3	Q53S33	Q86SX6
BPGM	P07738	Q9BXK1	BPNT1	O95861	Q8NE62
BRAF	P15056	P42772	BRAP	Q7Z569	P38398
BRD4	O60885	P0DTC4	BRIX1	Q8TDN6	O43159
BROX	Q5VW32	P23193	BRPF3	Q9ULD4-2	P07550
BST2	Q10589	Q10589	BTBD1	Q9H0C5	P61081
BTBD18	B2RXH4	Q9UBU9	BTBD2	Q9BX70	P40692
BTBD9	Q96Q07-2	P55212	BTD	P43251	P01008
BTN2A1	Q7KYR7	P22059	C1D	Q13901	Q99547, Q9Y2L1, P42285, Q01780, Q9NQT4, Q9H8H0
C1GALT1C1	Q96EU7	Q6ZMH5	C1orf54	Q8WWF1	Q14703
C22orf39	Q6P5X5	Q96RU7	C3orf38	Q5JPI3	P11142
C3orf70	A6NLC5	P31749	C6orf120	Q7Z4R8	Q9Y2A9
C8orf33	Q9H7E9	P78563	CA4	P22748	Q9Y6R1
CA5B	Q9Y2D0	P11182	CADM4	Q8NFZ8	Q9UMR7
CAMLG	P49069	Q9Y624	CAP2	P40123	P23528
CAPNS1	P04632	P20936	CAPZA2	P47755	Q14247
CARD10	Q9BWT7	O14908	CARNMT1	Q8N4J0	Q13547
CASP2	P42575	O95429	CASP8	Q14790	Q13546, Q9Y572
CASQ2	O14958	P62736	CBL	P22681	P08581
CBLB	Q13191	P46109	CBX3	Q13185	Q96KQ7, Q13185
CBX5	P45973	Q96KQ7, Q13185	CCAR2	Q8N163	P48431
CCDC12	Q8WUD4	O75152	CCDC124	Q96CT7	P06748
CCDC134	Q9H6E4	O75478	CCDC141	Q6ZP82-1	Q8N2W9

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
CCDC196	A0A1B0GWI1	Q9UBU9	CCDC32	Q9BV29	Q9NUX5
CCDC43	Q96MW1, Q86WV7	Q96T60	CCDC59	Q9P031	P98164
CCDC6	Q16204	Q13164	CCDC71	Q8IV32	Q13547
CCHCR1	Q8TD31-3	P21673	CCND2	P30279	P11802
CCNK	O75909-2	O15160	CCZ1	P86791	P51149
CD164	Q04900	P61073	CD44	P16070	P10451
CD55	P08174	P48960	CD82	P27701	P01033
CD9	P21926	P48509	CDC37L1	Q7L3B6	P07900
CDC40	O60508	P62306, P14678, P62304	CDC42BPA	Q5VT25	P60953
CDC42SE1	Q9NRR8	P60953	CDCA3	Q99618	Q9BY43
CDK1	P06493	Q99741	CDK11A	Q9UQ88-1	O96017
CDK12	Q9NYV4	P50750	CDK15	Q96Q40	P07900
CDKL5	O76039	P40692	CDKN2AIPNL	Q96HQ2	Q9H0D6
CDT1	Q9H211	Q99741	CENATAC	Q86UT8	Q14691
CENPH	Q9H3R5	P20700	CENPP	Q6IPU0	P49247
CENPQ	Q7L2Z9	P15336	CENPU	Q71F23-1, Q71F23	P53350
CEP112	Q8N8E3	P61981	CEP70	Q8NHQ1	P43897
CEP85	Q6P2H3-3	O00560	CEP89	Q96ST8	P61981
CEP97	Q8IW35	O43303	CER1	O95813	P35813
CERS5	Q8N5B7	Q9H244	CERT1	Q9Y5P4	Q13352
CETN3	O15182	Q01831	CFAP298	P57076	O00628
CFB	P00751	P61916	CGGBP1	Q9UFW8	O15160
CHAF1A	Q13111	Q13185	CHAMP1	Q96JM3	Q13185
CHGA	P26339	P00441	CHMP2B	Q9UQN3	Q9Y3E7
CHORDC1	Q9D1P4, Q9UHD1	P07900	CHRNG	P07510-2	O43597
CHTF8	P0CG13	P40937, Q9BVC3, P35249	CHTOP	Q9Y3Y2	Q14739
CIAO3	Q9H6Q4	P27540	CIART	Q8N365	P55212
CIITA	P33076	Q96EB6	CILP2	Q8IUL8	Q01523, P59665
CLASP1	Q7Z460	P35613	CLDN18	P56856	Q01453
CLDN20	P56880	P24593	CLDN22	Q8N7P3	P25025
CLDND2	Q8NHS1	Q9H2K0	CLEC4A	Q9UMR7	P22455
CLIP4	Q8N3C7	Q9NPC8	CLOCK	O08785	Q03164
CMIP	Q8IY22	P27986	CMKLR1	Q99788	Q9BUV8
CMPK1	P30085	Q6DKK2	CMTM6	Q9NX76	Q9NZQ7
CMTR1	Q8N1G2	O43143	CMYA5	Q8N3K9	O75923
CNIH1	O95406	Q6IN84	CNOT10	Q9H9A5	P47756
CNOT4	O95628-2	Q9UBH0	CNOT7	Q60809	P06730
CNTF	P26441	P42702	COBLL1	Q53SF7	P61981
COG7	P83436	Q9NZQ7	COL10A1	Q03692	Q8N9N5
COL12A1	Q99715	P08253	COL14A1	Q05707	P32119
COL18A1	P39060	P08253	COL3A1	P02461-1	P09486
COL8A1	P27658	O43559	COMMD10	Q9Y6G5	P31930
COMMD2	Q86X83	Q96PC5	COMMD8	Q9NX08	P0CG47
COPS6	Q7L5N1	P61081	COPS7B	Q9H9Q2	P61081
COPS8	Q99627	Q15831	COPS9	Q8WXC6	Q92905
CORO1A	P31146	P26640	COX6B1	P14854	P00403
COX6C	P09669	P00403	COX7C	P15954	P00403
CPT2	P23786	Q9HC36	CRCP	O75575	O15160, P19388, P61218, P62875

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
CREB3L2	Q70SY1	Q96BA8	CREBBP	Q92793	P29279
CREBL2	O60519	P35638	CREBZF	Q9NS37	O14867
CRK	P46108-1, P46108	Q13905	CRMP1	Q14194	P07900
CRTC2	Q53ET0	Q9UMX1	CRYBA1	P05813	O75593
CSF1R	P07333	P29323	CSNK1G1	Q9HCP0	P78368
CSNK1G2	P78368	Q9Y248, Q9BRT9, Q14691, Q9BRX5	CSNK2A2	P19784	P46100
CST7	O76096	P07711	CSTB	P04080	P13569
CSTF1	Q05048	Q99728	CSTF2T	Q9H0L4	Q9UMD9
CSTF3	Q12996	Q6P1J9	CTBP1	Q13363	Q96KQ7
CTDP1	Q9Y5B0	O94822	CTNNBL1	Q8WYA6	Q9UKN5
CTTNBP2NL	Q9P2B4	O14964	CUL5	Q93034	Q9Y314
CUTC	Q9NTM9	Q9NRD5	CWF19L1	Q69YN2	Q2TBE0
CYB5R2	Q6BCY4-2	Q9NW38	CYP3A5	P20815	P42858
DALRD3	Q5D0E6	P61978	DARS2	Q6PI48	Q8NFJ9
DAZAP1	Q96EP5	P13569	DBF4	Q9UBU7	P10809
DBN1	Q16643	P60484	DCAF11	Q8TEB1	P11182
DCAF5	Q96JK2	P26358	DCAKD	Q8WVC6	Q9Y2C2
DCDC2B	A2VCK2	O43559	DCP1A	Q9NPI6	Q92900
DCUN1D5	Q9BTE7	Q9NRD5	DDI1	Q8WTU0	Q92681
DDIT3	P35638	P17676	DDR1	Q08345	Q06124
DDR2	Q16832	Q16288	DDX19B	Q9UMR2	P45973
DDX21	Q9NR30	Q86U42	DDX24	Q9GZR7	O43159
DEDD	O75618	Q13217	DEK	P35659	Q9UER7
DENND2D	Q9H6A0	P55199	DENR	O43583	P25963
DHCR24	Q15392	P04578	DHFR	P00374	Q86XF0
DHRS7	Q9Y394	Q9NTJ5	DHX29	Q7Z478	P06748
DHX35	Q9H5Z1	Q9Y4Z2	DHX37	Q8IY37	O43159
DKK1	O94907	O75197	DLAT	P10515	P10515
DLEU1	O43261	Q92993	DNAAF5	Q86Y56	P46059
DNAJB14	Q8TBM8	P13569	DNAJB6	O75190-2	Q9BRX5
DNAJB9	Q9UBS3	Q13217	DNAJC19	Q96DA6	P46059
DNAJC7	Q99615	P51530	DNMBP	Q6XZF7, Q6XZF7-1	P50570, O00401
DOT1L	Q8TEK3	P42568, Q03111	DPAGT1	Q9H3H5	P42858
DPM1	O60762	Q13418	DPY30	Q9C005	O00487
DPYSL2	Q16555	Q8NFD5	DR1	Q01658	P60896
DRC3	Q8TE68	Q9UQB8	DSCC1	Q9BVC3	P00740
DST	Q03001	Q9UKG1	DTNBP1	Q96EV8	A1L4H1
DUSP5	Q16690	P28482	DYNC2LI1	Q8TCX1	Q14203
DYNLL1	P63167	Q6W0C5	DYNLT1	P63172	Q9BRX2
DYRK1A	Q13627	O00425, Q9NZI8, Q9Y6M1	E2F8	A0AVK6	P07550
ECM1	Q16610	Q9NQ94	EDAR	Q9UNE0	Q92838
EEF1A2	Q05639	P26641, P68104, P29692, P24534	EEF1B2	P24534	P26641, P68104
EFNA5	P52803	Q9UBD6	EGFL7	Q9UHF1	Q9UBX5
EHMT1	Q9H9B1	Q96KQ7, Q13185	EHMT2	Q96KQ7	Q13185
EIF1AD	Q8N9N8	Q9H668, Q2NKJ3	EIF1AX	P47813	P55010
EIF2B2	P49770	O14975	EIF3B	P55884	Q04637, P06730
EIF3J	O75822	Q04637	EIF4E	P06730	Q92900
EIF4ENIF1	Q9NRA8	P06730	ELOVL5	Q9NYP7	P09601

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
EMC2	Q15006	O43868	EMC4	Q5J8M3	P27824
EMC7	Q9NPA0	Q15363	EMP1	P54849	Q9Y5Q5
ENPP4	Q9Y6X5	P17931	ENTR1	Q96C92	Q9HC29
EOGT	Q5NDL2-3	Q9UKG4	EOLA1	Q8TE69	P02795
EP300	Q09472	P29279, Q13761	EPB41L1	Q9H4G0	Q9ULK5
EPC1	Q9H2F5	O96019	EPN1	Q9Y6I3	P42566
EPN2	O95208-2	Q86T03	EPPK1	P58107	Q9Y324
ERC1	Q8IUD2	P01222	ERCC6	Q03468	P00519
ERCC8	Q13216-1	Q03468	ERGIC2	Q96RQ1	P15907
ERGIC3	Q9Y282	Q96AA3	ERH	P84090	Q9Y5B0
ERVFRD-1	P60508	Q01453	ERVK3-1	Q96A10	O95429
ESR2	Q92731	P29279	ETFRF1	Q6IPR1	O14561
ETV6	P41212	O75197	EVI2A	P22794	Q9Y5Z9
EVI2B	P34910-2	P22732	EXOC4	Q96A65	P04601
EXOC5	O00471	Q9ULZ3	EXOSC7	Q15024	Q99547, Q13901, P42285, Q01780, Q9NQT4
EXT1	Q16394	Q03405	EXTL2	Q9UBQ6	P0DTC5
EYA3	Q6P4T3	P16104	F2	P00734	P05067
F2RL1	P55085	Q9BVG9	F2RL2	O00254	Q8NI35
FAM110B	Q8TC76	P49674	FAM114A2	Q9NRY5	P49638
FAM118B	Q9BPY3	Q9Y639	FAM120C	Q9NX05	P07910
FAM136A	Q96C01	Q12887	FAM221B	A6H8Z2	Q86US8
FAM222B	Q8WU58	Q9UBE8	FAM241B	Q96D05	O43914
FAM3C	Q92520	Q9UNA3	FAM78A	Q5JUQ0	P42858
FAM98B	Q52LJ0	P0DTC9	FAN1	Q16658	P54278
FANCE	Q9HB96	Q00597	FANCF	Q9NPI8	O15360, O15287
FAR1	Q8WVX9	P13569	FASN	P49327	Q16665
FASTKD1	Q53R41	P07237	FAXDC2	Q96IV6	Q12846
FBF1	Q8TES7-6	P63165	FBXL19	Q6PCT2-2	Q6ZPD8
FBXL20	Q96IG2	P63208	FBXL5	Q9UKA1	Q92905
FBXO11	Q86XK2	O95863	FBXO25	Q8TCJ0	P60709
FBXO3	Q9UK99	P04608	FBXO30	Q8TB52	Q9UM54
FBXO45	P0C2W1	P22692	FBXW11	Q9UKB1	Q13127
FCF1	Q9Y324	P48163	FGF1	P05230	P22455
FGF14	Q92915-2	P49638	FGFR3	P22607	Q00994
FICD	Q9BVA6	P42858	FIP1L1	Q6UN15	O95639
FKBP3	Q00688	Q00987	FKBP8	Q14318	P03431
FMC1	Q96HJ9	Q9NUJ1	FMNL3	Q6ZPF4	O75044
FMO3	P31513	Q9NTJ5	FNDC3A	Q9Y2H6	O43511
FNDC9	Q8TBE3	P78382	FNTA	Q04631	P01116-2
FNTB	P49356	P01185	FOSL1	P15407	P15336
FRG1	Q14331	Q9HCG8	FRMD3	A2A2Y4	Q9Y6X1
FRMD4A	Q9P2Q2	O14907	FRS2	Q8WU20	Q06124
FSD2	A1L4K1	Q00994	FUBP3	Q96I24	P09651
FZD4	Q9ULV1	P42858	FZD7	O75084	P41221
G2E3	Q7L622	O76024	GAB1	Q13480	P46109
GAB2	Q9UQC2-1	P62993-1	GABPA	Q06546	P08047
GABRE	P78334	O60427	GAK	O14976	O95630, Q96FJ0
GALNT2	Q10471	P40855	GAN	Q9H2C0	Q99426

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
GAP43	P17677	Q16822	GARIN6	Q8NEG0	P07237
GATAD2B	Q8WXI9	Q92769, Q13547, Q9UBB5	GBP3	Q9H0R5	P32455
GCM1	Q9NP62	P23771	GCSH	P23434, Q53XL7	Q13287
GDAP2	Q9NXN4	Q9Y3M2	GEMIN7	Q9H840	P62304
GET1	O00258	P21964	GGH	Q92820	P0DTC8
GGT7	Q9UJ14	Q5SRI9	GID4	P38263	Q5SW96
GIGYF1	O75420	Q86UK7	GLA	Q9UHE5	Q9H2K0
GLI2	P10070, Q0VGT2	Q9UMX1	GLI3	P10071	Q9UMX1
GLIS3	Q8NEA6	Q9UMX1	GLMN	Q92990	P15907
GLO1	Q04760	P13569	GLRA1	P23415	Q9Y5V3
GLRX3	O76003	Q9H2F3	GLS	O94925	Q9NXA8
GMFB	P60983	Q86X55	GMNN	O75496	P06493
GNAS	P04896	P07550	GNMT	Q14749	P24311
GNPDA1	P46926	Q9Y303	GOLT1B	Q9Y3E0	O95197
GOPC	Q9HD26	P60484	GORASP2	Q9H8Y8	P11926
GP6	Q9HCN6, Q9HCN6-1	P06241	GPATCH11	Q8N954	P62306, P14678, P62304
GPC6	Q9Y625	P32249	GPR160	Q9UJ42	P17252
GPR18	Q14330	P42357	GPR183	P32249	O14786
GPR21	Q99679	P17302	GPR52	Q9Y2T5	P17931
GPR89A	B7ZAQ6	Q9H1C4	GPRC5D	Q9NZD1	Q99650
GPRIN1	Q7Z2K8	P21453	GRAMD1B	Q3KR37	P68400
GREM1	O60565	P35968	GRIN2B	Q13224	Q14160
GRK7	Q8WTQ7	P07900	GSG1	Q2KHT4	Q12933
GSK3B	P49841	Q9NYF0	GSPT1	P15170	Q92900
GSPT2	Q8IYD1	Q92900	GSTM4	Q03013	P05067
GTF2B	Q00403	P30740	GTF2E1	P29083	Q9Y6K9
GTF2H3	Q13889	P50613	GTF2H4	Q92759	Q96JC9
GTPBP8	Q8N3Z3	P10809	GUSBP1	Q15486	P07550
H1-1	Q02539	P03496	H1-10	Q92522	Q9NR30
H1-2	P16403	P16104	H1-3	P16402	P17096
H1-4	P10412	P05455	H1-5	P16401	P19338
H1-6	P22492	Q15361	H2AC11	P0C0S8	O95760
H2AC13	P0C0S8	O95760	H2AC15	P0C0S8	O95760
H2AC16	P0C0S8	O95760	H2AC17	P0C0S8	O95760
H2AC18	Q6FI13	Q12824	H2AC19	Q6FI13	Q12824
H2AC20	Q16777	P08253	H2AC21	Q8IUE6	P17096
H2AC4	P04908	Q16695	H2AC8	P04908	Q16695
H2AZ1	P0C0S5	Q01831	H2BC10	P62807	Q96CW1
H2BC11	P06899	P02675	H2BC12	O60814	Q12824
H2BC13	Q99880	Q96CW1	H2BC15	U3KQK0	Q9H2G4
H2BC17	P23527	Q01831	H2BC18	H2BC18	Q07955
H2BC21	Q16778	P62805	H2BC4	P62807	Q96CW1
H2BC5	P58876	Q01831	H2BC6	P62807	Q96CW1
H2BC7	P62807	Q96CW1	H2BC8	P62807	Q96CW1
H2BC9	Q93079	P42858	H3C1	P68431	Q13185
H3C11	P68431	Q13185	H3C12	P68431	Q13185
H3C13	Q71DI3	P49736	H3C14	Q71DI3	P49736
H3C15	Q71DI3	P49736	H3C2	P68431	Q13185

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
H3C3	P68431	Q13185	H3C4	P68431	Q13185
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H3C8	P68431	Q13185	H4C1	P62805	Q13185
H4C13	P62805	Q13185	H4C16	P62805	Q13185
H4C2	P62805	Q13185	H4C3	P62805	Q13185
H4C4	P62805	Q13185	H4C5	P62805	Q13185
H4C6	P62805	Q13185	H4C8	P62805	Q13185
H4C9	P62805	Q13185	HASPIN	Q8TF76	Q07954
HAUS2	Q9NVX0	Q13426	HAUS3	Q68CZ6	A1L190
HBE1	P02100	Q6DKK2	HDAC2	Q92769	Q96KQ7
HDAC4	P56524	Q13761	HDGFL2	Q7Z4V5	Q9UER7
HDHD5	Q9BXW7	P10809	HECTD4	Q9Y4D8	P61981
HELB	Q8NG08	P19388	HELZ	P42694	Q8IWL3
HEXB	P07686	P06865	HHIP	Q96QV1	Q15465
HHIPL2	Q6UWX4	P09917	HIF3A	Q9Y2N7-4	Q16665
HIGD2A	Q9BW72	P40855	HINT1	P49773	P13569
HIP1R	O75146	Q9NZQ7	HIPK2	Q9H2X6	Q09472
HIRA	P54198	P84243	HIVEP1	P15822	P84022
HIVEP2	P31629	P63208	HLA-A	P04439	Q16288
HLA-DPB1	A0A1C3PI11, P04440	Q96CV9	HLA-DRA	P01903	Q9NZJ5
HMBOX1	Q6NT76	Q02548	HMGB1	P63158	P06746
HMGCS1	Q01581	P18827	HNRNPA0	Q13151	P09651
HNRNPA1L2	Q32P51	P09651	HNRNPH3	P31942	P09651
HNRNPL	P14866	P61978	HNRNPLL	Q8WVV9	P61978
HOXA10	P31260	P17947	HOXB5	P09067	Q9NWQ8
HOXB7	P09629	P78527	HOXC8	P31273	P50458
HP	P00738	P02647	HPF1	Q05164	O43791
HPGDS	O60760	Q13370	HPR	P77335	P77335
HPRT1	P00492	Q9H1K1	HS6ST1	O60243	Q9Y2A9
HSDL1	Q3SXM5	Q00169	HSF2	Q03933	Q9Y315
HSPA1L	P34931	Q13233	HSPB7	Q9UBY9	Q9GZV8
HTR3A	P46098	P17302	HUNK	P57058	P23528, P53667
HUS1	O60921	Q03405	HYPK	Q9NX55	P42858
IAPP	P10997	P10997	IER3IP1	Q9Y5U9	P08034
IFITM2	Q6IBB0	Q14973	IFITM3	Q01628	Q9NXB9
IFNGR1	P15260	Q9NPF0	IFT140	Q96RY7	Q7Z4L5, Q8NEZ3, Q96FT9, Q9HBG6, Q9P2L0
IFT20	Q8IY31-3	Q13286	IFT43	Q96FT9-2	P17252
IFT80	Q9P2H3	Q9Y266	IFT81	Q8WYA0	P36941
IGF1	P05019	P08833	IGFBP3	P17936	Q86XT9
IGSF6	O95976	Q9HD47	IGSF8	Q969P0	P21926
IKZF4	Q9H2S9	P48730	ILK	Q13418	P49327, Q13085
IMPA1	P29218-3	P42858	IMPACT	Q9P2X3	P20618
IMPDH2	P12268	P0DTC4	INHBA	P08476	P31749
INO80D	Q53TQ3	Q9Y265	INSYN2A	Q6ZSG2	O15111
INTS13	Q9NVM9	Q13951	INTS8	Q75QN2	P67775
IPO8	O15397	Q9UKV8	IQCN	Q9H0B3	P14373
IRF2	P14316	P51610	IRF2BP2	Q7Z5L9	Q8N6T7
IRF2BPL	Q9H1B7	P14316	IRGQ	Q8WZA9	Q9H2D1

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
ISCA1	Q9BUE6	Q96RU8	ISCA2	Q86U28	P01160
ITGB3BP	Q13352	Q969Y2	ITGB8	P26012	O43914
ITM2B	Q9Y287	Q86XT9	IVD	P26440	Q15831
JRK	O75564-2	O60341	KANSL1	Q7Z3B3	Q9BVI0
KANSL2	Q9H9L4	Q9BVI0	KANSL3	Q9P2N6	Q9BVI0
KAT6A	Q92794	P67870	KATNAL2	Q8IYT4	P03372
KATNB1	Q9BVA0	Q06187	KATNBL1	Q9H079	Q9UBU9
KBTBD6	Q86V97	Q6NZI2	KBTBD7	Q8WVZ9	Q86UK7
KBTBD8	Q8NFY9	Q9UF56	KCNAB2	Q13303	P05067
KCNIP4	Q6PIL6	P08123	KCNS2	Q9ULS6	P0DPK4
KCNV2	Q8TDN2	P11142	KCTD13	Q8WZ19	O15160
KCTD4	Q8WVF5	Q9UER7	KCTD7	Q96MP8-2	Q9UBU9
KDM4A	O75164	P25205, P33993	KDM5A	P29375, P29375-2	Q13547
KHDRBS1	Q07666	P19174	KHSRP	Q92945	P05412
KIAA0825	Q8IV33	Q96CV9	KIAA0930	Q6ICG6	Q14191
KIF1C	O43896	P61981	KIF24	Q5T7B8	P51955
KIF2C	Q99661	Q96ES7	KIFC1	Q9BW19	Q14974
KIN	O60870	Q9BUU2	KLC3	Q6P597	Q9BTA9
KLF12	Q9Y4X4	Q96KQ7	KLF14	Q3SY56	P36957
KLHDC3	Q9BQ90	Q15843	KLHL10	Q6JEL2	Q9UF56
KLHL11	Q9NVR0	Q9NRD5	KLHL12	Q53G59	O95050
KLHL8	Q9P2G9-2	P11086	KMT2A	Q03164	O00255
KMT2C	Q8NEZ4	P61964	KMT2D	O14686	P62937
KRT34	O76011	Q9NQ94	KRT8	P05787	Q9Y6K9
KRTAP1-5	Q9BYS1	P06850	KRTAP2-3	P0C7H8	Q8WUK0
KSR2	Q6VAB6	P07900	L2HGDH	Q9H9P8	O15431
LACTB2	Q53H82	Q8IUQ4	LAMTOR3	Q9UHA4	Q9Y2Q5, Q0VGL1, Q6IAA8, O43504, Q8NBW4
LAMTOR5	O43504	P68104	LANCL2	Q9NS86	Q9Y2W6
LARP1	Q6PKG0	P59595	LARP7	Q4G0J3	P23193
LATS1	O95835	P46937	LBR	Q14739	Q13185
LCLAT1	Q6UWP7	Q92504	LCOR	Q96JN0, Q96JN0-2	Q13363
LDAF1	Q96B96	Q05329	LEMD1	Q68G75	Q15125
LEO1	Q8WVC0	P23193	LHX4	Q969G2	Q9Y5N6
LIG1	Q96JA1	P22681	LIMK1	P53667	Q13177
LINC-PINT	A0A455ZAR2	Q08050	LMNB1	P20700	P02545-1
LMO2	P25791-3	Q13541	LONRF2	Q1L5Z9	P99999
LOX	P28300	Q9UBX5, P15502	LOXL4	Q96JB6	O60341
LPAR1	Q92633	A0PJW6	LPCAT1	Q8NF37	O14975
LPP	Q93052	P05120	LRAT	O95237	Q7Z5P4
LRCH3	Q96II8	Q9UM54	LRP2BP	Q9P2M1	P38936
LRRC20	Q8TCA0	P60709	LRRC40	Q9H9A6	Q96JA1
LRRK1	Q38SD2	P00519	LRRN4	O75427	A1L3X0
LRRTM2	O43300	Q9BT09	LSM14B	Q9BX40	Q9NTG7
LSM5	Q9Y4Y9	Q9H4B4	LSR	Q86X29	Q96FT7
LSS	P48449	P09917	LTBP2	Q14767	Q9UBX5
LUC7L2	Q9Y383	Q9GZQ8	LYAR	Q9NX58	P19525
LYG2	Q86SG7	Q86YN1	LYPLA1	Q6IAQ1, O75608	Q08379
LYRM7	Q5U5X0	Q9H1K1	LZTS3	O60299	Q13627

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
MAB21L1	Q13394	O00470	MAB21L2	Q9Y586	Q9UBB9
MAD1L1	Q9Y6D9	Q6NZI2	MAD2L1	Q13257	O43511
MAGEA4	P43358	Q9NUX5	MAGOH	P61326	O76075
MAGT1	Q9H0U3	Q15363	MAK	P20794	Q9UM11
MAP1B	P46821	Q9GZQ8	MAP1LC3B2	A6NCE7	Q9NT62
MAP3K12	Q12852	P00533	MAP3K7	O43318	O15111, O14920
MAPK8	P45983-4	Q9BY84	MARCKS	P29966	P13569
MASP2	O00187, O00187-PRO_0000027598	29108, P05155	MAX	P61244	P29692
MCL1	Q07820	Q13794	MCM9	Q9NXL9	P52701, P20585
MDM4	O15151	P06400	MEAF6	Q9HAF1	P40937
MED13	Q9UHV7	Q15648	MED18	Q9BUE0	Q63HM1
MEF2A	Q02078	P56524	MEP1B	Q16820	P14780
MEPE	Q9NQ76	P98164	METAP2	P50579	Q93034
METTL15	A6NJ78	P46059	MFAP4	P55083	Q16625
MFN1	Q8IWA4	Q16611	MFSD10	Q14728	P45973
MFSD14B	Q5SR56	Q9NRZ7	MGST3	O14880	Q00059
MICAL2	O94851	P54252	MICOS10	Q5TGZ0	P55056
MICU2	Q8IYU8	Q9H4I9, Q9BPX6, Q8NE86	MIER1	Q8N108	Q96KQ7
MINAR1	Q9UPX6	Q9Y5P4-2	MINDY1	Q8N5J2-3	P55212
MIP	Q99797	Q9BQP7	MIPEP	Q99797	Q9BQP7
MISP	Q8IVT2	P17707	MITD1	Q8WV92	P04618
MLF1	P58340	P51587	MLLT10	P55197	P29692
MMP2	P08253	P39060	MMP3	P08254	P09238
MMS19	Q96T76	P51530	MOB4	Q9QYW3	Q05193
MPL	P40238	O60674	MPP2	Q08050	Q13416
MPST	P25325	O96006	MPV17	B5MC10	Q8N0S2
MPZL1	O95297	Q06124	MR1	Q95460-1	P61769
MRPL1	Q9BYD6	P67809	MRPL18	Q9H0U6	Q14197
MRPL20	Q9BYC9	P40429	MRPL46	Q9H2W6	Q9BQP7
MRPL50	Q8N5N7	Q14197	MRPL51	Q4U2R6	Q14197
MRPL58	Q14197	Q00059	MRPL9	Q9BYD2	P21673
MRPS14	O60783	P40429	MRPS15	P82914	P62753
MRPS18C	Q9Y3D5	Q9UPR3	MRPS21	P82921	P84103
MRPS23	Q9Y3D9	O15160	MRPS24	Q96EL2	Q9P031
MRPS33	Q9Y291	Q00839	MRPS35	P82673	Q00839
MRPS9	P82933	P04618	MRTFB	Q9ULH7	P11831
MT-ATP8	P03928	P25090	MT1A	P04731	P04637
MT2A	P02795	P09429	MTA1	Q13330	Q96KQ7
MTA2	O94776	Q96KQ7	MTCL2	O94964-4	O14503
MTERF3	Q96E29	P22732	MTFR1	Q15390	Q9NR28
MTFR2	Q6P444	Q9UBB9	MTMR3	Q13615	Q9NQT4
MTPAP	Q9NVV4	Q9UHD2	MX2	P20592	Q96KQ7
MXD1	Q05195	Q96ST3	MYBPHL	A2RUH7	P36957
MYCBP2	O75592	P22692	MYD88	Q99836	Q99836
MYH11	P35749	P13569	MYH7	P12883	Q15475
MYL11	Q96A32	O76024	MYL12A	P19105	P35579
MYL4	P12829	P19320	MYL6	P60660	O43809
MYO5A	Q9Y4I1	P03372	MYOZ2	Q9NPC6	P12814

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MZT1	Q08AG7	Q6ZMU5	N4BP3	O15049	P40937
NAA50	Q9GZZ1	Q6NYC1	NADK	O95544	Q49AN0
NAE1	Q13564	Q8TBC4, P61081	NAMPT	P43490	P43490
NAP1L1	P55209	Q09472	NAP1L2	Q9ULW6	P42858, Q96HR3
NAP1L4	Q99733	P04608	NAV2	Q8IVL1	P63104
NBN	O60934	Q13315	NCAPD2	Q15021	P13569
NCBP3	Q53F19	Q86U42	NCF2	P19878	Q92845
NCK1	P16333	Q13177	NCOA1	Q15788	Q13569
NCOA2	Q15596	O00482-1	NCOA3	Q9Y6Q9	Q09472
NCOA6	Q14686	Q92793	NDE1	Q9NXR1, Q9CZA6	P43034
NDFIP1	Q9BT67	P43088	NDST2	P52849	Q03405
NDUFA1	O15239	O75380	NDUFA12	Q9UI09	Q8WTR4
NDUFA2	O43678	P29590	NDUFA5	Q16718	Q9P0J0
NDUFA8	P51970	Q9P0J0	NDUFAF3	Q9BU61	Q9NPB3
NDUFB2	Q9Y6T4	O00311	NDUFB9	Q9Y6M9	Q6UWR7
NDUFS3	O75489	Q9P0J0	NDUFV3	P56181	Q9P0J0
NECAB1	Q8N987	O00244	NEDD1	Q8NHV4	Q8IW35, O43303
NEDD8	Q15843	P68104	NEK1	Q96PY6	P04004
NEK3	P51956	P32971	NEK9	Q8TD19	P53350
NES	P48681	P08670	NEXN	Q0ZGT2	P60709
NF1	P21359	P05067	NFAT5	O94916	Q16236
NFATC3	Q12968	Q9UMX1	NFIB	O00712	P08651, Q14938, Q12857
NFIC	P08651	O00712	NFIX	Q14938	O00712
NFU1	Q9UMS0	P21741	NFYC	Q13952-2	P32519
NGDN	Q8NEJ9	Q9NY61	NGF	P01138-PRO_0000019600, P01138	P08138
NGFR	P07174, P08138	P01138	NHSL1	Q5SYE7	P16333
NHSL2	Q5HYW2	P53350	NIF3L1	Q9GZT8	P42772
NIN	Q8N4C6	P49841	NIPAL4	Q0D2K0	P21964
NKIRAS1	Q9NYS0	P56559	NOC2L	Q9Y3T9	P40429
NOL4L	Q96MY1	Q13363	NOL7	Q9UMY1	O00410
NOL9	Q5SY16	Q8IZL8	NOP14	P78316	Q9UER7
NOP9	Q86U38	P61073	NOPCHAP1	Q8N5I9	P52294
NOSTRIN	Q8IVI9	Q05193	NPFF	O15130-2	O76024
NPIPB3	Q92617	P59634	NPM3	O75607	P40429
NPTN	Q9Y639	O43929	NR1D1	P20393	P06727
NR2C2	Q6PHZ7	Q14957	NRP1	P97333	P08648
NSMF	Q9EPI6	P27361	NSUN4	Q96CB9	O95861
NTM	Q9P121-3	Q96P65	NUAK1	O60285	Q93008
NUBP1	P53384	Q9Y5Y2	NUDCD1	Q96RS6	O43143, Q92620, Q14562
NUDT21	O43809	P61978	NUP133	Q8WUM0	Q9UBP0
NUP214	P35658	Q9UBU9	NUP35	Q8NFH5	P04798
NUP54	Q7Z3B4	Q8IY31	NUPR1	O60356	O15534
NUS1	Q96E22	P11142	NXPE1	Q8N323	P12830
NXPE3	Q969Y0	Q6IN84	OAT	P04181	P05067
OCIAD1	Q9NX40	Q9Y2C2	ODC1	P11926	Q92993

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OGN	P20774	Q9BT09	ORC2	Q13416	O43929, Q9UBD5, Q13415, Q9Y5N6, O43913
ORC4	O43929	Q13416, Q9UBD5, Q13415, Q9Y5N6, O43913	ORC5	O43913	O43929, Q13416, Q9UBD5, Q13415
ORMDL1	Q9P0S3	Q8NBQ5	ORMDL2	Q53FV1	Q92911
OSR1	O95747	P02675	OSR2	Q8N2R0	P25788
OST4	P0C6T2	A6NI15	OSTC	Q9NRP0	P80370
OSTM1	Q86WC4	P17931	OXNAD1	Q96HP4	P10809
P2RY12	Q9H244	P53794	PAAF1	Q9BRP4	P60896
PABPC4L	P0CB38	P16104	PADI3	Q9ULW8	Q9Y496
PAIP1	Q9H074, Q9H074-3	P11940	PALS1	Q8N3R9	P46937
PAQR5	Q9NXK6	O95470	PAQR6	Q6TCH4	P50876
PARP14	Q460N5	O43914	PARP2	Q9UGN5	Q9NTX7
PATL1	Q86TB9	Q8IZH2	PAWR	Q96IZ0	Q01094
PBXIP1	Q96AQ6	O15554	PCBD1	P61457	P35680
PCDH7	O60245	Q9Y337	PCGF1	Q9BSM1	Q03111
PCLAF	Q15004	P38936	PCMTD2	Q9NV79	Q93034, Q9UBF6
PCNA	P12004	P52701, P20585	PDCD1LG2	Q9BQ51	Q9NZQ7
PDCD2L	Q9BRP1	P51813	PDCD6	O75340	P35968
PDCL	Q13371	P63211	PDCL3	Q9H2J4	Q13371
PDE6D	O43924	P36404	PDHB	P11177	P10515
PDZD11	Q5EBL8	Q04656	PDZD8	Q8NEN9	P23945
PDZK1	Q5T2W1	Q58EX2	PELI2	Q9HAT8	Q12891
PELP1	Q8IZL8	P19838	PER2	O15055	P40692
PERP	Q96FX8	O76024	PES1	O00541	P42568, Q9UHB7
PEX13	Q92968	O00628	PEX26	Q7Z412	Q969S8
PEX3	P56589	Q9Y5Y5, P40855	PFDN4	Q9NQP4	Q13164
PFDN5	Q99471	Q13163, Q13164	PFDN6	O15212	Q13164
PFKFB3	Q16875	P61981	PFKFB4	Q16877	P61981
PFN2	P35080	P60709	PGA4	P0DJD7	P15151
PHACTR4	Q8IZ21	P36873	PHAF1	Q9BSU1	P54252
PHB1	P35232	P40763	PHC3	Q8NDX5	Q13363
PHF12	Q96QT6	Q13547	PHF13	Q86YI8	Q9UMR3
PHF21A	Q96BD5, Q96BD5-2	Q92769, Q13547	PHF5A	Q7RTV0	O75533
PHOX2A	O14813	O43464	PHRF1	Q9P1Y6	P67870
PI16	Q6UXB8	P27449	PIGN	O95427	P43119
PIH1D2	Q8WWB5	O60469	PIK3R2	O00459	P03496
PILRA	Q9UKJ1	P29350	PIMREG	Q9BSJ6	P04156
PIP5K1A	Q99755	Q9H9Z2	PITX2	Q99697-3	Q12948
PKDCC	Q504Y2	Q9H5K3	PKIA	P61925	O43663
PKN1	Q16512	P26641	PKP3	Q9Y446	Q9Y496
PKP4	Q99569	Q8NI35	PLAGL2	Q9UPG8	O94955
PLCG1	P19174	P16234	PLD2	O14939	P23528
PLEKHA3	Q9HB20	O43148	PLEKHG2	Q9H7P9	Q13526
PLEKHG7	Q6ZR37	P51149	PLEKHM3	Q6ZWE6	P61981
PLN	P26678	Q9H2K0	PLOD2	O00469	P0DTC8
PLPP4	Q5VZY2	P17302	PLPP6	Q8IY26	P16471
PMS1	P54277	P40692	PMS2	P54278	P40692
PNPLA6	Q8IY17	Q8TDQ0	PNPT1	Q8TCS8	Q8IWL3

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PNRC2	Q9NPJ4	Q92900	PODXL2	Q9NZ53	P32971
POGLUT2	Q6UW63	P0DTC8	POGZ	Q7Z3K3	Q13185
POLD1	P28340	Q13485	POLD3	Q15054	O75152
POLE2	P56282	P52594	POLG	P18247	P06730
POLR1C	O15160	P23025	POLR1F	Q3B726	Q9NYV6
POLR1H	Q9P1U0	O95602, P19388, P61218	POLR2F	P61218	Q9Y5B0
POLR2H	P52434	O15160, O95602, P19388, P61218, P62875	POLR2K	P53803	O95602, P19388
POLR3K	Q9Y2Y1	O15160, P19388, P61218, P62875, P52434	POM121	Q96HA1-2	P25963
POM121C	A8CG34	Q14974	POMK	Q9H5K3	O75197
POP5	Q969H6	Q15116	POTEE	Q6S8J3	Q8NCH0
POTEF	A5A3E0	P00813	POTEI	P0CG38	Q9HB29
POTEJ	P0CG39	P13569	PPA2	Q9H2U2-3, Q9H2U2-6	P42858
PPIA	P62937-2	Q9BRX2	PPIAL4G	P0DN37	P60709
PPME1	Q9Y570	P67775, P63151	PPP1CB	P62140	Q9UQK1
PPP1R10	Q96QC0	P36873	PPP1R12B	O60237	O75344
PPP1R3E	Q9H7J1	P31946	PPP2R2D	Q66LE6	O43768
PPP2R5E	Q16537	O96017	PPP3CC	P48454	P16298
PPP3R1	P63098	P10415	PPP4R2	Q9NY27	Q9Y570
PRCC	Q92733	P38646	PRDM1	O75626	Q12857
PRDM2	Q13029	Q13363	PRG4	Q96GM1	Q9BVC3
PRIMPOL	Q96LW4	O95714	PRKAR2A	P13861	P24588
PRKRA	O75569	P03496	PRLR	P16471-1, P16471-7	P01236
PRMT6	Q96LA8	Q9Y3A6	PRPF19	Q9UMS4	Q13573
PRPF3	O43395	Q8WW38	PRR4	Q96NY8	Q15223
PRSS37	A4D1T9	O00462	PSG1	P11464	O15294
PSMA7	O14818	Q16665	PSMB4	P28070	Q9Y5K5
PSMB5	P28074	Q9Y5K5	PSMB7	Q99436	Q9Y5K5
PSMC1	P62191	O00487	PSMC6	P62333	P46109
PSMG4	Q5JS54	Q99436, P28066, Q9Y244, P28065, Q969U7, P49720, P25787, P25786	PTGES3	Q15185	Q8TAT5
PTK7	Q13308	Q9UMR7	PTPN1	P18031	Q16288
PTPN14	Q15678	P46937	PTPN21	Q16825	P51813
PTPRA	P18433	P12931	PTPRM	P28827	O60469
PTRH1	Q86Y79	Q9NRD5	PTRHD1	Q6GMV3	Q9H244
PYGO1	Q9Y3Y4	P61769	QPCT	Q16769	Q9Y5Y2
QRICH1	Q2TAL8	Q14938	R3HDM2	Q9Y2K5	P61981
RAB18	Q9NP72	P13569	RAB40AL	P0C0E4	Q93034
RAB41	Q5JT25	P53041	RAB4A	P20338	Q96JA1
RABGAP1	Q9Y3P9	P40763	RABGGTB	P53611	O94955
RABIF	P47224	Q9NZU5	RABL6	Q3YEC7-3	P17252
RAC3	P60763	P52565	RAD23B	P54727	Q01831, P41208
RAD52	P43351	Q06609	RAD54L2	Q9Y4B4	Q13185
RADIL	Q96JH8	P02654	RALA	P11233	P13569
RAMAC	Q9BTL3	O43148	RANBP6	O60518	O43808, O14975
RANBP9	Q96S59	P01275	RAP1B	P61224	Q12967
RAPGEF6	Q8TEU7	P61981	RAPH1	Q70E73	Q99962
RARS2	Q5T160	P21462	RASAL2	Q9UJF2	P49674

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
RBFOX2	O43251	Q9NWB1	RBL1	P28749	O43524
RBM17	Q96I25	P21673	RBM19	Q9Y4C8	P16104
RBM28	Q9NW13	P46087	RBM33	Q96EV2	O75152
RBM4B	Q9BQ04	P07910	RBM7	Q9Y580	P61978
RBMS2	Q15434	O00425	RBP2	P29375, P29375-2	Q13547
RBPMS	Q93062	Q53HV7	RBPMS2	Q6ZRY4	O75593
RBX1	P62877	P61081	RCC1	P18754	O14980
RCL1	Q9Y2P8	Q08379	RCN1	Q15293	Q969Y2
RECQL	P46063	P04798	RELL1	Q8IUW5	P26572
RERE	Q9P2R6	Q96KQ7	REST	Q8VIG1	Q96MT3
RFFL	Q8WZ73	P29353	RFX7	Q2KHR2	P61981
RGPD2	P0DJD1	P25103	RGS20	O76081-6	Q8WUK0
RHOXF2	Q9BQY4	P63165	RIMS4	Q9H426	Q96FA3
RIOK1	Q9BRS2	P14678	RMI1	Q9H9A7	P54132, Q13472
RMRP	EBI-9875627	O14746	RNF10	Q8N5U6	P51668
RNF11	Q9Y3C5	Q13569	RNF138	Q8WVD3	P07550
RNF145	Q96MT1	P0DTC5	RNF146	Q9NTX7	Q86WJ1
RNF169	Q8NCN4	Q08379	RNF38	Q9H0F5-2	O76024
RNF8	O76064	P62837	RNLS	Q5VYX0	P23634
RNPC3	Q96LT9	P62306, P14678, P62304	RPA1	P27694	P23025
RPA4	Q13156	Q01831	RPIA	P49247	P49247
RPL22	P35268	Q96CW1	RPL26L1	Q9UNX3	Q9P031
RPL27	P61353	Q99558	RPL30	P62888	Q15361
RPL31	P62899	P05455	RPL36AL	Q969Q0	P03372
RPL37	P61927	P63244	RPL38	P63173	P63244
RPL41	P62945	P63244	RPL5	P46777	P40429
RPL9	P32969	Q99558	RPLP2	P05387	Q9NR30
RPP25	Q9BUL9	Q00403	RPP38	P78345	P06748
RPS10	P46783	Q9H2U1	RPS11	P62280	Q04637
RPS12	P25398	P09429	RPS13	P62277	Q99558
RPS15A	P62244	Q15843	RPS23	P62266	Q15843
RPS25	P62851	P63244	RPS27	P42677	Q00987
RPS27A	P62990	P51668	RPS4X	P62701	Q9UPY3
RRAGA	Q7L523	P61916	RRAGB	Q5VZM2-2, Q5VZM2	Q9Y2Q5, Q0VGL1, Q6IAA8, O43504, Q8NBW4, Q9UHA4
RRBP1	Q9P2E9	P13569	RRN3	Q9NYV6	Q8NFM7
RRP8	O43159	P19525	RSL24D1	Q9UHA3	Q9UJW3
RUBCN	Q92622	Q13895	RUFY2	Q8WXA3	Q9Y639
RUNX1	Q01196	Q9H2X6	RUNX1T1	Q06455	P08047
RUNX2	Q13950	P17480	RYBP	Q8N488	Q13185
RYR2	Q92736	Q00987	S100A11	P31949	P01375
S1PR2	O95136	Q9NWM8	SAP130	Q9H0E3	Q9UER7
SAP25	Q1EHW4	Q96ST3	SAPCD2	Q86UD0	O75496
SAT2	Q96F10	P21673	SAXO2	Q658L1	Q14192
SBF1	O95248	Q13614	SC5D	O75845	Q8TDU9
SCAF8	Q9UPN6	Q08379	SCAMP1	O15126	P49638
SCAP	Q9NP31	O14543	SCCPDH	Q8NBX0	P40855
SCFD1	Q8WVM8	P16234	SCLT1	Q96NL6	A1L190
SCMH1	Q96GD3	P35227	SCO1	O75880	Q14061

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
SCOC	Q9UIL1-3	Q9BRV8	SCRIB	Q14160	P46937
SCUBE3	Q8IX30	P37173	SDF2	Q99470	P04578
SDHAF4	Q5VUM1	Q9NRP4, P31040	SEC11C	Q9BY50	P09601
SEC14L1	SEC14L1	Q07955	SEC24B	O95487	Q15436
SEC61A2	Q9H9S3	P04049	SEC62	Q99442	O43914
SEL1L	Q9UBV2	Q16288	SELENOK	Q9Y6D0	Q9H2K0
SEM1	P60896	P51587	SEMG1	P04279	Q6N021
SENP3	Q9H4L4	Q8IZL8	SENP5	Q96HI0	P06748
SEPTIN10	Q9P0V9	Q15019	SEPTIN7	Q16181	Q15019
SEPTIN9	Q9UHD8-1	Q16665	SERP1	Q9Y6X1	Q9H2K0
SERPINB2	P05120	Q9NRY4	SERPINB7	O75635	Q9NYY3
SERPINB8	P50452	Q15696	SESTD1	Q86VW0	Q15561
SETD2	Q9BYW2	P00740	SF3A2	Q15428	Q09161
SFT2D2	O95562	Q8NBQ5	SFXN1	Q9H9B4	Q7KZN9
SGPL1	O95470	P18827	SGSM3	Q96HU1	Q8NC56
SHLD1	Q8IYI0	O76024	SHLD2	Q86V20-2	Q12888
SHPRH	Q149N8-1	P61088	SHQ1	Q6PI26	Q9Y265
SIGMAR1	Q99720	P14416	SIK3	Q9Y2K2	P57059
SIM2	Q14190	Q15185	SIMC1	Q8NDZ2	Q8IWZ6
SIN3A	Q96ST3	Q92769, Q13547	SIPA1L1	O43166	Q96FJ0
SKP1	P63208	Q9UF56	SLC10A1	Q14973	O60906
SLC12A6	Q9UHW9	P17931	SLC16A1	P53985	P35613
SLC16A6	O15403	O76024	SLC1A1	P43005	Q6IAN0
SLC1A4	P43007	Q15758	SLC25A15	Q9Y619	Q15116
SLC25A24	Q6NUK1	Q05940	SLC25A29	Q8N8R3	P35613
SLC25A40	Q8TBP6	P02452	SLC25A43	Q8WUT9	P10809
SLC25A46	Q96AG3	Q8IVP5	SLC26A2	P50443	P17931
SLC31A1	O15431	Q96K12	SLC33A1	O00400	Q13304
SLC38A9	Q8NBW4	Q9Y2Q5, Q0VGL1, Q6IAA8, O43504, Q9UHA4	SLC39A1	Q9NY26	P21964
SLC39A5	Q6ZMH5	P22455	SLC39A7	Q92504	Q6UWP7
SLC41A1	Q8IVJ1	P30518	SLC48A1	Q6P1K1	P21964
SLC4A7	Q9Y6M7	Q14160	SLF2	Q8IX21	P25786
SMAD2	Q15796	Q9HAU4	SMAD3	P84022	O15397
SMARCD2	Q92925	O96019, Q15532	SMCO4	Q9NRQ5	P25025
SMDT1	Q9H4I9	Q9BPX6, Q8NE86	SMG7	Q92540	Q92900
SNAPC3	Q92966	O76024	SNRNP25	Q9BV90	Q9Y2W6
SNRNP27	Q8WVK2	P39748	SNRNP70	P08621	Q09161
SNTG2	Q9NY99	P60484	SNX10	Q9Y5X0	P17252
SNX12	Q9UMY4	O60493	SNX14	Q9Y5W7	P61073
SNX16	P57768	O95619	SNX27	Q96L92	P27037
SNX3	O60493	P56539	SOCS4	O14544	P46937
SOD2	P04179	P04179	SOX4	Q06945	P04637
SOX5	P35711	O00712, P08651, Q12857	SP2	Q9UKF7-2	O60906
SPATA22	Q8NHS9	Q00403	SPATA46	Q5T0L3	Q99685
SPCS3	P61009	P27824	SPDYA	Q5MJ70	P46527
SPECC1	Q5M775	P03372	SPIB	Q01892	P42858
SPIN2B	Q9BPZ2	Q01831	SPIN4	Q56A73	Q96LA8
SPINDOC	Q9BUA3	P31930	SPN	O75398	P49841

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SPP1	Q9BX95	Q8WTR4	SPPL2A	Q8TCT8	O76024
SPRING1	Q9H741	P40855	SPRY4	Q8WW59	P24534
SPRYD4	Q8WW59	P24534	SPTLC1	Q6NUL7	P60896
SREBF1	P36956-PRO_0000314029	P20700, P02545-1	SREBF2	Q12772	P08047
SRF	P11831	Q969V6	SRGAP1	Q7Z6B7	P61981
SRI	P30626	P30626	SRP14	P37108	Q96JC9
SRPRB	Q9Y5M8	P09601	SRRM2	Q9UQ35	P04608
SSH2	Q76I76	P63104	SSMEM1	Q8WWF3	P78382
SST	P61278	P05067	SSU72	Q9NP77	O60216
ST7	Q9Y561	Q9NPI5	STEAP1	Q9UHE8	Q13200
STEEP1	Q9H5V9	Q9Y3E7	STK24	Q9Y6E0	Q9BRV8
STK35	Q8TDR2	P07900	STMN1	P16949	P46527
STMN2	Q93045	Q9UBB5	STRC	A1L378	O76024
STRN4	Q9NRL3	Q9BRV8	STT3B	Q8TCJ2	Q14165
STX18	Q9P2W9	Q13190, Q9NYM9	STXBP2	Q15833	Q13277
SUB1	P53999	Q9NR30	SUN1	O94901	P04578
SUPT7L	O94864, O94864-2	Q96ES7	SURF6	O75683	P78563
SUZ12	Q15022	Q9UBC3	SV2A	Q02563	P0DPI0
SYN3	Q17R54	O75928-2	SYPL1	Q16563	P63027
SYTL3	Q4VX76	P51159	TACC2	B2RWP4	P22735
TACO1	Q9BSH4	P21673	TAF1C	Q15572	Q01105
TAFA4	Q96LR4	P08648	TANC2	Q9HCD6	O14907
TARDBP	Q13148	P18847	TARS2	Q9BW92	P0DTC5
TAS2R19	P59542	P15151	TAS2R5	Q9NYW4	Q5QGT7
TBC1D2B	Q9UPU7	P09917	TBC1D9B	Q66K14	P21462
TBCD	Q9BTW9	P36404	TBL1X	O60907	O15379
TBP	P20226	Q00403	TBPL1	P62380	O76024
TBX15	Q96SF7	O94955	TCEA1	P23193	Q8WVC0
TCEAL4	Q96EI5	Q93009	TCEAL9	Q9UHQ7	Q6VN20, Q96S59
TCF7L2	Q9NQB0	Q13761	TCP10L	Q8TDR4	Q13153
TCP11L1	Q9NUJ3	P50458	TDP2	O95551	O14641
TEAD4	Q15561	P46937	TECPR2	O15040	P35813
TET2	Q6N021	P19544	TET3	Q8BG87, O43151	O15294
TEX2	Q8IWB9	Q9NP91	TFAM	Q00059	Q14197
TFB1M	Q8WVM0	Q00059	TFG	Q92734	O75593
TGIF1	Q15583	P53350	THAP12	O43422	Q13158
THBS1	P07996-PRO_0000035842	P16671	THBS3	P49746	O60216
THUMPD1	Q9NXG2	Q96PZ0	TIE1	P35590	P04629
TIGD1	Q96MW7	P42858	TIGD7	Q6NT04	Q9BZF9
TIMM23	O14925	Q9BVV7	TIMM44	O43615	Q00403
TIMM8B	Q9Y5J9	Q99558	TIMP3	P35625	P16035
TINAGL1	Q9GZM7	O75386	TIPRL	O75663	P67775
TIRAP	P58753-2	P29466	TJP1	Q07157	P59637
TJP3	O95049	Q8NI35	TLE4	Q04727	O95231
TLK2	Q86UE8	Q92985	TM4SF20	Q53R12	Q15836
TM9SF3	Q9HD45	Q9H244	TMA16	Q96EY4	Q96CW1
TMC6	Q7Z403	P16284	TMCC1	O94876	Q9BRI3
TMED4	Q7Z7H5	Q9Y3A6	TMED7	Q9Y3B3	P31371

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
TMEFF1	Q8IYR6	Q13627	TMEM106A	Q96A25	Q16585
TMEM107	Q6UX40	Q8N661	TMEM120B	A0PK00	Q05329
TMEM131	Q92545	P0DTC3	TMEM165	Q9HC07	P13569
TMEM230	Q96A57	P27824	TMEM234	Q8WY98	Q8N130
TMEM254	Q8TBM7	Q9H2K0	TMEM30A	Q8VEK0	P05067-4
TMEM41A	Q96HV5	Q7L5A8	TMEM45A	Q9NWC5	P02652
TMEM50A	O95807	P21964	TMEM60	Q9H2L4	Q9UBD6
TMEM67	Q5HYA8	P59665	TMPO	P42166	Q93009
TMPPE	Q6ZT21	Q9H2H9	TMX3	Q96JJ7-2	P36957
TNFAIP8	O95379	Q96JC9	TNFAIP8L1	Q8WVP5	Q6NZI2
TNFAIP8L3	Q5GJ75	Q9Y3M2	TNFSF4	P23510	P43489
TNKS1BP1	Q9C0C2	O14965	TNRC6B	Q9UPQ9	P11940
TOMM22	Q9NS69	P0DOE7	TOMM40	O96008	P56705
TOMM7	Q75MR5	P50542	TOP1	P11387	P00519
TOR1A	O14656-2	Q96FZ7	TPM1	P09493-10, P09493	Q9Y6D9
TPRKB	Q9Y3C4	Q96S44	TRAF3IP1	Q8TDR0	Q13114
TRAF7	Q6Q0C0	Q9NWB7	TRAM1	Q9Y6Q9	Q09472
TRAPPC6B	Q86SZ2	Q9P2M4	TRERF1	Q96PN7	P46937
TRIM28	Q13263	O15397	TRIM32	Q13049	P61088
TRIM39	Q9HCM9, Q9HCM9-2	P51668	TRIM45	Q9H8W5	Q9BQP7
TRIM65	Q6PJ69	P09917	TRMT10C	Q7L0Y3	Q9BQ52
TRNAU1AP	Q9NX07	P36957	TRNT1	Q96Q11-3	O76024
TRUB1	Q8WWH5	Q96DP5	TSACC	Q96A04	Q13153
TSEN2	Q8NCE0	Q92989	TSG101	Q99816	P21453
TSHZ2	Q9NRE2	Q9UBB9	TSNAX	Q99598	Q00994
TSPAN31	Q12999	Q9BRI3	TSPAN9	O75954	P21926
TSR2	Q969E8	O75928-2	TSSK6	Q9BXA6	P07900
TSTD2	Q5T7W7	Q9UGI0	TTC28	Q96AY4	P61981
TTC33	Q6PID6	P02649	TTK	P33981	P08173
TTN	Q8WZ42	O75923	TUBA1B	P68363	Q15398
TUBB1	Q9H4B7	P00480	TULP4	Q9NRJ4	Q93034, Q9UBF6
TUSC3	Q13454	Q14165	TWF1	Q12792	P67870
TWIST1	Q15672	O60885	TXNDC17	Q9BRA2	Q96T51
TXNDC2	Q86VQ3	P35638	TXNDC9	O14530	O00628
TXNL4B	Q9NX01	O14503	U2AF2	P26368	P61978
UBAC1	Q9BSL1	Q09327	UBAP2	Q5T6F2	Q8NFF5
UBAP2L	Q14157	P10636-8	UBE2D1	P51668	Q9P0P0
UBE2D4	Q9Y2X8	Q9P0P0	UBE2E3	Q969T4	Q96NR8
UBE2H	A4D1L5	Q13895	UBE2N	P61088	Q14527
UBE2R2	Q712K3	O00560	UBE2V2	Q15819	Q14527, P61088
UBE2W	Q96B02	Q6ZNA4	UBL5	Q9I9K6	P24864
UBN2	Q6ZU65	P54198	UBTD2	Q8WUN7	P07902
UCK2	Q9BZX2	Q9NWZ5	UCP2	P55851	P20930
UFSP2	Q9NUQ7	Q96FT7	UGDH	O60701	P13569
UGGT2	Q9NYU1	P11021	ULBP1	Q9BZM6	Q6ZMH5
ULBP2	Q9BZM5	P32971	UNC50	Q53HI1	Q9H2K0
UPF1	Q92900	P03496	UQCR10	Q9UDW1	P31930, O14949
UQCRQ	O14949	P31930, Q9UDW1, P14927, P22695	USE1	Q9NZ43	O95249

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
USP1	O94782	Q8TAF3	USP13	Q92995	O60216
USP19	O94966	P07900	USP24	Q9UPU5	O75695
USP36	Q9P275	P06748	USP54	Q70EL1-9	O60733
USP6	EBI-11750223	Q9UBB9	UTP18	Q9Y5J1	P0DTC9
UTP3	Q9NQZ2	P02787	UTRN	P46939	Q6NZI2
UVRAG	Q9P2Y5	P46937	VARS1	P26640	P26641, P68104
VARS2	P26640	P26641, P68104	VASP	P50552	P22736
VAV2	P52735-1, P52735	Q03135	VBP1	P61758	Q13164
VDAC2	P45880	P21796	VEGFC	P97953-1	O60462
VHL	P40337	Q6VMQ6	VIPAS39	Q9H9C1	P12273
VIT	Q6UXI7	P11021	VPS26A	O75436	O60493
VPS39	Q96JC1	P51149	VPS52	Q8N1B4	P46087
VPS53	Q5VIR6	P51665	VPS8	Q8N3P4	P21462
WASHC4	Q2M389	O75396	WBP4	O75554	P62306
WDFY3	Q8IZQ1	Q13501	WDR45B	Q5MNZ6	P51813
WDR7	Q9Y4E6	P21281	WDR70	Q9NW82	Q06330
WDR88	Q6ZMY6-2	Q00613	WNK1	Q9H4A3	Q15149
WNT2B	Q93097	Q70UQ0	WNT3	P56703	P56704
WNT4	P56705	Q5T9L3	WTAP	Q15007-2	P54252
XBP1	P17861	Q16665	XPNPEP3	Q9NQH7	P11362
XPO1	O14980	P04618, P18754	XRCC1	P18887	O96017
XRCC3	O43542	Q86YC2	YAP1	P46937	Q8N3R9
YBX2	Q9Y2T7	Q14197	YEATS4	O95619	P30291
YIPF4	Q9BSR8	Q14973	YLPM1	P49750	P52294
YTHDF1	Q9BYJ9	P08311	YTHDF2	Q9Y5A9	Q99558
YY1AP1	Q9H869	Q16539	ZBTB14	O43829	Q96T60
ZBTB39	O15060	O60341	ZBTB49	Q6ZSB9	Q13114
ZC3H14	Q6PJT7	O75152	ZC4H2	Q9NQZ6	P55040
ZCCHC10	Q8TBK6	O94782	ZCCHC7	Q8N3Z6	P63279
ZCRB1	Q8TBF4	P62306, P14678	ZDHHC13	Q8IUH4	Q9H8X2
ZDHHC17	Q8IUH5	Q8IVB4	ZDHHC20	Q5W0Z9-4	O76024
ZDHHC9	Q9Y397	Q9H8X2	ZEB1	P37275	Q92769
ZFAND5	O76080	P0CG48	ZFAND6	Q6FIF0	Q12933
ZFAT	Q9P243	Q9UER7	ZFHX4	Q86UP3	P48431
ZKSCAN3	Q9BRR0	Q6IPR3	ZMPSTE24	O75844	P02545-1
ZNF133	P52736	P01100	ZNF134	P52741	P55081
ZNF20	P17024	P35240	ZNF219	Q9P2Y4	Q04721
ZNF264	O43296	O15397	ZNF268	Q14587	O15431
ZNF280C	Q8ND82	Q13185	ZNF302	Q9NR11	Q13263
ZNF320	A2RRD8	Q13263	ZNF350	Q9GZX5	P38398
ZNF384	EBI-11743072	P14373	ZNF395	Q9H8N7	P61981
ZNF397	Q8NF99	Q8NHY2	ZNF407	Q9C0G0	Q04721
ZNF460	Q14592	O15397	ZNF462	Q96JM2	Q96KQ7, Q13185
ZNF488	Q96MN9-2	Q00403	ZNF490	Q9ULM2	Q9UGI0
ZNF496	Q96IT1	Q9Y4Z2	ZNF521	Q96K83	Q13547
ZNF536	O15090	P48431	ZNF543	Q08ER8	Q13263
ZNF557	Q8N988	Q13263	ZNF585B	Q9PST7	P04637
ZNF592	Q92610	Q13547	ZNF597	Q96LX8	P51610
ZNF609	O15014	Q13547	ZNF625	Q96I27-2	O76024
ZNF69	Q9UC07-2	Q969Y2	ZNF7	P17097	O15397

Input	UniProt Id	Interacts with
ZNF70	Q9UC06	Q8IXZ2
ZNF74	Q16587	O15397
ZNF8	P17098	Q13185
ZNF816	Q0VGE8	Q9NS56
ZNF852	Q6ZMS4	Q9Y4X5
ZPR1	O75312	P68104
ZSCAN26	Q16670	O76024
ZSWIM7	Q19AV6	P14927
ZXDC	Q2QGD7	O76024

Input	UniProt Id	Interacts with
ZNF724	A8MTY0	P45973
ZNF786	Q8N393	Q9NS56
ZNF808	Q8N4W9	Q13263
ZNF827	Q17R98	Q92769, Q13547
ZNHIT6	Q9NWK9	P00740
ZSCAN22	P10073	Q00403
ZSCAN9	O15535	P17252
ZW10	O43264	P08173

Input	ChEBI Id	Interacts with
BOLA3	Q53S33	33737
GLRX	P35754	16856
RAB4A	P20338	15996
THBS1	P07996	28815

Input	ChEBI Id	Interacts with
FGF1	P05230	29036
MASP2	O00187-PRO_0000027598	29108
RCC1	P18754	17552

7. Identifiers not found

These 2604 identifiers were not found neither mapped to any entity in Reactome.

AATBC	ABCA11P	ABHD13	ABRA	ACYP1	ADARB2-AS1	AFF2	AGAP1-IT1
AKIRIN1	ALKBH3-AS1	ALOX12P2	ALPK2	AMZ1	ANAPC1P2	ANGPTL1	ANK2-AS1
ANKRD36C	ANKRD61	ANKRD7	ANXA2R	ANXA2R-AS1	AOPEP	AP1AR	AP5M1
APRG1	ARAP1-AS2	ARF4-AS1	ARHGAP11A-SCG5	ARHGAP26-AS1	ARHGAP26-IT1	ARHGAP27P1-BPTFP1-KPNA2P3	ARHGEF9-IT1
ARL13A	ARL5A	ARMH1	ASAP1-IT2	ASDURF	ASTN2-AS1	ATF7-NPFF	ATP11A-AS1
ATP5MGL	ATXN2-AS	AVIL	BCYRN1	BMS1P1	BMS1P4-AGAP5	BNC2	BOLA3-DT
BRD10	BRWD1-AS2	BTBD19	BTBD7	BZW1	C10orf67-AS1	C11orf58	C11orf91
C16orf95-DT	C18orf21	C1QTNF7	C1QTNF7-AS1	C1orf53	C20orf203	C21orf62	C2CD2
C2orf74-DT	C2orf80	C2orf88	C3orf49	C8orf44-SGK3	C9orf40	CA5BP1	CACNA1C-AS4
CACNA2D1-AS1	CACNA2D4	CACTIN-AS1	CARD11-AS1	CBR1-AS1	CCDC144NL-AS1.1	CCDC15	CCDC200
CCDST	CCNYL6	CD99P1	CDC14C	CDC20-DT	CDR2-DT	CDYL-AS1	CEACAM16-AS1
CECR3	CELF2-AS1	CELF3	CEP128	CEP170.1	CEP295NL	CFAP20DC-AS1	CFAP298-TCP10L
CGREF1	CHD1-DT	CHRM3-AS1	CHRM3-AS2	CHURC1-FNTB	CIMAP2	CLASRP	CLDN24
CLDN34	CLN8	CLYBL-AS2	CNN3-DT	COL18A1-AS1	COL18A1-AS2	CPEB2-DT	CPVL-AS1
CRADD-AS1	CREB3L2-AS1	CROCC	CRYBB2	CRYBB2P1	CRYZ	CSDC2	CSMD2
CSMD2-AS1	CTD-2350J17.1	CTNNA1-AS1	CXorf65	CYCSP52	CYP2T1P	CYTOR	CYYR1-AS1
CZIB	DANT2	DAPK1-IT1	DDN-AS1	DDT	DDTL	DDX11L16.3	DDX53
DENND6A-DT	DEPDC1	DHRS4L1	DHRS4L2	DIAPH1-AS1	DIAPH2-AS1	DICER1-AS1	DINOL
DIP2A-IT1	DIP2C-AS1	DLEC1	DLEU7	DLGAP1-AS3	DMTF1	DNAH9	DOCK4-AS1
DPY19L3-DT	DTD1	DTD1-AS1	DTD2	DUSP28	EBLN2	EBLN3P	ECE1-AS1
EDDM13	EEF1AKMT4	EFCAB8	EFR3A	EGOT	EHD4-AS1	EHHADH-AS1	ELK4
ELMOD2	ELOA-AS1	ELOF1	EPB41L4A	EPN2-AS1	EPS15-AS1	EQTN	ERC2-IT1
ERI3-IT1	ERLNC1	ERV3-1	ESAM-AS1	ETDA	ETDC	ETS1-AS1	EVA1A-AS
EXPH5	EZR-AS1	FAM157C	FAM180A	FAM180B	FAM184B	FAM186B	FAM193A
FAM87B	FAM95A	FARS2-AS1	FCRLB	FEZ1.1	FGD5-AS1	FGF12-AS1	FGF14-IT1
FILIP1L	FLJ13224	FLJ42393	FLVCR1-DT	FMC1-LUC7L2	FMO6P	FMR1-AS1	FOXD4L5
FOXJ3	FOXN3-AS2	FOXP1-AS1	FREM3	FTX	GABPB1-IT1	GAS5	GDF11
GGNBP2	GHRLOS	GNAS-AS1	GNB1-DT	GNG12-AS1	GOLGA6C	GOLGA6D	GOLGA8A
GOLGA8B	GPC6-AS1	GPR135	GPR171	GPR179	GPR22	GPR34	GPR75
GPR82	GPR87	GPRC5D-AS1	GREB1L	GTF2IRD2	GTPBP10.1	GYPE	H2BC12L.1
H2BP1	HAPSTR1	HCG25	HDAC4-AS1	HEATR4	HEATR5A-DT	HEATR9	HECW1-IT1
HIGD1B	HMGA2-AS1	HMGB4	HMGN1	HOXC10	HSD11B1-AS1	HSPA2-AS1	HSPA5-DT
IBA57-DT	IDNK	IFNK	IGF2BP2-AS1	IGSF23	INSYN2B	INTS4P1	INVS
ITGA6-AS1	ITPK1-AS1	JKAMP	JMJD1C-AS1	JPX	JRKL	KANSL1-AS1	KANTR
KAZALD1	KCNIP4-IT1	KCNJ13	KCNQ1OT1	KIAA1586	KIAA1614-AS1	KIF1C-AS1	KIZ-AS1
KLF7	KLHDC7A	KPNB1-DT	KRBA2	KRTCAP2	KY	LAMA5-AS1	LARGE-AS1
LCMT1-AS2	LENG8-AS1	LETM2	LIMS1-AS1	LINC00102	LINC00205	LINC00216	LINC00294
LINC00309	LINC00310	LINC00324	LINC00484	LINC00595	LINC00598	LINC00598.1	LINC00612
LINC00632	LINC00645	LINC00852	LINC00862	LINC00954	LINC01119	LINC01128	LINC01176
LINC01201	LINC01252	LINC01310	LINC01347	LINC01362	LINC01364	LINC01416	LINC01482

LINC01505	LINC01535	LINC01567	LINC01572	LINC01597	LINC01619	LINC01828	LINC01981
LINC02018	LINC02035	LINC02042	LINC02060	LINC02126	LINC02128	LINC02249	LINC02250
LINC02284	LINC02432	LINC02443	LINC02465	LINC02540	LINC02576	LINC02603	LINC02617
LINC02626	LINC02649	LINC02654	LINC02802	LINC02863	LINC02912	LINC02915	LINC02930
LINC02955	LINC02983	LINC02984	LINC02997	LITAFD	LPP-AS1	LPP-AS2	LRGUK
LRP3	LRRC2	LRRC37A3	LRRC37B	LRRC4	LRRC42	LRRN3	LRTM1
LSP1P4	LSP1P5	LYRM4-AS1	LYSMD3	MAGI1-AS1	MAGI1-IT1	MANSC4	MAP3K13
MAP3K5-AS1	MAP9	MAPK8IP3-AS1	MAPKAPK5-AS1	MARCHF11-AS1	MARCHF4	MATCAP1	MBLAC1
MBNL1-AS1	MCL1.1	MCTS1	MEF2C-AS1	MEGF10	MEIOB	MESTIT1	METTL25
MEX3D	MGC15885	MGC4859	MINAR2	MINCR	MIR101-1	MIR103A2	MIR103B2
MIR1204	MIR1206	MIR1244-4	MIR1251	MIR1255A	MIR1256	MIR1272	MIR1279
MIR1288	MIR1291	MIR1296	MIR133A1HG	MIR181A2	MIR181B2	MIR197	MIR200CHG
MIR205HG	MIR218-1	MIR2467	MIR2909	MIR302CHG	MIR3064	MIR30E	MIR3137
MIR3150BHG	MIR3162	MIR3165	MIR3173	MIR3174	MIR3175	MIR3184	MIR3187
MIR3188	MIR3192	MIR3198-1	MIR3198-2	MIR3651	MIR3654	MIR3655	MIR3662
MIR3679	MIR3680-1	MIR3680-2	MIR3684	MIR374A	MIR378G	MIR378J	MIR3917
MIR3940	MIR3975	MIR3978	MIR423	MIR4253	MIR4263	MIR4295	MIR4305
MIR4312	MIR4324	MIR4426	MIR4427	MIR4435-1	MIR4435-2	MIR4444-1	MIR4530
MIR4638	MIR4639	MIR4642	MIR4644	MIR4647	MIR4659A	MIR4659B	MIR4680
MIR4774	MIR4775	MIR4793	MIR5004	MIR5047	MIR5090	MIR5094	MIR545
MIR548K	MIR548L	MIR548U	MIR550A1	MIR550B1	MIR554	MIR5579	MIR558
MIR571	MIR604	MIR6075	MIR612	MIR6126	MIR617	MIR619	MIR621
MIR632	MIR636	MIR6501	MIR6515	MIR664A	MIR6739	MIR6764	MIR6800
MIR6820	MIR6865	MIR6874	MIR7-1	MIR7111	MIR7705	MIR7854	MIR8066
MIR92A1	MIR933	MIR937	MIRLET7G	MIX23	MORC2-AS1	MRAP-AS1	MROH9
MRPS31P5	MRTFA-AS1	MSANTD2	MSANTD3-TMEFF1	MSL3P1	MTUS1-DT	MXRA5	MYLK-AS2
MYO1H	MYRFL	N4BP2L2-IT2	NACA2	NAIP	NAPSB	NAV3	NBR2
NCOA7-AS1	NDFIP2	NEAT1	NECTIN1-AS1	NGFR-AS1	NHS-AS1	NKAPP1	NLGN1-AS1
NME6	NME9	NORAD	NPIPB11	NPIPB12	NPIPB13	NPIPB2	NPIPB4
NPIPB5	NPIPB8	NPSR1-AS1	NPTX2	NRARP	NREP	NT5DC3	NUCB1-AS1
NUDT16-DT	NUDT17	NUP210L	NUTM2B-AS1	None.1	None.10054	None.1007	None.10071
None.10072	None.10073	None.10075	None.10077	None.10086	None.10087	None.10093	None.10094
None.10095	None.10098	None.10125	None.10126	None.10137	None.10141	None.10148	None.10149
None.10157	None.10181	None.10192	None.10196	None.10199	None.10200	None.10201	None.10203
None.10204	None.10216	None.10217	None.10222	None.10230	None.10233	None.10234	None.10242
None.10251	None.10252	None.10254	None.1026	None.10263	None.10264	None.1027	None.10289
None.10291	None.10292	None.10297	None.10320	None.10322	None.10342	None.10346	None.10368
None.10374	None.10382	None.1060	None.10622	None.10624	None.10630	None.10635	None.10638
None.10642	None.10643	None.10646	None.10647	None.10648	None.10655	None.10656	None.10665
None.10670	None.1068	None.10695	None.107	None.10719	None.10720	None.10729	None.10734
None.10735	None.10743	None.10748	None.10750	None.10756	None.10757	None.10758	None.10759
None.10767	None.10770	None.10786	None.10801	None.10804	None.10807	None.10809	None.10810
None.10812	None.10816	None.10826	None.10832	None.10833	None.10836	None.10850	None.10852
None.10854	None.10859	None.1086	None.10860	None.1087	None.10873	None.10875	None.10877
None.10883	None.10886	None.10887	None.10889	None.10890	None.10891	None.10892	None.10897
None.10900	None.10926	None.10950	None.10953	None.10954	None.10955	None.10960	None.10961
None.10968	None.10969	None.10973	None.10974	None.10979	None.1098	None.10985	None.11
None.11012	None.11017	None.11021	None.11022	None.11023	None.1104	None.11054	None.11063
None.1108	None.11083	None.1109	None.11111	None.11113	None.11115	None.11116	None.11118
None.11120	None.11129	None.1113	None.11157	None.11158	None.11167	None.11171	None.11175

None.11182	None.11183	None.11187	None.11189	None.11190	None.11191	None.11193	None.11197
None.11198	None.11206	None.11218	None.11231	None.11245	None.11246	None.11247	None.11248
None.11251	None.11255	None.11256	None.11257	None.11289	None.11292	None.11294	None.11296
None.11297	None.113	None.11302	None.1134	None.11356	None.1137	None.11391	None.1141
None.11465	None.11476	None.11477	None.11483	None.1149	None.11499	None.11500	None.11501
None.11511	None.11512	None.11524	None.11534	None.11548	None.11554	None.11576	None.11577
None.11578	None.11588	None.1159	None.11612	None.11613	None.11614	None.11619	None.11620
None.11630	None.11631	None.11634	None.11642	None.11652	None.1168	None.11700	None.11702
None.1171	None.11730	None.11757	None.11759	None.11778	None.11786	None.11796	None.11797
None.1180	None.11806	None.11809	None.11810	None.11816	None.11817	None.11819	None.1182
None.11828	None.11829	None.11830	None.11831	None.11832	None.11833	None.11834	None.11865
None.11884	None.11908	None.11912	None.11928	None.11930	None.11934	None.11938	None.11942
None.11943	None.11944	None.11945	None.11951	None.1196	None.11963	None.11965	None.11978
None.1198	None.11986	None.11987	None.11988	None.11990	None.12	None.12005	None.12015
None.12016	None.12018	None.12019	None.12030	None.12035	None.12036	None.12040	None.12041
None.12049	None.1206	None.12093	None.12094	None.12102	None.12112	None.12113	None.12114
None.12119	None.12142	None.12147	None.1215	None.1216	None.12160	None.12161	None.12164
None.12168	None.12169	None.12170	None.12171	None.1218	None.12181	None.12187	None.12193
None.12241	None.12249	None.12253	None.12256	None.12281	None.12285	None.12296	None.12297
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None.12331	None.12332	None.12333	None.12339	None.1234	None.12344	None.12345	None.1235
None.1236	None.12363	None.12368	None.1237	None.12382	None.1239	None.12402	None.12404
None.12419	None.12441	None.12453	None.12454	None.12461	None.12463	None.12465	None.12469
None.12473	None.12474	None.1249	None.12499	None.12500	None.12533	None.12535	None.12539
None.12541	None.12545	None.12552	None.12566	None.12568	None.12574	None.12577	None.12583
None.12604	None.12610	None.12611	None.12619	None.12655	None.1266	None.12662	None.12676
None.12681	None.12691	None.12697	None.12704	None.12728	None.12755	None.12768	None.12772
None.12778	None.12784	None.12786	None.12800	None.12802	None.12812	None.12816	None.12818
None.12841	None.12856	None.12860	None.12861	None.1287	None.12872	None.12897	None.12919
None.1292	None.12942	None.12958	None.12978	None.12979	None.12981	None.13007	None.13015
None.13016	None.13020	None.13021	None.13027	None.13030	None.13043	None.13044	None.13052
None.13053	None.13101	None.13118	None.13121	None.13122	None.13130	None.13134	None.13196
None.13201	None.13202	None.13209	None.13212	None.13221	None.13223	None.13226	None.13227
None.13228	None.13232	None.13233	None.13234	None.13235	None.13236	None.13238	None.13240
None.13241	None.13242	None.13243	None.13244	None.13277	None.13291	None.13293	None.13307
None.13308	None.13309	None.13310	None.13318	None.1332	None.13337	None.13350	None.13368
None.13383	None.13384	None.13386	None.13413	None.13416	None.13418	None.1343	None.13446
None.13483	None.1350	None.1352	None.13521	None.13600	None.13602	None.13608	None.13615
None.13627	None.13636	None.13650	None.13652	None.13653	None.13656	None.13665	None.13681
None.13697	None.13698	None.13709	None.13719	None.13724	None.13725	None.13735	None.13736
None.13748	None.13749	None.13753	None.13755	None.13770	None.13779	None.1378	None.13788
None.13789	None.13791	None.13792	None.13815	None.13829	None.13836	None.13837	None.13840
None.13844	None.13849	None.13852	None.13853	None.13854	None.13883	None.13889	None.13897
None.13920	None.13925	None.13947	None.13948	None.13970	None.13974	None.13977	None.13987
None.13988	None.13989	None.13994	None.13998	None.14	None.14006	None.14007	None.14013
None.14019	None.1402	None.14039	None.14041	None.14051	None.14059	None.1406	None.14085
None.14091	None.141	None.1410	None.1411	None.1414	None.14162	None.14167	None.14168
None.14169	None.14171	None.1418	None.14186	None.14187	None.14188	None.1419	None.14202
None.14203	None.14204	None.14205	None.14207	None.14209	None.14210	None.14211	None.14212
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None.1439	None.1440	None.1456	None.147	None.1478	None.148	None.1481	None.15

None.1503	None.1518	None.1527	None.1529	None.1531	None.1532	None.1540	None.155
None.156	None.1563	None.1566	None.1568	None.1582	None.1587	None.1591	None.1594
None.1608	None.1609	None.1616	None.1618	None.1622	None.1647	None.1664	None.1665
None.1666	None.1675	None.1677	None.1683	None.1684	None.1685	None.1686	None.1687
None.1693	None.1694	None.174	None.1793	None.1809	None.1811	None.1834	None.1857
None.1858	None.1861	None.187	None.1872	None.1886	None.1893	None.1894	None.1897
None.1907	None.1918	None.193	None.1941	None.195	None.1957	None.1969	None.1970
None.1976	None.1977	None.1982	None.1988	None.1989	None.2001	None.2004	None.2036
None.2039	None.2040	None.2042	None.2058	None.206	None.2084	None.2086	None.2087
None.2089	None.2103	None.2114	None.2137	None.2156	None.2157	None.2158	None.2164
None.2169	None.2181	None.2183	None.2197	None.2216	None.2220	None.2222	None.2233
None.224	None.2260	None.2262	None.2263	None.2282	None.2283	None.2284	None.2285
None.2297	None.231	None.2328	None.2330	None.2332	None.2353	None.2360	None.2377
None.2378	None.2379	None.2380	None.2390	None.2393	None.2397	None.2398	None.2407
None.2421	None.244	None.2444	None.2445	None.2446	None.2447	None.2460	None.2463
None.2464	None.2466	None.2474	None.2475	None.2481	None.2484	None.2486	None.2487
None.2489	None.2497	None.2499	None.250	None.2502	None.2507	None.2508	None.2534
None.2565	None.2569	None.2571	None.2575	None.258	None.2581	None.2585	None.2615
None.2626	None.2630	None.2633	None.2635	None.2637	None.2645	None.2647	None.2648
None.2649	None.2650	None.2653	None.2654	None.2670	None.2678	None.2694	None.2703
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None.2777	None.2786	None.2790	None.2792	None.2796	None.2797	None.2805	None.2806
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None.2863	None.287	None.2888	None.2889	None.289	None.2895	None.2897	None.2901
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None.2971	None.2972	None.2975	None.2976	None.2977	None.2978	None.2980	None.2981
None.2985	None.2987	None.3006	None.3008	None.3012	None.3016	None.302	None.3021
None.3026	None.3043	None.3056	None.3060	None.3068	None.307	None.3071	None.308
None.3088	None.3100	None.3103	None.3106	None.312	None.3126	None.3127	None.3130
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None.3193	None.3195	None.3199	None.320	None.3200	None.3201	None.3206	None.3215
None.3243	None.3244	None.3245	None.3258	None.326	None.3261	None.3276	None.3279
None.3292	None.3297	None.3349	None.3353	None.336	None.3364	None.337	None.338
None.339	None.3394	None.3396	None.3407	None.3436	None.3446	None.3453	None.3454
None.3457	None.3458	None.3465	None.3466	None.3469	None.347	None.3479	None.3481
None.3489	None.3491	None.3497	None.3503	None.3510	None.3530	None.3539	None.3541
None.3544	None.3552	None.3553	None.3576	None.3578	None.3585	None.3586	None.359
None.36	None.3625	None.363	None.3647	None.3648	None.366	None.3663	None.3668
None.367	None.3682	None.3684	None.3695	None.3696	None.3699	None.371	None.372
None.3728	None.3740	None.3745	None.3757	None.376	None.3774	None.3783	None.3787
None.3788	None.3807	None.3809	None.3816	None.3817	None.3823	None.3843	None.3882
None.3885	None.3887	None.3897	None.392	None.3928	None.393	None.3932	None.394
None.395	None.396	None.3961	None.3962	None.3987	None.4011	None.4018	None.4039
None.4040	None.4048	None.406	None.4062	None.4071	None.4092	None.4100	None.4101
None.4106	None.4111	None.4115	None.4117	None.4125	None.4147	None.4149	None.4152
None.4153	None.4156	None.4158	None.4159	None.4160	None.4161	None.4165	None.4168
None.4169	None.4171	None.4176	None.4187	None.4191	None.4198	None.4199	None.4200
None.4201	None.4212	None.4214	None.4221	None.4227	None.4228	None.4239	None.424
None.4244	None.4256	None.4262	None.4278	None.4293	None.4294	None.4301	None.433
None.4342	None.4345	None.4358	None.4362	None.4375	None.4378	None.4379	None.4381
None.4385	None.4393	None.4394	None.4411	None.4416	None.4417	None.4420	None.4432

None.4439	None.4480	None.4510	None.4511	None.4522	None.4524	None.4536	None.4537
None.4538	None.4542	None.4561	None.4568	None.4591	None.4592	None.4593	None.4602
None.4605	None.4618	None.4619	None.4620	None.4621	None.4633	None.4658	None.4675
None.4678	None.4689	None.4693	None.4694	None.4706	None.4719	None.4723	None.4724
None.4727	None.4728	None.4729	None.4738	None.4762	None.481	None.485	None.4898
None.4921	None.4923	None.4942	None.4953	None.4955	None.4965	None.4969	None.4970
None.4981	None.4997	None.4998	None.500	None.5000	None.5010	None.5015	None.5016
None.5017	None.5020	None.5021	None.5039	None.5043	None.5044	None.5045	None.5061
None.5095	None.5104	None.5105	None.5106	None.5113	None.5119	None.512	None.5148
None.5160	None.5167	None.5168	None.5169	None.5180	None.5191	None.5211	None.5213
None.5242	None.5246	None.5248	None.5249	None.5250	None.5251	None.5268	None.527
None.5272	None.5278	None.5290	None.5297	None.5308	None.533	None.5344	None.5357
None.5359	None.5360	None.5362	None.5381	None.5386	None.5388	None.5390	None.5410
None.5415	None.5428	None.5438	None.5440	None.5459	None.5513	None.5514	None.5516
None.5525	None.5527	None.5529	None.5530	None.5534	None.5537	None.554	None.5543
None.5544	None.5559	None.5570	None.5582	None.5604	None.5606	None.5609	None.5610
None.5644	None.5645	None.5646	None.5648	None.5651	None.5653	None.5656	None.5661
None.5664	None.5683	None.5689	None.570	None.5719	None.5722	None.5724	None.5725
None.5726	None.5727	None.5728	None.5743	None.5744	None.5750	None.5757	None.5772
None.5783	None.5788	None.5842	None.5846	None.5847	None.5848	None.5849	None.5852
None.5863	None.5864	None.5865	None.5866	None.5874	None.5875	None.5877	None.5881
None.5882	None.5886	None.5887	None.5888	None.5907	None.5910	None.5915	None.5916
None.5918	None.5924	None.5931	None.5932	None.5952	None.5955	None.5956	None.5959
None.5962	None.5970	None.5982	None.5987	None.5988	None.5991	None.5992	None.5996
None.5997	None.6012	None.6028	None.6039	None.6058	None.6063	None.6069	None.6070
None.6076	None.6079	None.608	None.6087	None.6088	None.6089	None.6093	None.610
None.6102	None.6108	None.6115	None.6119	None.6178	None.6193	None.6195	None.6203
None.6223	None.6232	None.6244	None.6270	None.6274	None.6277	None.6286	None.6287
None.6310	None.6321	None.6329	None.6336	None.634	None.6348	None.6374	None.6389
None.6392	None.6393	None.6401	None.6414	None.6415	None.6423	None.6424	None.6427
None.6451	None.646	None.6465	None.6488	None.649	None.6498	None.6500	None.6516
None.6536	None.6540	None.6549	None.6562	None.6563	None.6576	None.6595	None.6647
None.6652	None.6653	None.6663	None.6677	None.669	None.6695	None.6711	None.6712
None.6725	None.6727	None.6733	None.6734	None.6754	None.6761	None.6767	None.6779
None.6782	None.6799	None.6800	None.683	None.6831	None.6838	None.6839	None.6841
None.6847	None.6849	None.6850	None.6876	None.6877	None.6878	None.6883	None.6885
None.6913	None.6916	None.6918	None.6934	None.6948	None.6953	None.6956	None.6962
None.6971	None.698	None.700	None.701	None.7014	None.7015	None.7024	None.7025
None.7026	None.7036	None.7040	None.7041	None.706	None.7067	None.7071	None.7076
None.7077	None.7079	None.709	None.7116	None.7121	None.7140	None.715	None.7161
None.7167	None.7179	None.7207	None.7217	None.7223	None.7227	None.7228	None.723
None.7244	None.7253	None.7256	None.7257	None.7259	None.7260	None.7261	None.7273
None.7324	None.734	None.7340	None.7346	None.736	None.7362	None.7364	None.7368
None.7374	None.7382	None.7387	None.7388	None.7405	None.7410	None.7426	None.7429
None.7447	None.7454	None.7460	None.7461	None.7462	None.7465	None.7480	None.7486
None.7488	None.7490	None.7493	None.7497	None.7502	None.7503	None.7504	None.7506
None.7511	None.7514	None.7515	None.7516	None.7518	None.7527	None.7530	None.754
None.7550	None.758	None.7594	None.7596	None.7601	None.7602	None.7612	None.7614
None.7615	None.7619	None.7683	None.7701	None.7703	None.7706	None.7718	None.7719
None.7720	None.7730	None.7736	None.774	None.7743	None.7757	None.7770	None.7771
None.7795	None.7802	None.7814	None.7841	None.7870	None.7871	None.7872	None.7874

None.7877	None.7879	None.788	None.7890	None.7891	None.7892	None.7896	None.7901
None.7917	None.7926	None.7927	None.7933	None.7939	None.7940	None.7954	None.7955
None.7965	None.7971	None.7972	None.7973	None.7977	None.7978	None.7979	None.7980
None.7981	None.7987	None.7988	None.7990	None.7992	None.7993	None.7994	None.7995
None.8004	None.8014	None.8035	None.8038	None.8041	None.8047	None.8051	None.807
None.8071	None.8089	None.810	None.8102	None.8109	None.8111	None.8132	None.8142
None.8146	None.815	None.8158	None.8161	None.8181	None.8182	None.8185	None.8189
None.8196	None.820	None.821	None.8213	None.8223	None.8224	None.8228	None.8237
None.825	None.8250	None.8255	None.8257	None.8265	None.8267	None.8268	None.8277
None.8284	None.8293	None.8297	None.8300	None.8305	None.8306	None.8312	None.8315
None.8316	None.8320	None.8323	None.8329	None.8333	None.8348	None.8351	None.8358
None.838	None.8380	None.84	None.8405	None.8411	None.8459	None.8477	None.8492
None.85	None.8506	None.851	None.8510	None.8517	None.8518	None.8528	None.8547
None.8553	None.8563	None.8564	None.8566	None.8587	None.8589	None.8597	None.860
None.8601	None.8602	None.8605	None.8606	None.8626	None.865	None.8675	None.868
None.8697	None.870	None.8707	None.8709	None.8738	None.8742	None.8747	None.875
None.8768	None.8769	None.8770	None.8778	None.8799	None.880	None.8800	None.8801
None.8802	None.8803	None.8805	None.882	None.8829	None.883	None.8835	None.8837
None.8840	None.8848	None.8850	None.8858	None.8865	None.8866	None.8878	None.8882
None.8892	None.8896	None.8904	None.8906	None.8907	None.8918	None.892	None.8935
None.894	None.8951	None.8956	None.897	None.8972	None.8973	None.8989	None.9002
None.9005	None.901	None.9014	None.9016	None.902	None.9020	None.9026	None.9035
None.9036	None.9043	None.9045	None.905	None.906	None.9064	None.9067	None.9070
None.9084	None.9087	None.9088	None.9100	None.9103	None.9107	None.9113	None.9114
None.9125	None.913	None.914	None.9146	None.915	None.9150	None.9151	None.9162
None.9206	None.9222	None.9245	None.9248	None.9249	None.9253	None.9270	None.929
None.9294	None.9299	None.9314	None.9316	None.9331	None.9340	None.9342	None.9344
None.9354	None.9356	None.9357	None.9388	None.9397	None.9417	None.9437	None.9440
None.9441	None.9449	None.9452	None.9466	None.9468	None.9485	None.9493	None.9495
None.9496	None.9499	None.9504	None.9505	None.9510	None.9525	None.9533	None.9537
None.9560	None.9561	None.958	None.9591	None.9594	None.9599	None.9610	None.9731
None.9736	None.9748	None.9754	None.9758	None.9763	None.9765	None.9766	None.9767
None.9768	None.9774	None.9797	None.9798	None.9799	None.9800	None.9801	None.9816
None.9819	None.9839	None.984	None.9846	None.9847	None.9860	None.9861	None.9878
None.998	OGFR	OIP5-AS1	OMG.1	OSBPL10-AS1	OTUD6B	OVOL3	PANCR
PAPPA-AS1	PARAIL	PARM1-AS1	PARP11-AS1	PART1	PATE4	PBOV1	PBX1-AS1
PCA3	PCAT1.1	PCDH11Y.1	PCDHGB3	PCDHGB7	PCDHGB8P	PCP4	PDE9A-AS1
PDXP-DT	PDZD9	PGAM4	PHACTR1	PHTF2	PIGBOS1	PIK3IP1-DT	PINLYP
PIWIL4-AS1	PKD1L1	PKDREJ	PKIA-AS1	PLAC4	PLCXD1	PLD5	PLGLB1
PLGLB2	PLPBP	PLS3-AS1	PMF1-BGLAP	POM121L9P	POMZP3	POU5F2	PP2D1
PPIAL4C	PPP1R14B	PPP1R9A-AS1	PRANCR	PRB2	PRDM4-AS1	PRDX6-AS1	PRH2
PRICKLE2	PRKX-AS1	PRKY	PROX2	PRSS48	PSMD6-AS2	PTCHD4	PTER
PTGES3L	PVT1	PWAR1	PWRN1	PWWP3A	PYROXD2	PZP	RAB6D
RABGAP1L-AS1	RAET1K	RAPGEF4-AS1	RARRES1	RBIS	RBM12B	RBM12B-AS1	RBM43
RC3H1	RCBTB1	REXO6P	RFXAP	RGMB-AS1	RGPD1	RHOXF1	RLIG1
RN7SK	RN7SL1	RN7SL2	RN7SL3	RNA5-8SN1	RNASEH2B	RNF113B	RNF148
RNF180	RNF212	RNF216-IT1	RNFT2	RNPEP	RNU1-2	RNU105B	RNU6ATAC
RNVU1-19	ROR1-AS1	RPL13AP17	RPL13AP20	RPL13AP6	RPL23AP7	RPP21.1	RPS6KA2-IT1
RPS6KC1	RPSAP52	RRS1-DT	RSPH10B.1	RSPH4A	RTL8A	RTP4	RWDD3-DT
RWDD4	RXYLT1-AS1	SAYSD1	SCAF4	SCARNA1	SCARNA18B	SCARNA2	SCARNA20

SCARNA5	SCARNA9	SCARNA9.1	SCG5-AS1	SCPPPQ1	SCRG1	SDAD1-AS1	SDCBP2-AS1
SDHAP1	SEMA5A-AS1	SEMA6B	SENCR	SEPTIN7P9	SEPTIN9-DT	SETBP1	SETSIP
SFSWAP	SH3PXD2A-AS1	SHANK2-AS1	SHANK2-AS3	SHBG	SHISAL2B	SHOC1	SLC10A5
SLC16A1-AS1	SLC16A4	SLC24A3-AS1	SLC25A36	SLC49A4	SLC66A3	SLC6A16	SLC7A6OS
SLCO4A1-AS1	SLCO5A1	SLFNL1	SMAD1-AS2	SMANTIS	SMC2-DT	SMCO2	SMCR8
SMIM15	SMIM18	SMLR1	SMN1.1	SMYD3-AS1	SMYD5	SND1-IT1	SNHG1
SNHG15	SNHG16	SNHG17	SNHG21	SNHG3	SNHG32	SNN	SNORA11B
SNORA11F	SNORA16B	SNORA21B	SNORA22	SNORA22B	SNORA26	SNORA28	SNORA2C
SNORA36B	SNORA3B	SNORA40B	SNORA48	SNORA50A	SNORA57	SNORA58B	SNORA59A.1
SNORA66	SNORA70B	SNORA70G	SNORA73B	SNORA74A	SNORA74C-1	SNORA74C-2	SNORA74D
SNORA7B	SNORA80A	SNORA84	SNORD107	SNORD108	SNORD116-30	SNORD117	SNORD118
SNORD12	SNORD121B	SNORD14E	SNORD15A	SNORD15B	SNORD18A	SNORD23	SNORD24
SNORD38B	SNORD3D	SNORD42A	SNORD42B	SNORD4B	SNORD53B	SNORD62A	SNORD62B
SNORD64	SNORD71	SNORD73B	SNORD8	SNORD92	SNORD99	SOCS2-AS1	SORL1-AS1
SOX11	SPACA5	SPAG17	SPATA13-AS1	SPATA31G1	SPATA31H1	SPDYE1	SPDYE10.1
SPDYE11	SPDYE14	SPDYE15	SPDYE16	SPDYE17	SPDYE18	SPDYE2	SPDYE2B
SPDYE3	SPDYE5	SPDYE6	SPDYE8	SPIN2A	SPINK4	SPINK8	SPINT3
SPINT4	SPMIP3	SQLE-DT	SRGAP2-AS1	SRGAP3-AS2	SRI-AS1	SRRM5	SSBP3-AS1
ST20-MTHFS	ST7-OT4	STAG3L4	STAG3L5P	STARD9	STK32A	STK38L	STMP1
STX18-AS1	STX18-IT1	SUMO4	SYNDIG1L	SYNJ2-IT1	SYNJ2BP-COX16	SYNPR-AS1	TAF12-DT
TALAM1	TAS2R14.1	TATDN1	TBC1D19	TCF4-AS1	TCF7L1-IT1	TEKTIP1	TEX13B
TEX38	TFAP2A-AS2	THBS3-AS1	THOC7-AS1	TIMP4	TLCD3A	TLDC2	TMCO6
TMED11P	TMEM123	TMEM135	TMEM14EP	TMEM161B	TMEM161B-DT	TMEM167A	TMEM170A
TMEM18	TMEM268	TMEM274P	TMEM35B	TMEM39A	TMEM41B	TMEM64	TMEM81
TMPRSS11D	TMPRSS11E	TMPRSS5	TNRC6B-DT	TOX	TPM3P9	TRAF3IP2-AS1	TRAPPC2B
TRIM66	TRIM67	TSBP1	TSPAN13	TSPEAR	TSTD3	TTC3-AS1	TTC34
TTC36	TTC39B	TTC39C	TTN-AS1	TTN-AS1.1	TVP23A	TVP23C	TXNDC12-AS1
UBAP1L	UFC1	ULK4	UQCC4	URGCP-MRPS24	USF3	USP34-DT	USP7-AS1
VTRNA1-3	WDR38	WDR49	WFDC3	WWC2-AS1	XNDC1N	XRRA1	YPEL4
ZBED3	ZBTB37	ZC3H11C	ZC3H12B	ZC3H12C	ZFAND3	ZFX	ZHX2
ZNF142	ZNF236	ZNF275.1	ZNF283	ZNF337-AS1	ZNF362	ZNF385D	ZNF460-AS1
ZNF567-DT	ZNF582-DT	ZNF623	ZNF652	ZNF66	ZNF8-DT	ZNF8-ERVK3-1	ZNF83
ZNRF3-AS1	ZSCAN16-AS1	ZSWIM1	gene				